

LM01 Market Organization and Structure

1. Introduction	3
2. The Functions of the Financial System	3
2.1 Helping People Achieve Their Purposes in Using the Financial System.....	3
2.2 Determining Rates of Return	5
2.3 Capital Allocation Efficiency	5
3. Assets and Contracts	5
3.1 Classifications of Assets and Contracts.....	5
4. Securities	6
4.1 Fixed Income.....	6
4.2 Equity.....	7
4.3 Pooled investments	7
5. Currencies, Commodities, and Real Assets.....	7
5.1 Commodities	7
5.2 Real Assets	7
6. Contracts.....	8
6.1 Forward Contracts	8
6.2 Futures Contracts.....	8
6.3 Swap Contracts.....	8
6.4 Option Contracts.....	8
6.5 Other Contracts.....	9
7. Financial Intermediaries	9
7.1 Brokers, Exchanges, and Alternative Trading Systems	9
7.2 Dealers.....	9
7.3 Arbitrageurs	10
8. Securitizers, Depository Institutions and Insurance Companies	10
8.1 Depository Institutions and Other Financial Corporations	11
8.2 Insurance Companies	11
9. Settlement and Custodial Services	11
10. Positions and Short Positions	12
10.1 Short Positions.....	12
11. Leveraged Positions.....	13
12. Orders and Execution Instructions.....	15
12.1 Execution Instructions.....	16
13. Validity Instructions and Clearing Instructions.....	17

13.1 Stop orders	17
13.2 Clearing Instructions	17
14. Primary Security Markets.....	18
14.1 Public Offerings	18
14.2 Private Placements and Other Primary Market Transactions.....	19
14.3 Importance of Secondary Markets to Primary Markets	19
15. Secondary Security Market and Contract Market Structures.....	19
15.1 Trading Sessions	19
15.2 Execution Mechanisms	21
15.3 Market Information Systems.....	21
16. Well-Functioning Financial Systems	21
17. Market Regulation	22
Summary.....	23

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1. Introduction

This reading covers the functions of the financial system, the various assets used by financial analysts, the role of financial intermediaries, different positions one can take like short and long, various types of orders, market participants, primary and secondary markets and, finally, the characteristics of a well-functioning financial system.

2. The Functions of the Financial System

The financial system includes markets and financial intermediaries that help transfer financial assets, real assets, and financial risk between entities from one place to another, and from one point in time to another.

The six purposes people use the financial system for are as follows:

- to save money for the future.
- to borrow money for current use.
- to raise equity capital.
- to manage risks.
- to exchange assets for immediate and future deliveries.
- to trade on information.

Three main functions of the financial system are to:

- achieve the purposes for which people use the financial system.
- discover the rates of return that equate aggregate savings with aggregate borrowings.
- allocate capital to the best uses.

2.1 Helping People Achieve Their Purposes in Using the Financial System

People often use a single transaction to achieve more than one of the six purposes when using the financial system. For example, an investor who buys the stock of a bank may be saving for the future, or trading based on research that the stock is undervalued, or trying to benefit from the central bank's policy to slash interest rates in the medium term. Each of the six purposes listed earlier are discussed in detail below:

Saving

Saving is moving money from the present to the future. By saving, we choose not to spend now and make that money available in the future. One common example is people saving for retirement. The financial system offers various instruments such as bank deposits, stocks, and bonds for this purpose.

Borrowing

Entities like people, companies, and governments often want to spend money now but do not have money. People borrow to buy homes, cars, and education, while companies borrow to fund new projects. Governments borrow to provide better infrastructure, rural development, or other such benefits for its citizens. The financial system facilitates

borrowing by aggregating from savers the funds that borrowers require. In simple terms, these are known as loans.

Raising Equity Capital

Companies raise money for projects by selling equity ownership interests. Instead of taking a loan, they sell a certain percentage of ownership in the company to raise funds. The financial system brings together the companies in need of money and entities providing money in the form of investment banks. Investment banks help companies issue equities, analysts value the securities that companies sell, regulators and standards-setting bodies ensure meaningful financial disclosures are made.

Managing Risk

Entities face financial risks related to exchange rates, interest rates, and raw material prices and might want to hedge these risks.

Example of financial risk management:

Consider a sugarcane producer (typically farmers) and a sugar-refining firm. The sugar-refining firm purchases sugarcane from the farmers and processes them to produce sugar. The sugarcane season typically lasts 150 days in a year but is based on a variety of factors such as amount of rainfall, temperature, pests, etc. Both the farmer and refining firm are worried about what the prices will be when the sugarcane is ready. The farmer fears it will be lower due to overproduction or poor quality of crop, while the refining firm fears it will be higher because of demand, global commodity prices, and production worldwide. By entering in to a forward contract (discussed in detail in the derivatives reading), they eliminate the uncertainty related to changing prices.

Exchanging Assets for Immediate Delivery (Spot Market Trading)

People often trade one asset for another if the value of the other asset is more to them. Examples include currencies, carbon credits, and gold. The financial system facilitates these exchanges when liquid spot markets exist, which removes substantial transaction costs.

Information-Motivated Trading

Information-motivated traders aim to profit from information that they believe allows them to predict future prices. Unlike pure investors, information-motivated traders strive to leverage their information to earn extra return in addition to the normal return expected by investors.

Active investment managers are information-motivated traders who, after a thorough analysis, buy under-valued and sell over-valued securities. Pure investors and information-motivated traders differ in their motives and not so much in the risk they take. The primary motive of the latter is to profit from the superior information they possess.

2.2 Determining Rates of Return

Saving, borrowing, and selling equity are all means of moving money through time. While savers move money from the present to the future, borrowers and issuers of equity move money from the future to the present.

Money can travel forward in time if an equal amount of money is traveling in the other direction. Think of it this way: the instruments in which savers invest are those created by the borrowers. For instance, a bond or a stock that a saver invests in is issued by a government or a company. The company is moving money to now, while the investor is saving it for later.

How much savers save or move consumption to the future is related to the expected return on investments. If the rates are high, investors will want to save more. Similarly, if the cost of borrowing is less for borrowers now, they will want to move more money from the future to the present, i.e., borrow more. The total amount of money saved must equal the total amount of money borrowed to achieve a balance. It will create an imbalance if either one of them is too high or low. If rate of return is low, savers will want to save less now than how much borrowers will want to borrow.

Equilibrium interest rate is the interest rate at which the aggregate supply of funds equals the aggregate demand for funds. Different securities have different equilibrium rates based on their characteristics which are usually a function of risk, liquidity, and time. For instance, investors demand a higher rate of return for equities than debt, long-term investments than short-term investments, or illiquid securities than liquid securities.

2.3 Capital Allocation Efficiency

Primary capital markets are the markets in which companies and governments raise capital. Economies are considered allocationally efficient when capital (money) is allocated to the most productive uses, i.e., projects with the highest NPV or internal rate of return (IRR). Investors actively seek information on the various investment opportunities available before making investment decisions.

3. Assets and Contracts

3.1 Classifications of Assets and Contracts

Classification criteria:		
Based on the underlying	Financial assets: Means by which individuals hold claim on real assets and future income generated by these assets, e.g., securities like stocks and bonds.	Real assets: Include physical assets like real estate, equipment, commodities, and other assets.

Based on the nature of claim by financial securities	Debt securities: Periodic interest payments made on borrowed funds that might be collateralized.	Equity securities: Represent ownership positions and claim on the future cash flows of the business.
Based on where the securities are traded	Publicly traded: These securities trade in public markets through exchanges or dealers and are subject to regulatory oversight.	Privately traded: These securities are not traded in public markets. They are often not subject to regulation.
Based on delivery	Spot market: Markets for immediate delivery of assets.	Forward market: Contracts that call for future delivery of assets and include forwards, futures, and options.
Based on the underlying of the derivative contract	Financial derivative contract: These contracts draw their value from financial assets like equities, equity indexes, debt, and other assets.	Physical derivative contract: These contracts draw their value from real assets like commodities.
Based on issuance of security	Primary market: Issuers sell securities directly to investors.	Secondary market: Investors buy and sell securities among themselves.
Based on maturity	Money market: Securities with maturities of one year or less.	Capital market: Securities that have more than one year maturity or equities that do not have any maturity.
Based on the type of investment markets	Traditional investment markets: Includes all publicly traded debt and equities.	Alternative investment markets: Includes hedge funds, private equity, commodities, real estate, and precious gems that are hard to trade and value.

4. Securities

Securities can be broadly classified into: fixed income, equities and shares in pooled investment vehicles.

4.1 Fixed Income

Refers to debt securities where the borrower is obligated to pay interest and principal at a pre-determined schedule. They might be collateralized, i.e., investors have claim of certain physical assets in case of a default.

The different types are:

- Bonds: Long-term debts.

- Notes: Intermediate-term debts.
- Bank borrowings: Long- to short-term involving revolving credit lines and other debt instruments.
- Convertible: Debt can be exchanged for a specified number of equity shares.

4.2 Equity

Refers to ownership claims by investors in companies.

The different types are:

- Common shareholders: They have a residual claim over any assets and income after all the senior securities have been paid.
- Preferred shareholders: They are paid scheduled dividends before the common shareholders.
- Warrants: They give the holder a right to buy the firm's security at a price, called the exercise price, within a specified time period. (similar to options)

4.3 Pooled investments

Pooled investments include mutual funds, trusts, exchange traded funds (ETFs), and hedge funds. They issue securities to represent the shared ownership in the assets. Money from several investors is pooled together to be managed by a professional money manager according to a specific investment strategy. The advantage of investing in pooled vehicles is to benefit from the investment management services of managers and from diversification opportunities. Pooled vehicles may be open-ended or close-ended.

5. Currencies, Commodities, and Real Assets

Currencies

Currencies are monies issued by national monetary authorities. Reserve currencies such as dollar and euro are currencies that national central banks around the world hold in large quantities. Currencies trade in foreign exchange markets, spot markets, forward markets, or futures markets.

5.1 Commodities

Commodities include precious metals, energy products, industrial metals, agricultural products, and carbon credits. They trade in spot, forward, and futures markets. They are traded in spot markets for immediate delivery and in forwards and futures markets for future delivery.

5.2 Real Assets

Real assets are tangible assets such as real estate, machinery, and airplanes which are normally held by operating companies. Real assets are unique, illiquid, and costly to manage. They are attractive to investors for two reasons:

- Low correlation with other investments.

- Income and tax benefits to investors.

Real estate investment trusts (REITs) and master limited partnerships (MLPs) securitize real assets and facilitate indirect investment in real assets. Since these securities are more homogeneous and divisible than the real assets they represent, they are often more liquid and more suitable as investments.

6. Contracts

A **contract** is an agreement between traders to perform some action in the future that can either be settled physically or in cash.

Based on the underlying asset, contracts can be further classified into:

- Physical contract: If contracts are based on physical assets like crude oil, wheat, gold, or any other commodity, then it is a physical contract.
- Financial contract: If contracts are based on financial assets such as indexes, interest rates, and currencies, then they are called financial contracts.

Contracts for Difference (CFD) allow people to speculate on the price of an underlying asset. The buyer benefits if the price of the underlying asset increases. These are derivative contracts because their value is derived from the underlying asset. They are generally settled in cash.

The major types of contracts (also termed as derivatives) are:

6.1 Forward Contracts

A forward contract is an agreement to trade the underlying asset at a future date at a pre-specified price. It is not standardized and is not traded on exchanges or in dealer markets.

6.2 Futures Contracts

A futures contract is a standardized forward contract for which amount, asset characteristics and delivery date are the same. Standardization ensures higher liquidity.

6.3 Swap Contracts

A swap contract is an agreement to swap payments of one asset for the other. The different types are:

- Interest rate swap: Floating rate payments are swapped for fixed-rate payments for a specified period.
- Currency swap: Currency amount swapped for another currency for a specified period.
- Equity swap: Returns earned on one investment are swapped for the other.

6.4 Option Contracts

Contracts that give the holder a right, but not the obligation, to buy/sell an underlying security at a specified price at or before a specific date. The different types are:

- Call options: Buyer gets the right but not the obligation to buy the underlying security. The seller of the call option gets the premium upfront but has to sell the security if the buyer exercises his option to buy.
- Put options: Buyer gets the right but not the obligation to sell the underlying security. The seller of the put option gets the premium upfront but has to buy the security if the buyer exercises his option to sell.

6.5 Other Contracts

Credit default swaps:

Contracts that offer insurance to bondholders. They make payments to a bondholder if a borrower defaults on its bonds.

7. Financial Intermediaries

Financial intermediaries help entities achieve their financial goals. They provide products and services which help connect buyers to sellers. There are several types of intermediaries:

7.1 Brokers, Exchanges, and Alternative Trading Systems

Brokers:

- They find counterparties for transactions (other entities willing to take the opposing side in a transaction) and do not indulge in trade with their clients directly.

Block brokers:

- Provide similar services as brokers, except that their clients have large trade orders that might potentially impact the security prices if the trade is executed without proper care.

Investment banks:

- They provide advice for corporate actions like mergers and acquisitions and help firms raise capital by issuing securities such as common stock, bonds, preferred shares, etc.

Exchanges:

- They provide places where traders can meet.
- They regulate traders' actions to ensure smooth execution of the trades.

Alternative trading systems (ATS):

- They serve the same trading function as exchanges but have no regulatory oversight.
- ATS where client orders are not revealed are also known as dark pools.

7.2 Dealers

- They trade directly with their clients by taking the opposite side of their trades.
- They provide liquidity by buying or selling from their own inventory and earning profits on the spread between the transactions.

7.3 Arbitrageurs

Arbitrageurs trade when they can identify opportunities to buy and sell identical or essentially similar instruments at different prices in different markets.

Example of an arbitrage opportunity:

Consider a stock of HLL Corp. that trades on two exchanges in a country. If a trader buys the stock from one exchange at a lower price and sells on another at a higher price, then an arbitrage opportunity exists as you can profit at the same time due to differences in prices. If the same instrument (like HLL in the example above) is bought and sold in different markets at different prices, it is pure arbitrage.

If markets are efficient, pure arbitrage opportunities rarely exist. When it does happen, the arbitrageur will engage in transactions that will quickly eliminate this arbitrage. However, buying an instrument in one form and selling it in another form is called **replication**. It is common for arbitrage opportunities to exist between similar instruments. Example: Buy stock and sell overpriced calls for the same stock.

8. Securitizers, Depository Institutions and Insurance Companies

Securitizers

Securitization is the process of buying assets, placing them in a pool, and then selling assets that represent ownership of the pool. One common example is that of mortgage-backed securities or mortgage pass-through securities.

Securitization example:

Take the example of a mortgage bank that gives mortgage loans to a thousand homeowners. Each mortgage loan is like an asset on the bank's balance sheet. If the mortgage bank combines the thousand individual mortgage loans into a pool and sells shares of the pool to investors as securities, then this process is called securitization. The mortgage bank acts as the intermediary as it connects investors who want to buy mortgages with homeowners who want to borrow money. The interest and principal payments from the homeowners are paid to the investors of these securities.

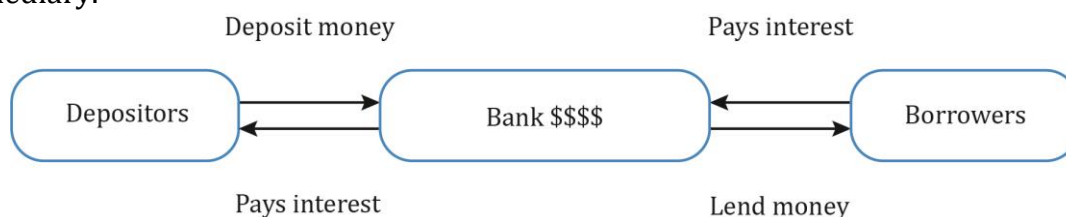
Benefits of Securitization

- Improves liquidity in the mortgage markets as it allows investors to indirectly invest in mortgages that they would otherwise not buy. The risks associated with MBS are more predictable than that of individual mortgages, therefore MBS are easier to price and sell when investors need to raise cash.
- Reduces cost of borrowing for homeowners. Higher liquidity means that investors are willing to pay more for securitized mortgages. This results in higher mortgage prices and lower interest rates.
- Diversification of portfolio for individual investors who wish to invest in mortgages but cannot service it efficiently.
- Losses from default and early prepayments are more predictable.

Besides mortgages, other assets that are securitized include car loans, credit card receivables, bank loans, airplane leases etc.

8.1 Depository Institutions and Other Financial Corporations

Depository institutions include commercial banks, savings and loan banks, credit unions and similar institutions that raise funds from depositors and other investors and lend it to borrowers. The diagram below explains the function of a depository institution as a financial intermediary.



Depositors (or investors) deposit their money in the banks. Banks pay interest to the depositors for using their money and offers services, such as check writing. The banks, in turn, lend this money to borrowers in need of the money. The borrowers pay an interest to the bank. The interest a bank earns from borrowers is usually higher than the interest it pays to the depositors, that is how the bank makes money. The bank is a financial intermediary here as it connects depositors with borrowers. Banks also raise funds by selling equity or issuing bonds of the bank.

8.2 Insurance Companies

Insurance companies help people and companies offset risks by issuing insurance contracts. The contracts make a payment to the party that buys the contracts in case an event occurs. Examples of insurance contracts include life, auto, home, fire, medical, theft, and disaster.

Example of an insurance contract:

Assume you own a car and wish to insure the car against any damages. You buy car insurance from an insurance company and pay a premium at periodic intervals (annually). By doing this, you have transferred the risk of car ownership to the insurance company. In case the car is involved in an accident, the insurance company pays for the damages.

9. Settlement and Custodial Services

A **clearinghouse** helps clients settle their trades. In futures markets, they guarantee contract performance and, hence, eliminate counterparty risk. By requiring participants to post an initial margin and maintain the margin, the clearinghouse ensures there are no defaults. In other markets they may act as escrow agents, transferring money from the buyer to the seller while transferring securities from the seller to the buyer.

Depositories or **custodians** hold securities for their clients so that investors are insulated from loss of securities through fraud or natural disasters.

10. Positions and Short Positions

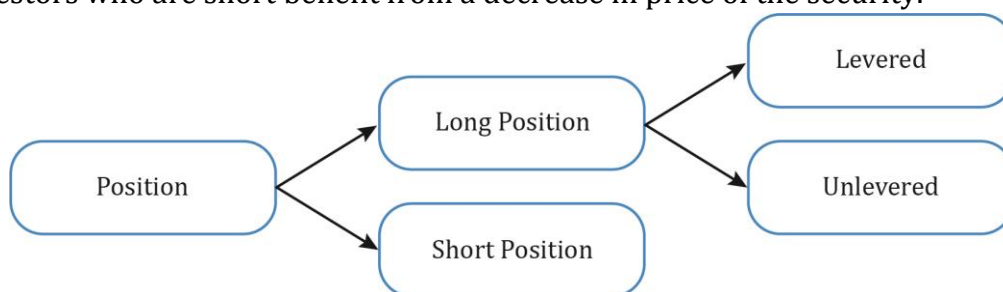
An investor's position in a security may either be a long position or a short position.

Long positions

- These are created when a trader owns an asset or has a right or obligation under a contract to purchase an asset.
- Investors who are long benefit from an increase in price of the security.
- A long position can be levered or unlevered.

Short positions

- These are created when traders borrow an asset and sell it, with the obligation to replace the asset in the future.
- Investors who are short benefit from a decrease in price of the security.

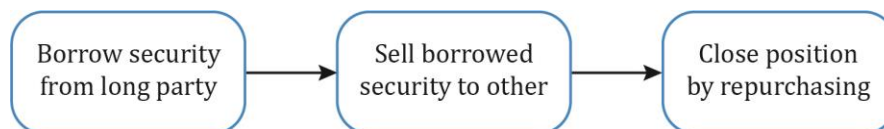


We will now look at each of these positions in detail.

10.1 Short Positions

Short positions are created when traders sell contracts or stocks they do not own. It is similar to borrowing an asset you do not own.

How to create a short position in a security:



Example of a short position:

Assume you research a stock – XYZ Corporation – and forecast its price to go down in the short term. The holder of a short position benefits when the security price goes down. To profit from this view, you borrow securities from a long party and sell the borrowed XYZ stock to other traders when it is trading at \$50. The stock falls to \$40 in line with your forecast. You then close the position by repurchasing and delivering it to the long party, profiting \$10 per share in the process.

The potential gain is bounded, in our example, to a maximum of \$50. That is, the maximum profit you can earn is \$50 if the stock falls from \$50 to \$0. Conversely, the potential loss is unbounded. If the stock's price increases instead of falling, then the short seller incurs a loss and theoretically, there is **no maximum limit** to the loss. This makes a short position very

risky. For a long position, the reverse happens. If you own XYZ stock and the stock's price increases, there is no limit to the maximum profit you can make. However, the loss if the stock falls is limited to \$50.

To secure the security loans given to short sellers, security lenders require that proceeds of the short sale be posted as collateral (\$50 in the example above). The security lender then invests the proceeds in short term securities and pays interest on collateral to short sellers at rates known as **short rebate rates**. Security lenders lend their securities because the short rebate rates they pay on the collateral are lower than the interest rates they receive from investing the collateral.

Short Position: sell the stock (owe the asset)

Maximum gain = 100 % of investment

Maximum loss = unbounded

Long Position: buy the stock (own the asset)

Maximum gain = unlimited

Maximum loss = 100% of investment

11. Leveraged Positions

In some markets, traders are allowed to buy securities by borrowing some percentage of the purchase price. The leverage ratio is a measure of the amount borrowed relative to the total value of the asset. It shows how many times larger a position is than the equity that supports it.

$$\text{Leverage ratio} = \frac{\text{Value of the position}}{\text{Value of the equity investment in it}}$$

The borrowed money is called the margin loan and the interest paid is called the call money rate. Traders who buy securities on margin are subject to margin requirements. The **initial margin requirement** is the minimum percentage of the purchase price that must be paid by the trader (called trader's equity).

Traders usually borrow money from their brokers. The advantage of buying securities on margin is that it increases the amount of profit a trader makes if the share price goes up. If the share price falls to a certain level (the margin call price) the trader will receive a call from the broker (lender) and will be asked to add more money to his account. The minimum amount of equity to be maintained in the positions is called the **maintenance margin requirement**. Traders receive a **margin call** when equity falls below the maintenance margin requirement.

$$\text{Margin call price} = P \times \left(\frac{1 - \text{Initial Margin}}{1 - \text{Maintenance Margin}} \right)$$

Example

Your broker allows you to purchase stocks on margin. The initial margin requirement is 40% and the maintenance margin requirement is 25%. You purchase a stock for \$50 using \$20 of

your money and you borrow the rest from the broker. The interest rate on borrowed money is 5%. What is the leverage ratio? At what rate will you receive a margin call?

Solution:

You borrow \$30, your equity is \$20 and the total value of the asset is \$50.

The leverage ratio is $\frac{50}{20} = 2.5$.

Margin call price = Price $\times \frac{1 - \text{Initial margin}}{1 - \text{maintenance margin}} = 50 \times \frac{1 - 0.4}{1 - 0.25} = 40$.

If the stock price comes down to 40, you still owe the \$30 and your equity has come down to \$10. This is 25% of \$40 (the asset price). If the stock price falls below \$40 the equity becomes less than 25%, the maintenance margin. In this situation, the broker (lender) will ask you to add money to your account such that your equity is at least 25%.

Example

We continue with the earlier example where your initial margin requirement is 40%. You believe stock X will go down in price and decide to short sell 500 shares at the current price of \$30. How does the margin requirement impact you?

Solution:

Proceeds from short sale = $500 \times \$30 = \$15,000$. Just like long buyers buy on margin, even short sellers are required to post a margin amount as a security. If the price goes up, then it is a loss for the short seller (you); to mitigate this risk of loss, the broker requires margin traders to maintain a minimum amount of equity in their positions called the **maintenance margin** requirement. The margin amount required here is $0.4 \times 15,000 = \$6,000$.

The total return to the equity investment in a levered position considers:

Profit or loss on the position

- Margin interest paid
- + Dividends received
- Sales commission

To calculate the return percentage on a leveraged position, we need to divide the total profit by the initial investment. This is illustrated below:

Example

What is the overall return in percentage terms given the following data?

Purchase price = 30

Sales price = 32

Shares purchased = 500

Leverage ratio = 2

Call money rate = 5%

Dividend = \$0.50 per share

Commission = \$0.02 per share

Solution:

Trader's equity = $\frac{\text{Total amount of investment}}{\text{Leverage ratio}} = \frac{30}{2} = 15$ per share i.e. the remaining 15 is borrowed.

Initial investment: (Equity + Commission) x (Number of shares purchased) = 15.02 x 500 = 7,510

Trader's remaining equity after the sale can be computed as:

Proceeds from the sale:	16,000 (32 x 500)
Payoff loan	-7,500 (15 x 500)
Margin interest paid	-375 (0.05 x 15 x 500)
Dividends received	250 (0.5 x 500)
Sales commission paid	-10 (0.02 x 500)
Remaining equity	8365

Total profit = equity at end – initial investment = 8,365 – 7,510 = 855

Total return = $\frac{\text{Total profit}}{\text{Initial investment}} = \frac{855}{7510} = 11.38\%$

12. Orders and Execution Instructions

Brokers, dealers and exchanges arrange the trades between buyers and sellers by issuing orders.

All orders specify the following basic information:

- What instrument to trade (name of the stock, ETF, bond, etc.)
- How much to trade (quantity such as 500 shares of Microsoft Corp.)
- Whether to buy or sell (example: sell Oracle stock)

Most orders have additional instructions:

- Execution instruction: How to fill the order.
- Validity instruction: When the orders may be filled.
- Clearing instruction: How to arrange the final settlement.

In many markets, dealers are willing to buy/sell from traders. The dealer creates the market. Some important terms:

- Bid and ask price: The prices at which dealers are willing to buy are called bid prices. The prices at which dealers are willing to sell are called ask prices. The ask prices are usually higher than the bid prices.
- Bid and ask size: Traders often trade various quantities of a stock at various prices. The quantities for a bid offer are called bid sizes and the quantities for an ask offer are called ask sizes.

- The highest bid in the market is called the best bid and lowest ask in the market is called the best ask. The difference between the best bid and best offer is the market bid-ask spread.
- Bid-ask spreads are an implicit cost of trading. Small bid-ask spread imply lower trading costs and vice-versa.

12.1 Execution Instructions

Execution instructions types are:

Market Orders:

- The order is immediately executed at the best price available.
- It executes the order quickly. However there can be substantial slippages in execution price if a stock is thinly traded.

Limit Orders:

- Sets a minimum execution price on sell orders and maximum execution price on buy orders.
- The order ensures that an investor never exceeds his price limit on a transaction.
- However, there is a possibility that the order may not execute at all if the markets are fast moving or there isn't enough liquidity.

All-or-Nothing Orders:

- These orders will be executed only if the entire quantity can be traded.
- Are beneficial when the trading costs depend on the number of executed trades and not on the size of the order.

Hidden Orders:

- These are large orders that are known only to the brokers or exchanges executing them until the trades are executed.

Iceberg Orders:

- A small visible portion of a large hidden order is executed first to gauge the market liquidity before the entire order is executed.

From a testability perspective, it is important to note the difference between a market order and a limit order.

	Market order	Limit order
Execution	Executed at the best available market price.	Sets a minimum execution price on sell orders and maximum execution price on buy orders.
Advantages	Quick execution when a trader believes that the prices are volatile.	Avoids slippages as the orders are executed at the pre-determined or better prices.

Disadvantages	Quick execution can lead to unfavorable trade prices and has trade price uncertainty.	In a volatile market, the order might be partially filled or not filled at all, making the possibility of missing out on trade.
Additional information	Trader sacrifices price certainty for immediate liquidity.	Types of limit orders: • <u>Marketable or aggressively priced</u>: Limit buy order above the best ask or a limit sell order below the best bid. It will be immediately executed. • <u>Making a new market or inside the market</u>: Limit price is between the best bid and the best ask.
		• <u>Behind the market</u>: Limit buy order with limit price below the best bid, and limit sell order with limit price above the best ask. If the limit prices are way behind the market, they are termed as far from the market limit orders.

13. Validity Instructions and Clearing Instructions

Validity instructions types are:

- **Day orders**: Orders that expire if unfilled for the trading day on which they are submitted.
- **Good-till-cancelled orders**: Orders that last until the buy or sell order is executed.
- **Immediate or cancel (fill or kill) orders**: These orders are to be immediately filled, i.e., when they are received by the broker or exchange. If it fails to execute, the order is canceled from the system.
- **Good-on-close (market-on-close)**: These orders can only be filled at the close of trading. Mutual funds often rely on this order type.

13.1 Stop orders

Also called stop-loss orders, this order comes with a trigger price. Stop-sell order executes only if the price is at or below the stop price or trigger price. Stop-buy order executes only if the price is at or above the stop price or trigger price.

13.2 Clearing Instructions

Clearing instructions tell brokers and exchanges how to arrange final settlement of trades. These instructions convey who is responsible for clearing and settling the trade.

14. Primary Security Markets

Primary markets are where issuers first sell their securities to investors. For example, when a private company goes public, its shares are issued first to the investors in the primary market before it starts trading in the secondary market.

14.1 Public Offerings

Issuers generally contract with an investment bank to help them sell their securities to the public. The investment bank builds the list of subscribers who will buy the security. This process is known as book building. Investment banks attract investors by providing investment information and opinion about the issuing company.

In an accelerated book build, issuers may issue securities with the help of an investment bank in only one or two days.

The two major types of offerings provided by investment banks are the underwritten offering and the best efforts offering.

- **Underwritten offering:** The investment bank guarantees the amount of shares and the price at which they will be sold (think of it as though the issuer has sold the entire issue to the investment bank, who then sells it to other investors through the book building process). This price is called the offering price. Assume the investment bank promises to sell 1,000,000 shares at \$20 and only 800,000 are sold. If the entire issue is not sold, the investment bank buys the remaining securities at the offering price, in this case it buys the remaining 200,000 shares. The issuer pays an underwriting fee of about 7% to the bank for these services.
- **Best efforts offering:** Unlike underwritten offering, in this case the investment bank only serves as a broker to bring investors to the issuer. Any securities not sold in an undersubscribed issue will remain as is.

An **IPO (Initial Public Offering)** is where issuers sell securities to the public for the first time.

- IPO could be oversubscribed or undersubscribed. If the offering price is low, more investors will be interested in subscribing than the number of shares issued (oversubscribed). Similarly, if the price is high, less number of investors will be interested, leading to the issue being undersubscribed.
- Investment banks have a conflict of interest in their dual role as agents and underwriters in choosing the right offering price. As an underwriter, it is in the interests of the investment bank to have the offering price as low as possible. But as agents for issuers, the offering price should be right to raise the required amount of money for the issuer.

A **seasoned or secondary offering** is where an issuer sells additional units of a previously issued security. As an example a company might have raised \$10 million through an IPO and

four years later wants to raise another \$15 million through a secondary offering. Note that the secondary offering is a transaction between the issuer and investors.

14.2 Private Placements and Other Primary Market Transactions

A **private placement** is where corporations sell securities directly to a small group of qualified (sophisticated) investors as opposed to the public. Private placement requires relatively low disclosure requirements because qualified investors are aware of the risks involved. It is less costly than a public offering.

In a **shelf registration**, corporations sell seasoned securities directly to the public on a piecemeal basis over time instead of selling it in a single transaction. They are sold in secondary markets. Consider a publicly traded company that announces the sale of 700,000 shares to a small group of qualified investors at €0.75 per share. This is an example of a private placement and not shelf registration because the company is not selling on a piecemeal basis.

In a **rights offering**, companies distribute the right to buy new stock at a fixed price to existing shareholders in proportion to their holdings. For example, a publicly traded Italian company is raising new capital. Its existing shareholders may purchase three shares for €3.07 per share for every 10 shares they hold.

14.3 Importance of Secondary Markets to Primary Markets

Primary markets are where entities raise money. Secondary markets are markets where investors trade (buy/sell) in securities. The cost of raising capital in primary markets is lower for corporations and governments whose securities trade in liquid secondary markets. In a liquid market, the transaction costs are low to buy/sell a security. Since investors value liquidity, they are willing to pay more for liquid securities. These high prices result in lower costs of capital for issuers.

15. Secondary Security Market and Contract Market Structures

Trading in securities takes place in a variety of structures. We will consider three aspects of market structure:

- Trading Sessions
- Execution Mechanisms
- Market Information Systems

15.1 Trading Sessions

The two categories of securities market based on when they are traded are as follows:

1. Call markets:

- Trade takes place only at specific times of the day where all the traders are present and all bid-ask quotes are used to arrive at one negotiated price.

- Markets are highly liquid when the market is in session and illiquid when the market isn't in session.
- Usually used for smaller markets or to determine the opening and closing prices at stock exchanges.

2. Continuous markets:

- Trades can occur at any time the market is open where the prices are either quote-driven or auction-driven.

The example below illustrates how a large order is filled in a continuous trading market.

Example

At the start of the trading day, the limit order book for stock X looks as follows:

Buyer	Bid Size	Limit Price (\$)	Offer Size	Seller
John	150	30		
Joe	80	31		
Jill	100	32		
		33	40	Sam
		34	60	Simon
		35	120	Sue

Tom submits an order to buy 150 shares, limit \$34. What is the impact on the limit order book?

Solution:

Tom has placed a marketable limit order. He will buy 40 shares from Sam and 60 shares from Simon as these satisfy the limit price criteria of at or below \$34. He will not buy from Sue as hers is a limit order of \$34. Only 100 shares are filled; 50 remain unfilled.

Average price = $0.4 \times 33 + 0.6 \times 34 = 33.6$

In the limit order book, Tom is a buyer with bid size of 50 at a price of \$34. Sam and Simon's orders are removed from the limit order book as they are filled. It looks like this:

Buyer	Bid Size	Limit Price (in \$)	Offer Size	Seller
John	150	30		
Joe	80	31		
Jill	100	32		
Tom	50	34		
		33	40	Sam
		34	60	Simon
		35	120	Sue

15.2 Execution Mechanisms

The three categories of the securities market based on how they are traded are as follows:

1. **Quote-Driven Markets:**

- Trade takes place at the price quoted by dealers who maintain an inventory of the security.
- Dealers provide liquidity in these markets and gain from the difference in bid-ask spread (high in opaque market).
- They are also called over-the-counter markets, price-driven, or dealer markets.

2. **Order-Driven Markets:**

- Trading rules match buyers to sellers, thus making them supply liquidity to each other.
- Trading rules uses two sets of rules:
 - Order matching rules: This establishes the order precedence based on price, their arrival time, and other factors.
 - Trade pricing rules: This determines the price of the transaction.

3. **Brokered Markets:**

- Brokers arrange trades between counterparties.
- Used for instruments that are unique or illiquid, like real estate or art pieces.

15.3 Market Information Systems

The two categories of the securities market based on when the information is disclosed are as follows:

1. **Pre-trade transparent:** Here trade information on quotes and orders is publically available prior to the trades.
2. **Post-trade transparent:** Here trade information on quotes and orders is publically available after the trade.

16. Well-Functioning Financial Systems

Why do we need a well-functioning financial system?

- So that investors can save (move money from the present to the future) and obtain a fair rate of return.
- Borrowers can borrow money easily (move money from the future to the present).
- Hedgers can offset their risks.
- Traders can trade currencies for commodities.

Four characteristics of a well-functioning financial system include:

- Well-developed markets trade instruments that help people solve their financial problems.
- Liquid markets with low cost of trading (operationally efficient markets) where commissions, bid-ask spreads and order price impacts are low.

- Timely and accurate financial disclosures that allow market participants to forecast the value of securities (support informationally efficient markets).
- Prices that reflect fundamental values (informationally efficient markets).

17. Market Regulation

The role of a market regulator is to ensure fair trading practices. The objectives of market regulation are to:

- Prevent fraud.
- Control agency problems by setting minimum standards of competence for agents.
- Promote fairness.
- Set mutually beneficial standards such as IFRS or U.S. GAAP.
- Prevent undercapitalized firms from exploiting their investors by making excessively risky investments.
- Ensure that long-term liabilities are funded.

Summary

LO: Explain the main functions of the financial system.

The curriculum outlines six purposes for why people use the financial system:

- To save money for the future.
- To borrow money for current use.
- To raise equity capital.
- To manage risks.
- To exchange assets for immediate and future deliveries.
- To trade on information.

Three main functions of the financial system are to:

- Achieve the purposes for which people use the financial system.
- Discover the rates of return that equate aggregate savings with aggregate borrowings.
- Allocate capital to the best uses.

LO: Describe classifications of assets and markets.

Classification criteria:		
Based on the underlying	Financial assets	Real assets
Based on the nature of claim by financial securities	Debt securities	Equity securities
Based on where the securities are traded	Publicly traded	Privately traded
Based on delivery	Spot market	Forward Market
Based on the underlying of the derivative contract	Financial derivative contract	Physical derivative contract
Based on issuance of security	Primary market	Secondary market
Based on maturity	Money market	Capital market
Based on the type of investment markets	Traditional investment markets	Alternative investment markets

LO: Describe the major types of securities, currencies, contracts, commodities, and real assets that trade in organized markets, including their distinguishing characteristics and major subtypes.

Securities can be broadly classified into:

- Fixed Income
- Equity

- Pooled investments

A contract is an agreement among traders to do something in the future. Contracts can be settled physically or in cash. Contracts can be further classified into physical or financial contracts based on the underlying asset. Examples of contracts are:

- Forward contract
- Futures contract
- Swap contract
- Options

Currencies are monies issued by national monetary authorities. Currencies trade in foreign exchange markets in the spot market, forward markets, or futures markets.

Commodities include precious metals, energy products, industrial metals, agricultural products, and carbon credits. They trade in spot, forward, and futures markets.

Real assets are tangible assets that are normally held by operating companies.

LO: Describe types of financial intermediaries and services that they provide.

Brokers, Exchanges, and Alternative Trading Systems:

- Brokers are agents who fill orders for their clients; they do not trade with their clients but search for traders who are willing to take the other side of their clients' orders.
- Investment banks provide advice and help companies raise capital by issuing securities such as common stock, bonds, preferred shares, etc.
- Exchanges provide places where traders can meet to arrange their trades.
- Dealers trade with their clients, i.e., by taking the opposite side of their clients' trades. One of the primary services a dealer provides is liquidity.
- Alternative trading systems (ATS) serve the same trading function as exchanges but have no regulatory oversight.

Depository institutions include commercial banks, savings and loan banks, credit unions and similar institutions that raise funds from depositors and other investors and lend them to borrowers.

Insurance companies help people and companies offset risks by issuing insurance contracts; the contracts make a payment to the party that buys the contracts in case an event occurs.

A clearinghouse helps clients settle their trades.

Depositories or custodians hold securities for their clients so that investors are insulated from loss of securities through fraud or natural disaster.

LO: Compare positions an investor can take in an asset.

Long positions are created when a trader owns an asset or has a right or obligation under a contract to purchase an asset.

Short positions are created when traders borrow an asset and sell it, with the obligation to replace the asset in the future.

In general, investors who are long benefit from an increase in the price of an asset and those who are short benefit when the asset price declines.

LO: Calculate and interpret the leverage ratio, the rate of return on a margin transaction, and the security price at which the investor would receive a margin call.

Leverage ratio = Value of the position / value of the equity investment in it

Margin call price = $P * (1 - \text{Initial Margin}) / (1 - \text{Maintenance Margin})$

The total return to the equity investment in a levered position considers:

Profit or loss on the position

- Margin interest paid
- + Dividends received
- Sales commission

To calculate the return percentage on a leveraged position, we need to divide the total profit by the initial investment.

LO: Compare execution, validity, and clearing instructions.

Execution Instructions indicate how to fill orders. The most common execution orders are:

- Market Orders
- Limit Orders
- All-or-Nothing Orders
- Hidden Orders
- Iceberg Orders

Validity instructions specify when an order should be executed. Different types of validity instructions include:

- Day orders
- Good-till-cancelled orders
- Immediate or cancel (fill or kill) orders
- Good-on-close (market-on-close)
- Stop orders (also called stop-loss orders)

Clearing instructions tell brokers and exchanges how to arrange final settlement of trades. These instructions convey who is responsible for clearing and settling the trade.

LO: Compare market orders with limit orders.

	Market order	Limit order
Execution	Executed at the best available market price.	Sets a minimum execution price on sell orders and maximum execution price on buy orders.

Advantages	Quick execution when a trader believes that the prices are volatile.	Avoids slippages as the orders are executed at the pre-determined or better prices.
Disadvantages	Quick execution can lead to unfavorable trade prices and has trade price uncertainty.	In a volatile market, the order might be partially filled or not filled at all, making the possibility of missing out on trade.

LO: Define primary and secondary markets and explain how secondary markets support primary markets.

Primary markets are where issuers first sell their securities to investors. The two major types of offerings are underwritten offering and best efforts offering.

- IPO (Initial Public Offering) is where issuers sell securities to the public for the first time.
- Seasoned or secondary offering is where an issuer sells additional units of a previously issued security.

Private placement is where corporations sell securities directly to a small group of qualified (sophisticated) investors as opposed to the public.

Secondary markets are where investors trade (buy/sell) in securities. The companies do not raise money from secondary markets. The cost of raising capital becomes low in primary markets when securities trade in liquid secondary markets.

LO: Describe how securities, contracts, and currencies are traded in quote-driven, order-driven, and brokered markets.

Quote-Driven Markets: Customers trade at the price quoted by dealers. They are also called over-the-counter markets, price-driven or dealer markets. Dealers provide liquidity in these markets.

Order-Driven Markets: Trading is based on the rules to match buyers to sellers. In order-driven markets, traders supply liquidity to each other. Orders are matched using an order-matching system run by the trading system such as exchange, or broker.

Brokered Markets: Brokers arrange trades between customers; used for instruments that are unique or illiquid.

LO: Describe characteristics of a well-functioning financial system.

Four characteristics of a well-functioning financial system include:

- Well-developed markets trade instruments that help people solve their financial problems.
- Liquid markets with low cost of trading (operationally efficient markets) where commissions, bid-ask spreads and order price impacts are low.
- Timely and accurate financial disclosures that allow market participants to forecast the value of securities (support informationally efficient markets).

- Prices that reflect fundamental values (informationally efficient markets).

LO: Describe objectives of market regulation.

Markets are regulated to prevent fraud, control agency problems, promote fairness, set mutually beneficial standards, prevent undercapitalized firms from exploiting their investors by making excessively risky investments, and ensure that long-term liabilities are funded.

LM02 Security Market Indexes

1. Introduction	2
2. Index Definition and Calculations of Value and Returns	2
2.1 Calculation of Single-Period Returns.....	2
2.2 Calculation of Index Values over Multiple Time Periods.....	3
3. Index Construction	3
3.1 Target Market and Security Selection	3
3.2 Index Weighting.....	3
4. Index Management: Rebalancing and Reconstitution	9
4.1 Rebalancing	9
4.2 Reconstitution	9
5. Uses of Market Indexes	9
6. Equity Indexes	9
7. Fixed-Income Indexes	10
7.1 Construction.....	10
7.2 Types of Fixed-Income Indexes	10
8. Indexes for Alternative Investments	11
8.1 Commodity indexes.....	11
8.2 Real Estate Investment Trust Indexes	11
8.3 Hedge Fund Indexes.....	11
Summary	13

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Version 1.0

1. Introduction

An index is an indicator, sign, or measure of something. Since an index is a single measure and reflects the performance of the entire security market, it makes it easy for investors to measure and track performance.

Security market indexes were first introduced as a simple measure to reflect the performance of the U.S. stock market. Dow Jones Average, the world's first security market index, was introduced in 1884 comprising only nine railroad and two industrial companies. Until then, investors gathered data of individual securities to assess performance.

Now, security market indexes have multiple uses that help an investor track performance of various markets, estimate risk, and evaluate the performance of an investment. Major indexes include S&P 500, FTSE, and Nikkei.

This reading defines what a security market index is, explains how to calculate the returns of an index, how indexes are constructed, the need for market indexes, and the types of indexes.

2. Index Definition and Calculations of Value and Returns

A security market index measures the value of different target markets such as security markets, market segments, and asset classes. The index value is calculated on a regular basis using actual or estimated prices of constituent securities. Constituent securities are the individual securities comprising an index.

Each index often has two versions based on how the return is calculated:

- A **price return index** or price index measures only the percentage change in price of the constituent securities within the index.
- A **total return index** considers the prices of constituent securities and the reinvestment of all income (dividend and/or interest) since inception.

The value of both versions will be the same at inception. However, as time passes, the value of the total return index will exceed the value of the price return index.

2.1 Calculation of Single-Period Returns

The price return and total returns for an index can be computed using the following formulae:

Price return of an index:

$$PR_I = (V_{PRI1} - V_{PRI0})/V_{PRI0}$$

where:

PR_I = price return of an index (in decimal)

V_{PRI1} = value of the price return index at the end of the period

V_{PRI0} = value of the price return index at the beginning of the period

Total return of an index:

$$TR_I = \frac{V_{PRI1} - V_{PRI0} + Inc_1}{V_{PRI0}}$$

where

TR_I = total return of the index portfolio

V_{PRI1} = value of the price return index at the end of the period

V_{PRI0} = value of the price return index at the beginning of the period

Inc_1 = income from all the securities in the index over the period

2.2 Calculation of Index Values over Multiple Time Periods

Once returns are calculated for each period, the calculation of index values over multiple periods is done by geometrically linking returns.

For example, if the value of a total return index at the start of period 1 is 100 and the total returns over three periods are: 16%, 11%, and -4%, the index value at the end of period three will be: $100 \times 1.16 \times 1.11 \times 0.96 = 123.61$.

3. Index Construction

Constructing and managing an index is similar to building a portfolio of securities. The difference is that an index is a paper portfolio but a real portfolio consists of actual securities. The following factors must be considered when constructing a security index:

- Target market. E.g., U.S. equities.
- Security selection. E.g., large cap securities.
- Weight allocated to each security in the index.
- Index rebalancing.
- Reconstitution.

3.1 Target Market and Security Selection

The target market determines the investment universe. It can be defined broadly (for example, all U.S. equities) or narrowly (for example, large cap telecom stocks in China). If the target market is U.S. equities, then the constituent securities for the index will come from the universe of U.S. equities. The target market may also be based on market capitalization, asset class, geographic region, industries, sizes, exchange, and/or other characteristics.

3.2 Index Weighting

Index weighting determines how much of each security to include in the index. This decision impacts index value. We will see four methods to determine the weight of the securities in an index:

- Price weighting
- Equal weighting
- Market-capitalization weighting
- Fundamental weighting

For each weighting method, there could be a price return index or a total return index.

Price-Weighted Index

The weight of each security is calculated by dividing its price by the sum of all prices. One example of a price-weighted index is the Dow Jones Industrial Average.

$$\text{Price-weighted index} = \frac{\text{Sum of stock prices}}{\text{Divisor (number of stocks in the index adjusted for splits)}}$$

Example

Consider three securities A, B, and C comprising an index with the following beginning of period (BOP) and end of period (EOP) values. Using a divisor of 3, compute a) the index value, b) the price return and the total return.

	Beginning of period price	Beginning of period weight	End of period price	Dividends/share
A	4	20%	2	0
B	6	30%	6	1
C	10	50%	14	2

Solution:

Using the above equation, value of the index at start of the period = $\frac{\text{Sum of the security values}}{\text{Divisor}}$

$$= \frac{20}{3} = 6.67$$

$$\text{Value of index at end of the period} = \frac{22}{3} = 7.33$$

$$\text{Price return} = \frac{7.33 - 6.67}{6.67} = 9.89\%$$

$$\text{Dividend return} = \frac{\text{Income}}{\text{Beginning of period price}} = \frac{3}{20} = 15\%$$

$$\text{Total return} = \text{Price return} + \text{Dividend return} \approx 25\%$$

The divisor is adjusted to remove the impact of stock splits, security addition or deletion.

Example

In the previous example, if there is a 2-for-1 split in stock C during the period, what is the impact on index value and return calculations?

Solution:

Initial divisor was 3 and end-of-index value = 7.33. End-of-period price of C is 7 after the split.

The divisor must be adjusted to prevent the stock split and the new weights from changing the value of the index.

$$\text{Value of index} = \frac{\text{Sum of constituent securities}}{\text{Divisor}} = \frac{2+6+7}{3}$$

$$7.33 = \frac{15}{\text{Divisor}}$$

$$\text{Divisor} = 2.05$$

Note that every time there is a stock split, the value of the divisor will decrease.

Advantage of price weighted index: Simplicity.

Limitations of price weighted index:

- Results in arbitrary weights for securities.
- If the price of a security is high, it will receive a relatively high weight, even though its market capitalization might be low.

Equal Weighted Index

The equal weighting method assigns an equal weight to each constituent security at inception.

An equal weighted index can be created by allocating an equal amount of money to all securities.

Let's say, you have \$180,000 to invest. You will invest \$60,000 each in shares of A, B, and C trading at \$4, \$6, and \$10 respectively. This would mean 15,000 shares of A, 10,000 shares of B, and 6,000 shares of C. However, at the end of the period, the index will no longer be equally weighted as share prices may have changed. So, it requires rebalancing (buy shares of depreciated stock, sell shares of appreciated stock) for the index to be equal weighted.

The return of an equal weighted index is calculated as a simple average of the returns of the index stocks.

$$\text{Equal weighted index} = \text{Initial index value} * \left(1 + \frac{\text{average of percentage change in prices}}{100}\right)$$

Example

Given the following data, compute the price return and total return.

	Beginning of period Price	End of period Price	Dividend/share
A	4	2	0
B	6	6	1
C	10	14	2

Solution:

Price return for A: -50%; B: 0%; C: 40%.

$$\text{Since weights are equal, price return} = \frac{-50+0+40}{3} = \frac{-10}{3} = -3.3\%.$$

Dividend return for A: 0%; B: 16.67%; C: 20%.

$$\text{Total dividend return} = \frac{0 + 16.67 + 20}{3} = \frac{36.67}{3} = 12.22\%.$$

$$\text{Total return} = \text{Price return} + \text{Dividend return} = -3.3 + 12.22 = 8.9\%.$$

Advantage of equal weighting: Simplicity.

Limitations of equal weighting:

- Securities with largest market value are underrepresented; those with lowest market value are overrepresented.
- Maintaining equal weights requires frequent rebalancing. If not rebalanced periodically, the chances of drifting away from the weights are high.

Market-Capitalization Weighted Index

In this method, the weight of each security is determined by dividing its market capitalization with total market capitalization.

$$\text{Weight of a security} = \frac{\text{Market cap of the security}}{\text{Total market cap of all index securities}}$$

$$\text{Market Capitalization index} = \frac{\text{current total market value of index stocks}}{\text{base year total market value of index stocks}} * \text{base year index value}$$

Example

The following data is given:

	Shares outstanding	Beginning of period price	End of period price	Dividends per share
A	500	4	2	0
B	100	6	6	1
C	100	10	14	2

1. Given the data, what divisor must be used such that the initial index value is 1,000?
2. Compute: 1) the final index value 2) the price return and total return.
3. Compute the price return if stock C has a market float of 40%.

Solution:

1. Sum of market capitalization of all securities = $500 \times 4 + 100 \times 6 + 100 \times 10 = 3,600$

$$\text{Initial index value} = 1,000 = \frac{3,600}{\text{divisor}}; \text{divisor} = 3.6.$$

This value of the divisor is used to calculate the index value anytime in the future.

2. The weights of the three securities are tabulated below:

	Price return	Market capitalization weights
A	$(2 - 4) / 4 = -0.5$	$2,000 / 3,600 = 0.56$

B	$(6 - 6) / 6 = 0$	$600 / 3,600 = 0.17$
C	$(14 - 10) / 10 = 0.4$	$1,000 / 3,600 = 0.28$

$$\text{Final index value} = \frac{500 \times 2 + 100 \times 6 + 100 \times 14}{3.6} = \frac{3,000}{3.6} = 833.33.$$

$$\text{Price return} = \frac{833.33 - 1000}{1000} = -16.67\%.$$

Price return can also be calculated as:

$$\text{Price return} = w_A \times \text{PR}_A + w_B \times \text{PR}_B + w_C \times \text{PR}_C = 0.56 \times (-50) + 0.17 \times 0 + 0.28 \times 40 = -16.8\%.$$

$$\text{Dividend return} = \frac{0 + 1 \times 100 + 2 \times 100}{3600} = 8.3\%.$$

$$\text{Total return} = -16.67 + 8.3 = \sim -8.3\%.$$

3. Assume the remaining 60% of stock C is not available for trading as the founding family owns them. Only 40% of shares are available for trading. To calculate the price return, instead of using 100%, only 40% of shares are used in calculation. In this case, 40 shares.

$$\text{The sum of market capitalization of all securities} = 500 \times 4 + 100 \times 6 + 40 \times 10 = 3,000$$

$$\text{Initial index value} = 1,000 = \frac{3,000}{\text{divisor}}; \text{divisor} = 3.$$

$$\text{Final index value} = \frac{500 \times 2 + 100 \times 6 + 40 \times 14}{3} = \frac{2,160}{3} = 720.$$

$$\text{Price return} = \frac{720 - 1000}{1000} = -28\%.$$

A float-adjusted market-capitalization weighted index weights each of its constituent securities by price and the number of its shares available for public trading, i.e., by excluding the shares held by the promoter group, etc.

Advantages of market-capitalization weighting: Constituent securities are correctly represented in proportion to their value in the market.

Limitations of market-capitalization weighting: Securities whose prices have risen or fallen the most see a big change in their weights. Stocks whose prices have increased are over weighted; similarly, stocks whose prices have fallen are underweighted.

Fundamental Weighted Index

Fundamental weighting addresses the disadvantages of using market capitalization as weights. Instead of using a stock's price as a measure, fundamental weighting uses measures such as book value, cash flow, revenue, earnings, and dividends to calculate the weight of each security. For instance, a stock with higher earnings yield (earnings/price) than the overall market will have more weight in a fundamental-weighted index than in a market-weighted index. This weighting method is biased towards value stocks. This is sometimes called a 'value tilt' and is illustrated in the example below.

Example

Compute the price return for the following index. Weight the securities based on earnings.

	Shares outstanding (in million)	Beginning of period price	Earning (in \$ million)	End of period price
A	500	4	20	2
B	100	6	20	6
C	100	10	20	14

Solution:

All the three companies have earnings of \$20 million and total earnings of \$60 million.

Earnings yield, earnings weight, and price return of the three companies:

	Earnings yield	Earnings weight	Price return
A	$20 / (500 \times 4) = 1\%$	$20 / 60 = 33.3\%$	$(2 - 4) / 4 = -0.5$
B	$20 / (100 \times 6) = 3.3\%$	$20 / 60 = 33.3\%$	$(6 - 6) / 6 = 0$
C	$20 / (100 \times 10) = 2\%$	$20 / 60 = 33.3\%$	$(14 - 10) / 10 = 0.4$

$$\text{Price return} = w_A \times PR_A + w_B \times PR_B + w_C \times PR_C$$

$$= 0.33 \times (-50) + 0.33 \times 0 + 0.33 \times 40 = -3.3\%.$$

All the three securities have equal weights here as the earnings are equal. Under the market capitalization method, A would have highest weight and B would have the lowest weight. In other words, a value stock like B (low P/E ratio or high earnings yield) has more weightage in the fundamental-weighted method than it would have in the market-capitalization method.

Summary of Results

The table below compares all the weighting methods.

	Number of shares	BOP price	EOP Price	Earnings	Dividends/share
A	500	4	2	20	0
B	100	6	6	20	1
C	100	10	14	20	2

Method	Price Return	Total Return
Price	10%	25%
Equal	-3.3%	8.9%
Market Cap	-16.7%	-8.3%
Fundamental	-3.3%	8.9%

The pros and cons of the different index weighting methods are shown below.

Method	Pros	Cons
Price	Simple	Arbitrary weights.
Equal	Simple	High market cap stocks are under-represented. Requires frequent rebalancing.
Market Cap	Securities held in proportion to their value	Influenced by overpriced securities.
Fundamental	Value tilt	Does not consider market value. Requires rebalancing.

4. Index Management: Rebalancing and Reconstitution

4.1 Rebalancing

Rebalancing means adjusting the weights of constituent securities in an index to maintain the weight of each security in the index. The weights do not remain constant as the prices of securities change. For weighting methods like price-weighted and market-weighted index, rebalancing is not necessary as the weight is determined by the price. However, as we saw in the case of equal-weighting method, the weights digress heavily when the price of a security appreciates/depreciates. If rebalancing happens too often, then the transaction costs will be high. If rebalancing does not happen often enough, then the portfolio will digress from equal weights.

4.2 Reconstitution

Reconstitution is the process of changing the constituent securities in an index. It is part of the rebalancing cycle. The frequency of reconstitution varies from index to index. When a constituent security no longer meets the necessary criteria it is removed from the index and a new security is added. For example, a stock might be part of a large-cap index but after an erosion of over 80% of its market cap it no longer meets the large cap criteria. This stock will be removed from the index and another one which meets the criteria will be added.

5. Uses of Market Indexes

Security indexes serve the following purpose:

- Index performance serves as a proxy of market sentiment.
- Investment management performance can be better evaluated in comparison with a suitable index that serves as a benchmark.
- Serves as a proxy for measuring and modeling returns, systematic risk, and risk-adjusted performance.
- Serves as a proxy for asset class performance in asset allocation models.
- Useful in creation of passive portfolios that track index funds and ETFs.

6. Equity Indexes

Equity indexes can be classified into:

Broad market index

- Provides a proxy for the overall market performance.
- Typically, 90% of the securities in the market are represented in the index.
- Example: Wilshire 5000 index

Multi-market index

- Constructed from several indexes of different countries.
- Countries included can be based on national markets, geographic region (Latin America index), development groups (emerging market index), etc.

Sector index

- Constructed to track performance of a specific economic sector such as finance, technology, energy, health care, etc., or on a national or global basis.

Style index

Constructed to track performance of securities that are classified based on characteristics like:

- Market capitalization: Securities are classified based on market capitalization to form indexes like large-cap, mid-cap, and small-cap indexes.
- Value/Growth: Includes securities based on value/growth criteria to form growth and value indexes. (uses price-to-earnings and dividend yields to classify securities)
- Combination of market capitalization and value/growth: Includes these combinations: Large-cap value, large-cap growth, mid-cap value, mid-cap growth, small-cap value, small-cap growth indexes.

7. Fixed-Income Indexes

7.1 Construction

Compared to equity indexes, fixed-income indexes are difficult to construct and replicate. They are challenging to construct because:

- There are a large number and variety of fixed-income securities ranging from zero coupon bonds to callable and puttable bonds. Pricing data is not always available.
- Many fixed-income securities are not liquid, i.e., not easy to replicate.

7.2 Types of Fixed-Income Indexes

Like equities, fixed-income securities can be classified based on the issuer, geographic region, maturity, type of issuer, market sector, style, credit quality, currency of payments, etc. The following table illustrates how the fixed-income securities can be organized based on various dimensions.

Dimensions of Fixed Income Indexes	
Market	Global Regional Country or currency zone
Type	Corporate Collateralized/securitized/mortgage backed Government agency Government
Maturity	Short term (e.g. < 1 year) Medium term (e.g. 7 - 10 years) Long term (e.g. 20 + years)
Credit Quality	Investment grade (e.g. S&P rating of BBB or above) High yield

8. Indexes for Alternative Investments

8.1 Commodity indexes

Commodity indexes consist of futures contracts on one or more commodities such as agricultural products (like wheat and sugar), precious metals (like gold), and energy (like crude oil). It is important to recognize the following points related to commodity indexes:

- Since commodity indexes are based on futures indexes, the performance of the index and the underlying commodities can be different.
- It is common to have multiple indexes with the same commodities but in different proportions or weights. For example, while one commodity index may have a higher weight for energy, the other may be overweight on agricultural products. This also leads to a different risk-return profile.

8.2 Real Estate Investment Trust Indexes

Real estate indexes represent markets for real estate securities (such as REITs) and the market for actual real estate. Examples of actual real estate investments include properties such as apartment buildings, retail malls, office buildings, etc. Real estate is a highly illiquid market with few transactions and non-transparent pricing. There are several types of real estate indexes: appraisal indexes, repeat sales indexes, and REIT indexes. This material is covered in detail under alternative investments.

8.3 Hedge Fund Indexes

Hedge fund indexes reflect the returns on hedge funds. Research organizations collect data on hedge fund returns and compile this information into indexes. Since hedge funds are not required by regulation to report their performance, the research firms rely on voluntary cooperation of hedge funds to report returns. Here are some important points to consider when evaluating hedge fund indexes:

- Constituents determine the index.

- Poorly performing hedge funds are less likely to report.
- Returns of hedge fund indexes are likely to be overstated/biased upward due to survivorship bias.

Summary

LO: Describe a security market index.

An index is a single measure that reflects the performance of the entire security market. It makes it easy for investors to measure and track performance.

LO: Calculate and interpret the value, price return, and total return of an index.

Price return index or price index measures only the percentage change in price of the constituent securities within the index.

$$PR_I = (V_{PRI1} - V_{PRI0}) / V_{PRI0}$$

Total return index reflects the prices of constituent securities and the reinvestment of all income (dividend and/or interest) since inception.

$$TR_I = (V_{PRI1} - V_{PRI0} + Inc_1) / V_{PRI0}$$

Calculation of index values over multiple periods is done by linking returns.

LO: Describe the choices and issues in index construction and management.

Index providers must consider the following:

- Which target market should the index represent? E.g., U.S. Equities.
- Which securities should be selected from that market? E.g., Large cap securities.
- How much weight should be allocated to each security in the index?
- When should the index be rebalanced?
- When should the security selection and weighted decision be re-examined?

Target market can be defined broadly or narrowly. It may also be based on asset class, geographic region, industries, sizes, exchange, and/or other characteristics.

LO: Compare the different weighting methods used in index construction.

Index weighting determines how much of each security to include in the index. This decision impacts index value. Various methods used to determine the weight of the securities in an index are:

Price Weighting: The weight on each security is determined by dividing its price by the sum of all prices.

Equal Weighting: Assign equal weight to each constituent security at inception.

Market-Capitalization Weighting: Weight of each security is determined by dividing its market capitalization with total market capitalization.

Fundamental Weighting: Instead of using a stock's price as a measure, fundamental weighting uses measures such as book value, cash flow, revenue, earnings, and dividends to calculate the weight of each security.

Method	Pros	Cons
Price	Simple	Arbitrary weights.
Equal	Simple	High market cap stocks are under-represented. Requires frequent rebalancing.
Market Cap	Securities held in proportion to their value	Influenced by overpriced securities.
Fundamental	Value tilt	Does not consider market value. Requires rebalancing.

LO: Calculate and analyze the value and return of an index given its weighting method.

$$\text{Price weighted index} = \frac{\text{Sum of Stock Prices}}{\text{No. of stocks in index adjusted for splits}}$$

Market Capitalization index

$$= \frac{\text{current total market value of index stocks}}{\text{base year total market value of index stocks}} * \text{base year index value}$$

$$\text{Equal weighted index} = \text{Initial index value} * \left(1 + \frac{\text{average of percentage change in prices}}{100}\right)$$

LO: Describe rebalancing and reconstitution of an index.

Rebalancing means adjusting the weights of an index's constituent securities. The weight of each security in an index should reflect the weighting method used. The weights do not remain constant as the prices of securities change.

Reconstitution is the process of changing the constituent securities in an index. It is part of the rebalancing cycle. The frequency of reconstitution varies from index to index.

LO: Describe uses of security market indexes.

The most important use of indexes is that they give a sense for how a particular security market performed over a particular period. Indexes also serve as:

- Indicators (gauges) of market sentiment.
- Proxies for measuring and modeling returns, systematic risk, and risk adjusted performance.
- Proxies for asset classes in asset allocation models.
- Benchmarks to evaluate the performance of a portfolio.
- Model portfolios for index funds and ETFs.

LO: Describe types of equity indexes.

Equity indexes can be classified into: broad market, multi-market, sector, and style indexes.

The broad market index tries to represent the entire market. Typically, 90% of the securities of the selected market are represented in the index.

The multi-market index includes indexes from different countries as they represent multiple security markets based on national markets, geographic region, development groups, etc.

The sector index focuses on a specific economic sector such as consumer goods, finance, energy, health care, technology, etc., on a national or global basis.

The style index contains securities based on certain characteristics like market capitalization, value, growth, or a combination of any of these.

The market-capitalization index contains securities based on market capitalization such as large cap, mid cap and small cap.

The value/growth index contains a group of stocks based on value/growth criteria.

The market-capitalization and value/growth index combine the three market capitalization groups with value/growth classification resulting in the following six basic index style categories: Large-cap value, large-cap growth, mid-cap value, mid-cap growth, small-cap value, small-cap growth.

LO: Describe types of fixed-income indexes.

Dimensions of Fixed Income Indexes	
Market	Global Regional Country or currency zone
Type	Corporate Collateralized/secured/mortgage backed Government agency Government
Maturity	Short term (e.g. < 1 year) Medium term (e.g. 7-10 years) Long term (e.g. 20+ years)
Credit Quality	Investment grade (e.g. S&P rating of BBB or above) High yield

LO: Describe indexes representing alternative investments.

Commodity indexes consist of futures contracts on one or more commodities such as agricultural products (like wheat and sugar), precious metals (like gold), and energy (like crude oil).

Real estate indexes represent markets for real estate securities (such as REITs) and the market for actual real estate.

Hedge fund indexes reflect the returns on hedge funds. Research organizations collect data on hedge fund returns and compile this information into indexes.

LO: Compare types of security market indexes.

Security market indexes represent asset classes and target markets that can be classified based on geographic location, sector, industry, economic growth, value stocks, growth stocks, etc. Some globally known indexes include Dow Jones Industrial average, S&P, Barclays Capital Global aggregate Bond Index, etc.

LM03 Market Efficiency

1. Introduction	2
2. The Concept of Market Efficiency	2
2.1 The Description of Efficient Markets	2
2.2 Market Value versus Intrinsic Value	2
3. Factors Affecting Market Efficiency Including Trading Costs	2
4. Forms of Market Efficiency	3
4.1 Weak Form	3
4.2 Semi-strong Form	3
4.3 Strong Form	4
5. Implications of the Efficient Market Hypothesis	4
6. Market Pricing Anomalies – Time Series and Cross-Sectional	5
6.1 Time-Series Anomalies	5
6.2 Cross-Sectional Anomalies	5
7. Other Anomalies, Implications of Market Pricing Anomalies	5
8. Behavioral Finance	5
Summary	7

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Version 1.0

1. Introduction

Market efficiency concerns the extent to which market prices incorporate available information. Investors are interested in market efficiency because if prices do not fully incorporate information, then opportunities exist to make abnormal profits. Governments and regulators are interested in market efficiency because market efficiency promotes economic growth.

2. The Concept of Market Efficiency

2.1 The Description of Efficient Markets

- An **informationally efficient market** is one in which asset prices reflect new information quickly and rationally.
- 'Quick' is relative to the time a trader takes to execute an order. If it takes 15 minutes for new information to be incorporated into security prices and trade execution time is 30 minutes, we can say the new information is incorporated quickly.
- Market prices should not react to information that is well anticipated; only unexpected information should move prices.
- In a perfectly efficient market investors should use a passive investment strategy because active investment strategies will underperform due to transaction costs and management fees.

2.2 Market Value versus Intrinsic Value

- **Market value** of an asset is its current price at which the asset can be bought or sold.
- **Intrinsic value** is the value that would be placed on an asset by investors if they had full knowledge of the asset's characteristics.
- In highly efficient markets, full information is available in the market and is reflected in asset prices. Therefore, market value = intrinsic value.
- However, if markets are not efficient, the two prices can diverge significantly.

3. Factors Affecting Market Efficiency Including Trading Costs

The following factors affect a market's efficiency:

- **Market Participants** – More participants increase efficiency.
- **Information availability and financial disclosure** – More information increases efficiency.
- **Limits to trading** – Limitations on arbitrage and short selling decrease efficiency.

Transaction Costs and Information-Acquisition Costs

Two types of costs are incurred by traders when trading on market inefficiencies: transaction costs and information-acquisition costs. These costs should be considered when evaluating a market's efficiency.

- **Transaction costs** – High costs decrease efficiency.

- **Information-acquisition costs** – High costs decrease efficiency.

4. Forms of Market Efficiency

The table below introduces three forms of market efficiency which are differentiated based on assumptions about the level of information in security prices.

Forms of Market Efficiency	Market Prices Reflect:		
	Past Market Data	Public Information	Private Information
Weak form	Yes	No	No
Semi-strong form	Yes	Yes	No
Strong form	Yes	Yes	Yes

Evidence that investors can consistently earn abnormal returns by trading on the basis of information would challenge the efficient market hypothesis.

4.1 Weak Form

In a weak-form efficient market:

- security prices fully reflect **all past market data**.
- non-market public and private information is not necessarily incorporated into the stock price.
- technical analysts cannot make abnormal returns on a consistent basis simply by analyzing historical market information.

Tests to check whether securities markets are weak-form efficient:

- Look at patterns of prices. Is there any serial correlation in security returns? If yes, the market is not weak-form efficient.
- Can trading rules or any technical analysis method involving historical data be used to make abnormal profits? If yes, then it contradicts weak-form efficiency.

4.2 Semi-strong Form

In a semi-strong-form efficient market:

- prices reflect **all publicly known and available information**. This includes financial data such as earnings and dividends, and trading data such as closing prices, volume, etc. Weak-form is a subset of semi-strong-form.
- prices adjust quickly and accurately to new public information.
- efforts to analyze publicly available information are futile.
- fundamental analysis will not lead to abnormal returns in the long run. Lots of fundamental analysts (active investors, portfolio managers) evaluating securities to buy/sell help the market in becoming semi-strong-form efficient.

Tests to check whether securities markets are semi-strong-form efficient:

- Researchers test for when markets are semi-strong-form efficient using event studies. Most research indicates that developed securities markets are semi-strong-form efficient while developing countries' markets may not be semi-strong-form efficient.

4.3 Strong Form

In a strong-form efficient market:

- prices reflect all **public and private information**. It encompasses semi-strong and weak form.
- investors will not be able to earn abnormal profits by trading on private information.

Tests to check whether securities markets are strong-form efficient:

- Researchers test whether a market is strong-form efficient by testing whether investors can earn abnormal profits by trading on non-public information.
- Most research indicates that markets are not strong-form efficient as regulations prohibit the use of private information (or insider trading).

5. Implications of the Efficient Market Hypothesis

We can draw the following implications of the efficient market hypothesis:

Form of Market Efficiency	Implication	Conclusion
Securities markets are weak-form efficient.	Investors cannot earn abnormal returns by trading on the basis of past trends in price.	Technical analysts assist markets in maintaining weak-form efficiency.
Securities markets are semi-strong-form efficient.	Analyst must consider whether the information is already reflected in security prices and how any new information affects a security's value.	Fundamental analysts assist markets in maintaining semi-strong-form efficiency.
Securities markets are NOT strong-form efficient.	Investors trading on private information can make abnormal profits.	Regulations try to prevent insider trading.

If markets are semi-strong form efficient, active portfolio managers cannot outperform the market on a consistent basis, therefore investors should invest passively.

The role of portfolio managers is not necessarily to beat the market, but to establish and manage portfolios consistent with their clients' objectives and constraints.

6. Market Pricing Anomalies – Time Series and Cross-Sectional

A market anomaly is something that challenges the idea of market efficiency. Some anomalies observed in the market are:

6.1 Time-Series Anomalies

- Calendar anomalies: The returns in January are higher than in any other month, especially for small firms. This phenomenon is known as the January effect.
- Momentum and overreaction anomalies: Momentum effect refers to the findings that stocks that have experienced high-returns in the short term tend to continue to generate higher return in subsequent periods.
Overreaction effect is based on the idea that investors often overreact to events or release of unexpected public information. For example, it has been observed that stocks that have had poor returns in the past three-to-five years (losers) tend to outperform the market in subsequent periods.

6.2 Cross-Sectional Anomalies

- Size effect: Small-cap stocks tend to perform better than large-cap stocks.
- Value effect: Value stocks (stocks with lower P/E, P/B or high dividend yields) tend to perform better than growth stocks.

7. Other Anomalies, Implications of Market Pricing Anomalies

- Closed-end investment fund discounts: Closed-End investment funds sell at a discount to NAV.
- Earnings surprise: Investors can earn abnormal profits by buying stock of companies with positive earnings surprise and selling those with negative earnings surprise.
- IPOs: Prices rise on listing day, but underperform in the long term.
- Predictability of returns based on prior information: Research has found that equity returns are related to prior information such as interest rates, inflation rates, stock volatility, and dividend yields.

In practice, it is not easy to trade and benefit from anomalies. Most research concludes that anomalies are not violations of market efficiency, but are the result of statistical methods used to detect anomalies.

Many anomalies might simply be a result of **data mining**. At times researchers carefully analyze data and form a hypothesis. This is the opposite of what should happen. Ideally, a hypothesis should be formed and then the data should be analyzed to accept or reject the hypothesis.

8. Behavioral Finance

Behavioral finance uses human psychology to explain investment decisions. Some irrational behavior and biases observed in the market are:

- Loss aversion: Investors dislike losses more than they like gains of the same amount.
- Herding: In herding, investors ignore their private information and act as other investors do.
- Overconfidence: Overconfident investors do not process information. They place too much confidence in their ability to process and analyze information and, thus, value a security.
- Information cascades: Information cascade is when people observe the actions of a handful of market participants and blindly follow their decisions. The informed participants act first and their decision influences the decisions of others.

Other behavioral Biases

- Representativeness: Investors with this bias will assess probabilities based on events seen before, or prior experiences, instead of calculating the outcomes.
- Mental accounting: Investors divide investments into separate mental accounts, they do not view them as a total portfolio.
- Conservatism: Investors tend to be slow to react to changes.
- Narrow framing: Investors focus on issues in isolation.

Behavioral Finance and Investors

Behavioral biases affect all investors irrespective of their experience. An understanding of behavioral finance will help individuals make better decisions, both individually and collectively.

Behavioral Finance and Efficient Markets

If investors must be rational for efficient markets, the existence of behavioral biases implies that the markets cannot be efficient. If the effects of the biases did not cancel each other out, then the markets could not be efficient. But, since investors are not making abnormal returns consistently, the markets can be considered efficient. Evidence supports market efficiency. In other words, markets can be considered efficient even if market participants exhibit seemingly irrational behavior.

Summary

LO: Describe market efficiency and related concepts, including their importance to investment practitioners.

In an informationally efficient market, asset prices reflect new information quickly and rationally. 'Quick' is relative to the time a trader takes to execute an order. In an efficient market, it is not possible to consistently achieve superior abnormal returns. Prices should only react to unexpected information. In an efficient market, passive investment strategy is preferred over active investment strategy.

LO: Contrast market value and intrinsic value.

Market value is the price at which an asset can be bought or sold. Intrinsic value is the value based on complete information. In highly efficient markets, complete information is available in the market which is incorporated in the stock price. Therefore, market value = intrinsic value.

LO: Explain factors that affect a market's efficiency.

- Market Participants
- Information availability and financial disclosure
- Limits to trading
- Transaction costs
- Information-acquisition costs

LO: Contrast weak-form, semi-strong-form, and strong-form market efficiency.

Forms of Market Efficiency	Past Market Data	Public Information	Private Information
Weak form	Yes	No	No
Semi-strong form	Yes	Yes	No
Strong form	Yes	Yes	Yes

LO: Explain the implication of each form of market efficiency for fundamental analysis, technical analysis, and the choice between active and passive portfolio management.

- If markets are weak-form efficient, then technical analysts cannot make abnormal returns on a consistent basis simply by analyzing historical market information.
- Fundamental analysis will not lead to abnormal returns in the long run if the market is semi-strong-form efficient.
- In a strong-form efficient market, investors will not be able to earn abnormal profits by trading on private information.

LO: Describe selected market anomalies.

Time Series anomalies:

- Calendar anomalies: The returns in January are higher than in any other month, especially for small firms. This phenomenon is known as the January effect.
- Momentum and overreaction anomalies: Investors overreact to events or release of unexpected public information.

Cross-sectional anomalies:

- Size effect: Small-cap stocks tend to perform better than large-cap stocks.
- Value effect: Value stocks tend to perform better than growth stocks.

Other anomalies:

- Closed-end fund discounts: Closed-End funds sell at a discount to NAV.
- Earnings surprise: Investors can earn abnormal profits by buying stock of companies with positive earnings surprise and selling those with negative earnings surprise.
- IPOs: Prices rise on listing day, but underperform in the long term.
- Predictability of returns based on prior information: Research has found that equity returns are related to prior information such as interest rates, inflation rates, stock volatility, and dividend yields.

LO: Describe behavioral finance and its potential relevance to understanding market anomalies.

Behavioral finance examines if investors act rationally, how investor behavior affects financial markets, and how cognitive biases may result in anomalies.

Some of the observed irrational behaviors include:

- Loss aversion: Traditional finance assumes that investors are risk averse. Behavioral finance suggests that humans are loss averse.
- Herding: Herding is where one set of investors follows another set of investors for no rational reason.
- Overconfidence: The overconfidence bias explains pricing anomalies. Overconfident investors do not process information. They place too much confidence in their ability to process and analyze information and, thus, value a security.
- Information cascades: Information cascade is when people observe the actions of a handful of market participants (or experts) and follow their decisions.
- Representativeness: Investors with this bias will assess probabilities based on events seen before, or prior experiences (instead of calculating the outcomes).
- Mental accounting: Investors divide money into different buckets, they do not view their assets as a whole but allocate based on goals.
- Conservatism: Investors tend to be slow to react to changes.
- Narrow framing: Investors focus on issues in isolation.

LM04 Overview of Equity Securities

1. Importance of Equity Securities	2
1.1 Equity Securities in Global Financial Markets	2
2. Characteristics of Equity Securities	2
2.1 Common Shares	2
2.2 Preference Shares	3
3. Private versus Public Equity Securities	3
4. Non-Domestic Equity Securities	4
4.1 Direct Investing	4
4.2 Depository Receipts	4
5. Risk and Return Characteristics	6
5.1 Return Characteristics of Equity Securities	6
5.2 Risk of Equity Securities	6
6. Equity and Company Value	7
6.1 Accounting Return on Equity	7
6.2 The Cost of Equity and Investors' Required Rates of Return	8
Summary	9

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1. Importance of Equity Securities

In this reading, we look at the different types of equity securities, how private equity securities differ from public equity securities, the risk involved in investing in equities, and the relationship between a company's cost of equity, its return on equity, and investors' required rate of return.

1.1 Equity Securities in Global Financial Markets

In 2008, the U.S. contributed about 21% to the global GDP, but its contribution to the total capitalization of global equity markets was around 43%.

Historically, equity markets have offered high returns relative to government bonds and T-bills but at higher risk. The volatility in equity markets was high during key crises such as World War I, World War II, the Tech Crash of 2000-2002, the Wall Street Crash, and the most recent credit crash of 2007-2008. In the recent crash, while the world markets fell by 53%, Ireland was the worst hit incurring losses of over 70%.

An important point to note is that equity securities are a key asset class for global investors.

2. Characteristics of Equity Securities

2.1 Common Shares

Common shares represent an ownership interest in a company and give investors a claim on its operating performance, the opportunity to participate in decision-making, and a claim on the company's net assets in the case of liquidation.

Common shareholders can vote on major corporate governance decisions such as election of its board of directors, the decision to merge with another company, selection of auditors etc. If a shareholder cannot attend the annual meeting in person, he can 'vote by proxy' i.e. have someone else to vote on his behalf.

Statutory voting versus cumulative voting

In statutory voting each share is entitled for one vote. In cumulative voting, a shareholder can cumulate his total votes and choose one particular candidate. For example, let's say that a shareholder holds 100 shares and is supposed to vote for the election of three board members' position. In statutory voting, he can vote 100 votes for each position while in cumulative voting, he can vote all the 300 votes to a single candidate thereby increasing his likelihood of winning. Cumulative voting is beneficial to minority shareholders.

Different classes (Class A and Class B)

A firm can have different classes of equity shares which may have different voting rights and priority in liquidation. For example: Class A shares would have more votes than Class B shares.

2.2 Preference Shares

Preference shares are a form of equity in which payments made to preference shareholders take precedence over payments to common shareholders.

Cumulative and non-cumulative preference shares

- Cumulative: If dividends are not paid out for year one and two, year three dividends would be sum of the third year's dividends plus the non-paid out dividend of years one and two.
- Non-cumulative: If dividends are not paid out for year one and two, and the firm decides to pay dividends in the third year, it would only have to pay third year dividends.

Participating and non-participating preference shares

- Participating: As the name implies, preferred shareholders participate in the firm's profit. Shareholders receive extra dividends than the pre-specified rate in case of higher profits. The shareholders also receive a higher proportion of firm's asset than the par value in case of liquidation.
- Non-participating: Shareholders receive only the pre-specified rate even if the firm earns higher profits. The shareholders only receive the par value in case of liquidation.

Convertible preference shares

- Convertible preference shares are those that can be converted to common stock and hence have lower risk and the inherent option to gain from a firm's future profits.

3. Private versus Public Equity Securities

Private equity refers to the sale of equity capital to institutional investors via private placement. The key characteristics of private equity are:

- Less liquidity as shares are not publicly traded.
- Price discovery can be biased as the security is not available for valuation by a broad base of public participants.
- Management can focus on long-term value creation as it doesn't have to worry about reporting results to market.
- Lower reporting costs due to lesser regulatory requirements.
- Potentially weaker corporate governance due to lesser regulatory requirements.
- Potential for generating high returns when investment is exited.

The types of private equity are:

Venture capital:

- Refers to capital provided to firms in early stages of development.
- The three stages of funding include: seed/startup capital, early stage, and mezzanine financing.

- Investors can range from family and friends to wealthy individuals and private equity funds.
- Investments are illiquid and require a commitment of funds for a relatively long period of time, typically 3 to 10 years.

Leveraged buyout:

- Large amount of debt relative to equity is used to buy out a firm.
- The large proportion of debt amplifies returns if the buyout turns out to be successful.
- Leveraged buyout performed by management is termed as **Management Buyout (MBO)**.
- The firm acquired either has to generate the adequate cash flows or sell assets to service the debt.

Private investment in public equity: A public company, which needs additional capital immediately, sells equity to private investors.

4. Non-Domestic Equity Securities

A market is said to be “**integrated**” with the global market if capital flows freely across its borders. However, some countries place restrictions on capital flows.

The key reasons why capital flows into a country’s equity securities might be restricted is:

- To prevent foreign entities from taking control of domestic companies.
- To reduce volatility of financial markets which can rise by the constant inflow and outflow of capital.
- To provide domestic investors the advantage of earning better returns.

The two ways to invest in the equity of companies in a foreign market are:

- Direct investing
- Depository receipts

4.1 Direct Investing

It refers to directly buying and selling securities in foreign markets. Some potential issues associated with direct investing are:

- Along with the stock performance, the returns are exposed to the currency risk as the trade is made in foreign currency.
- Investors must be aware of the investment environment and laws of the foreign land.
- The disclosure requirement of the foreign country might be low, impeding the analysis process.

4.2 Depository Receipts

A depository receipt (DR) is a security that trades like an ordinary share on a local exchange and represents an economic interest in a foreign company.

Process of creating a DR

A foreign company's shares are deposited in a local bank, which in turn issues receipts representing ownership of specific number of shares. The receipts then trade on a local exchange in local currency price. For example, a Japanese firm's shares are held by a UK bank, which then issues DR representing this stock to the UK citizens. The depository bank is responsible for handling dividends, stock splits, and other events.

Based on the foreign company's involvement, DR can either be:

- **Sponsored DR:** Foreign company is involved in issuance and holders of DR are given voting rights.
- **Unsponsored DR:** Foreign company is not involved in issuance and the bank retains the voting rights.

Based on the geography of issuance, DRs can either be:

- **Global depository receipt (GDR):**
 - DRs issued outside the company's home country and outside the U.S.
 - GDRs are issued by a depository bank which is located or has branches in the countries on whose exchanges the shares are traded.
- **American depository receipt (ADR):**
 - USD denominated DRs that trade like common shares on U.S. exchanges.
 - Some ADRs allow firms to raise capital and use shares to acquire other firms in the US.
- **Global registered shares (GRS):**
 - Shares traded on different stock exchanges in different currencies.
- **Basket of listed depository receipts (BLDR):**
 - Is an ETF representing a collection of DRs.

Types of ADRs

The table below shows the four types of ADRs:

	Level I	Level II	Level III	Rule 144A
Objectives	Broaden U.S. investor base with existing shares.	Broaden U.S. investor base with existing shares.	Broaden U.S. investor base with existing shares. Attract new investors.	Access qualified institutional buyers.
Raising capital on U.S. markets?	No	No	Yes, through public offerings.	Yes, through private placements or QIBs.

SEC Registration	Required	Required	More registration required	Not required
Trading places	Over-the-counter (OTC)	Stock exchanges	Stock exchanges	Private placement
Listing Fees	Low	High	High	Low
Earnings requirements	None	Size constraint is applicable.	Size constraint is applicable.	None

5. Risk and Return Characteristics

5.1 Return Characteristics of Equity Securities

There are two sources of total return for equities: capital gains (or price change) and dividend income. That is, how much the stock appreciates in price and how much dividend is paid by the company during that period. For investors who buy foreign securities directly or through depository shares, there is another source of income: foreign exchange gains or losses due to currency conversion.

5.2 Risk of Equity Securities

Risk is based on uncertainty of future cash flows. A stock's return is from the price change and dividends paid. Since a stock's price is uncertain, the expected future return is uncertain. The standard deviation of the equity's expected total return measures this risk.

The table below shows the risk characteristics of different types of equity securities.

Risk characteristics of different types of equity securities		
Common shares vs. preference shares. Preference shares are less risky.	Preference Shares - Dividends on preference shares are fixed as a percentage of the par value. - Dividends are paid before common shares. - On liquidation, preference shareholders get par value of the shares.	Common Shares - Returns are unknown as can be from capital gains (price appreciation) and dividends. - On liquidation, common shareholders have residual claim, i.e., they get paid after claims of debt and preferred shares have been met; hence the amount to be received is unknown. - Foreign investments are subject to currency exposure risk.

Cumulative vs. non-cumulative preference shares. Cumulative shares are less risky.	1. Any unpaid dividends are accumulated and paid before common stock dividends are paid.
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6. Equity and Company Value

Companies issue equity in primary markets to raise capital and increase liquidity. A company needs capital for the following reasons:

- to finance revenue-generating activities (organic growth). The capital is used to purchase long-term assets, invest in profit-generating projects, expand to new territories, or invest in research and development.
- to make acquisitions (inorganic growth).
- to provide stock-based and option-based incentives to employees.
- in some cases, if the company is cash-strapped, it needs the capital to keep it a going concern, fulfill debt requirements, and maintain key ratios.

The goal of a company's management is:

- to increase book value or shareholders' equity on a company's balance sheet. Management has control over the book value as it can increase net income or sell and purchase its own shares. If the company pays little or no dividends and retains the earnings, then book value increases. Book value = assets - liabilities.
- to ensure that the stock price rises (maximizing market value of equity). Management cannot directly influence what price a stock trades at. It depends on investors' expectations, analysts' view of the company's future cash flows, and market conditions, etc.

Book value is based on the current value of assets and liabilities (historic) whereas market value is based on what investors expect will happen in the future (intrinsic value). Book value and market value of equity are rarely equal. A useful ratio to compute and understand this relationship better is the price to book ratio (P/B).

6.1 Accounting Return on Equity

ROE is a key ratio to determine whether the management is using its capital effectively.

$$ROE_t = \text{Net Income} / \text{Average book value of equity} = NI_t / (BVE_t + BE_{t-1}) / 2$$

Sometimes the beginning book value of equity is used instead of average book value.

ROE can increase over time because of the following reasons:

- Increase in business profitability that increases net income relative to the increase in book value of the equity.
- Rapid decline in book value, i.e., net income declines at a slower rate compared to the decline in book value.

- Increase in leverage that increases net income and reduces book value of the equity, thereby increasing overall risk.

As only the first case is desirable in the above three cases, a proper analysis of the increase in ROE should be done. The DuPont formula can yield a better understanding of the sources of growth in the ROE ratio.

6.2 The Cost of Equity and Investors' Required Rates of Return

When investors purchase company shares, their minimum required rate of return is based on the future cash flows they expect to receive.

Cost of equity is the minimum expected rate of return that a company must offer its investors to purchase its shares (not easily determined).

- Cost of equity may be different from the investors' required rate of return.
- Because companies try to raise capital at the lowest possible cost, the cost of equity is often used as a proxy for the investors' minimum required rate of return.
- If the expected rate of return is not maintained, the share price falls.

Cost of equity can be estimated using methods such as the dividend discount model (DDM) and the capital asset pricing model (CAPM). These models are discussed in detail in other readings.

Summary

LO: Describe characteristics of types of equity securities.

There are two types of equity securities: common shares and preference shares.

Common shares represent an ownership interest in a company, including voting rights. In statutory voting, each share is entitled for one vote. In cumulative voting, a shareholder can cumulate his total votes and choose one particular candidate.

Preference shares get precedence over common shares while claiming a company's earnings in the form of dividends, and net assets upon liquidation. Dividends on preference shares can be cumulative, non-cumulative, participating, non-participating, or a combination of these. Convertible preference shares are those that can be converted to common stock.

LO: Describe the differences in voting rights and other ownership characteristics among different equity classes.

A firm can have different classes of equity shares, which may have different voting rights and priority in liquidation. For example: Class A shares would have more votes than Class B shares.

LO: Compare and contrast public and private equity securities.

Private equity refers to the sale of equity capital to institutional investors via private placement.

The types of private equity are:

- Venture capital
- Leveraged buyout
- Management buyout
- Private investment in public equity

LO: Describe methods for investing in non-domestic equity securities.

There are two ways to invest in equity of companies outside the local market: direct investing and depository receipts.

Direct Investment: Buy and sell securities directly in foreign markets in the company's domestic currency.

Depository receipt: A security that trades like an ordinary share on a local exchange and represents an economic interest in a foreign company.

Based on the foreign company's involvement a DRs can be sponsored or unsponsored. Based on the geography of issuance, DRs can be classified as

- Global depository receipt (GDR)
- American depository receipt (ADR)
- Global registered shares (GRS)

- Basket of listed depository receipts (BLDR)

LO: Compare the risk and return characteristics of different types of equity securities:

Risk characteristics of different types of equity securities		
Common shares vs. preference shares. Preference shares are less risky.	Preference Shares 1. Dividends on preference shares are fixed as a percentage of the par value. 2. Dividends are paid before common shares. 3. On liquidation, preference shareholders get par value of the shares.	Common Shares 1. Returns are unknown as they can be from capital gains (price appreciation) and dividends. 2. On liquidation, common shareholders have residual claim, i.e., they get paid after claims of debt and preferred shares have been met; hence the amount to be received is unknown. 3. Foreign investments are subject to currency exposure risk.
Cumulative vs. non-cumulative preference shares. Cumulative shares are less risky.	1. Any unpaid dividends are accumulated and paid before common stock dividends are paid.	

LO: Explain the role of equity securities in the financing of a company's assets.

Companies issue equity in primary markets to raise capital and increase liquidity. A company needs capital to finance revenue-generating activities, make acquisitions, and provide stock-based and option-based incentives to employees.

LO: Contrast the market value and book value of equity securities.

Book value is based on the current value of assets and liabilities (historic) whereas market value is based on what investors expect will happen in the future (intrinsic value). Book value and market value of an equity are rarely equal. A useful ratio to compute and understand this relationship better is the price-to-book ratio (P/B).

LO: Compare a company's cost of equity, its accounting return on equity, and investors' required rates of return.

Return on equity (ROE) is an important measure to determine whether the management is using the capital effectively. Both net income and the book value of equity in the formula below are affected by the management's choice of accounting methods related to depreciation, inventory, etc.

$$ROE_t = \text{Net Income} / \text{Average book value of equity} = NI_t / (BVE_t + BE_{t-1})/2$$

When companies raise money by issuing debt or equity securities, there is a minimum return that investors expect in return for their money, which is called the cost of capital. Cost of equity is the minimum expected rate of return that a company must offer its investors to purchase its shares.

LM05 Company Analysis: Past and Present

1. Introduction.....	2
2. Company Research Reports.....	2
3. Determining the Business Model.....	4
4. Revenue Analysis.....	11
5. Operating Profitability and Working Capital Analysis.....	15
6. Capital Investments and Capital Structure.....	21
Summary.....	25

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Version 1.0

1. Introduction

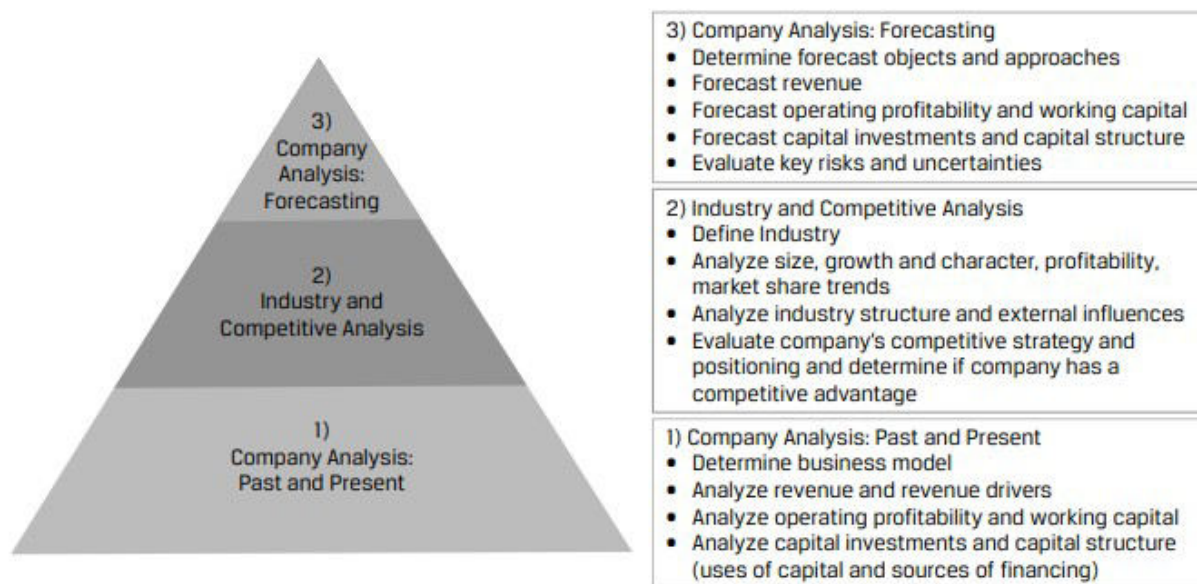
An insightful and well written company research report can help investors understand a company and make good investment decisions about the company's securities. This and the next two learning modules provide a framework to prepare a company research report.

This learning module covers:

- Elements of a company research report
- Determining a company's business model
- Evaluating a company's revenue and revenue drivers – including pricing power
- Evaluating a company's operating profitability and working capital management
- Evaluating a company's capital investments and capital structure

2. Company Research Reports

Exhibit 1 from the curriculum presents a framework for company and industry analysis. This module focuses on the foundational block – Company analysis: Past and Present.



Company research reports examine a company's past and present financial statements, as well as its industry and competitors, and forecast its future financial statements. Reports conclude with a valuation, an investment recommendation, and a discussion of investment risks.

The structure content and tone of a research report can vary. For example, reports for distribution to external clients (a sell-side report) often consist of an extensive initial report. The structure of an initial company research report is shown in Exhibit 2 of the curriculum.

The primary audience of this report is those who are not already knowledgeable about the company.

Front Matter	▪ Issuer name and security identifiers (e.g., symbol, CUSIP) ▪ Analysts' recommendation (buy, hold, sell) and target buy or sell prices ▪ Disclosures, disclaimers, and other legally required information
Recommendation	▪ Analysts' recommendation ▪ Summary of key reasons supporting the recommendation
Company Description	▪ Discussion of issuer's business model and strategy ▪ Explanatory charts and figures—for example, disaggregation of revenues and profits by product, geography, etc.
Industry Overview & Competitive Positioning	▪ Industry size, growth rate and key drivers, market share trends, profitability—historical and outlook ▪ Competitive analysis such as Porter's Five Forces ▪ Analysis of external industry influences such as political, economic, social, technological, legal, and environmental ("PESTLE" analysis) ▪ Evaluation of company's position and strategy in the industry
Financial Analysis and Model	▪ Evaluation of key drivers of revenue, costs, profitability, cash flows, and the issuer's uses and sources of capital ▪ Forecasts of key drivers and supporting discussion ▪ Historical and forecasted financial statements
Valuation	▪ Estimates of company and security values with target price ▪ Typically includes relative and present value approaches ▪ Discussion of key inputs and scenario and sensitivity analyses
Environmental, Social, and Governance (ESG) Considerations	▪ Evaluation of ESG indicators and risks ▪ Ownership structure and management composition ▪ Executive compensation
Risks	▪ Evaluation of material downside and upside risks and discussion of how they are considered in the financial analysis and valuation

Subsequent company research reports are usually shorter than initial reports. Their primary audience consists of people who are already familiar with the company and need an update based on new information (such as a quarterly report) or a change in the analyst's recommendation. The structure of a subsequent company research report is shown in Exhibit 3 of the curriculum.

Front Matter	▪ Analysts' names ▪ Issuer name ▪ Security and exchange identifiers (e.g., symbol, CUSIP) ▪ Analysts' recommendation: buy, hold, sell ▪ Current security price and analysts' target price ▪ Disclosures, disclaimers, and other legally required information
Recommendation	▪ Analysts' recommendation ▪ Summary of changes to the recommendation and support explanations
Analysis of New Information	▪ Comparison of quarterly actual results with expected results ▪ Analysts' interpretation of new information and changes to forecasts ▪ Historical and updated forecasted financial statements
Valuation	▪ Estimates of company and security values ▪ Discussion of changes from valuation in prior report
Risks	▪ Update of risk factors from prior report with discussion of changes

3. Determining the Business Model

The first step of company analysis is understanding the company's business model. A business model describes the company's operations and includes the elements listed below, which analysts investigate by answering several key questions.

Business Model Element	Key Questions for Analysts
The product(s) or service(s) the company sells	What is the firm selling?
Customers and key customer groups	To whom does the company sell?
Sales channels, including customer acquisition and product/service delivery mechanisms	How does the company reach potential and current customers and how does it deliver products?
How the product(s) or service(s) are priced and paid for	How much does the company charge and how are prices and payment terms structured?
Resource, supplier, and partner relationships need to operate effectively	What does the company buy and rely on?

For each of the key questions, there are numerous additional questions, such as determining whether customers are few and concentrated, whether suppliers and resources are specialized or difficult to replace, and so on.

To answer these questions, analysts rely on both issuer and third-party information sources.

Issuer sources (available freely if the company is public)

- Regulatory filings, especially the annual and quarterly reports
- Quarterly or semi-annual earnings conference calls
- Presentations and events for investors
- Press releases
- Issuer management, investor relations, or other personnel
- Company website or properties that the analyst may be able to visit as either a customer or an investor

Public third-party sources

- Free industry white papers or analyst reports from a consultancy
- Economic or industry indicators from governments and other organizations
- General news outlets
- Industry-specific news outlets
- Social media
- Miscellaneous sources available via search engines

Proprietary third-party sources

- Analyst reports and communications, including from “sell-side” or “Wall Street” analysts and credit rating agencies
- Reports and data from platforms such as Bloomberg and FactSet
- Reports and data from consultancies, often industry specific, such as Rystad in energy, IQVIA and Evaluate in biopharma, Gartner and IDC in information technology

Proprietary primary research

- Surveys, conversations, product comparisons, and other studies commissioned by the analyst or conducted directly

Instructor’s Note: The curriculum presents a case study of a fictitious company – ‘Warehouse Club Inc’ to help understand the company and industry analysis framework. This case study is used throughout this and the next two modules. We have presented the case as-is from the curriculum.

CASE STUDIES**Warehouse Club Inc.**

Elaine Nguyen is an analyst at Fyleton Investments. To find investment candidates for one of Fyleton’s funds, Nguyen ran a screen for companies in developed markets that exceeded a certain level of sales growth and return on invested capital over the last five years while also performing well in the last recession. One company that passed the screen is Warehouse Club Inc. (“Warehouse”). Neither Nguyen nor her colleagues have ever worked on Warehouse before. A portfolio manager suggested that Nguyen research the company and its industry for several days and then discuss initial findings with colleagues before writing a

formal research report.

Warehouse is a public company, so Nguyen compiles several issuer information sources, including the company's latest annual report and a transcript of management's presentation at a recent investor event. The following paragraphs and figures are excerpts from these sources.

Warehouse Club Inc. ("we," "our," "us," "the company") is a leading discount retailer, consistently offering customers more than 20% savings on a broad range of food and other merchandise compared to supermarket, department store, convenience store, and e-commerce competitors. Warehouse operated 149, 145, and 141 stores as of 31 December 2X19, 2X18, and 2X17, respectively.

We sell a core assortment of packaged food and beverages, fresh foods, and non-food merchandise such as apparel, appliances, electronics and entertainment, housewares, and sporting goods. In addition to our core assortment, our stores feature a changing assortment of seasonal or discounted items based on the latest consumer trends and availability from suppliers, which vary by store. We also offer ancillary products and services such as petrol stations co-located with our stores, in-store takeaway prepared foods, pharmacy and optician centers, and hearing aids. In addition to carrying the leading branded goods, we also sell packaged food and household goods under our exclusive private-label brand that we source from contract manufacturers, which account for over 15% of our net sales and carry higher than average profitability.

We operate a member-based model that requires customers to show their membership card to enter and shop in our stores. The annual membership fee is \$60 per household. We estimate that members' annual savings versus shopping at competitors for the same basket of goods amounts to over five times their annual membership fee. Our membership fee revenues were \$601, \$576, and \$553 million in fiscal years 2X19, 2X18, and 2X17, respectively. Besides providing us with a source of recurring revenue, membership engenders shopper loyalty and drives our industry-low levels of inventory theft of less than 20 basis points as a percentage of net sales, which we believe is at least one-fifth that of competitors. Total memberships were 10.2, 9.8, and 9.4 million as of 31 December 2X19, 2X18, and 2X17, respectively.

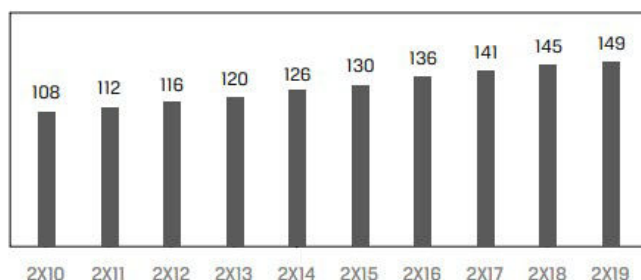
Our strategy is based on price and cost leadership. We stock a limited selection of high-quality branded and private-label products in a wide range of categories that produce high sales volumes and rapid inventory turnover. We limit items to fast-selling models, sizes, and colors and carry an average of approximately 4,000 unique products per store, less than a quarter of the assortment of our supermarket and supercenter competitors. Most items are sold only in bulk sizes. Wherever possible, we utilize direct-from-manufacturer distribution and move merchandise immediately to the sales floor, presenting on pallets in a no-frills warehouse atmosphere.

Our high-volume purchases of a limited number of products, direct from manufacturer distribution, and reduced handling of merchandise enable us to hold a low-cost position in the retail industry in terms of merchandise and labor costs. Our low costs and membership fee revenues enable us to operate profitably while charging significantly lower prices than other retailers. Low prices for quality products drive greater membership and net sales over time, leveraging our selling, general, and administrative expenses, reducing them as a percentage of net sales.

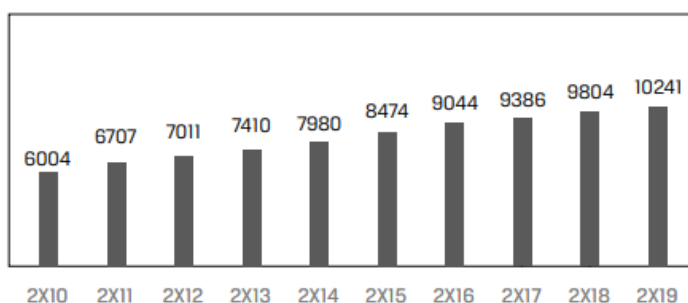
Our stores operate on a seven-day, 70-hour week with predictable shifts for employees. Because the hours of operation are shorter than other retailers, and due to other efficiencies inherent in a warehouse-type operation, labor costs are lower relative to the volume of sales. We look to maintain a large base of full-time employees with above-average wages and benefits to establish long tenures, as we believe this maximizes productivity and customer service at our stores. Our employee retention rate for employees who have been with us for at least a year is 90%, which we believe is substantially higher than the retail industry average.

Our retail operations, which represent substantially all our consolidated net sales, are our only reportable segment. We do not have significant sales outside our domestic geography: the United States. No single customer or supplier represents a material amount of revenues or merchandise costs, respectively.

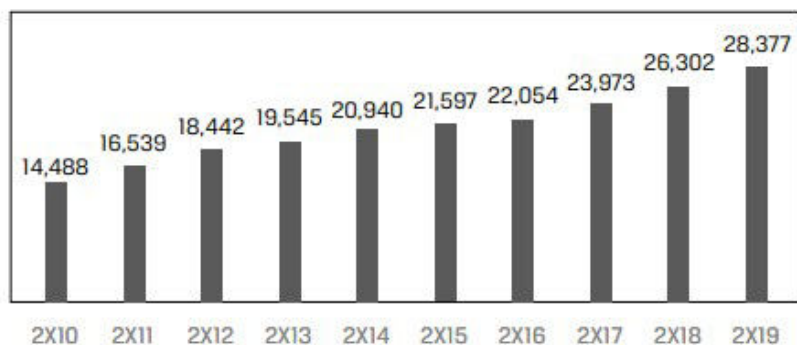
Warehouse Club Inc.
Number of Stores at Year-End, 2X10-2X19



Warehouse Club Inc.
Number of Members (thousands) at Year-End, 2X10-2X19



Warehouse Club Inc.
Annual Net Sales (USD millions), 2X10–2X19



	31 December 2X17	31 December 2X18	31 December 2X19
Operating income	791	830	865
Interest expense	(25)	(30)	(29)
Interest income	10	14	24
Income before taxes	776	814	860
Income tax expense	(178)	(187)	(198)
Net income	598	627	662
Shares outstanding	83	83	84
Earnings per share	7.20	7.55	7.88

During its presentation at a recent investor event, Warehouse management made the following statements that were not in the annual report:

- Three years ago, the company launched “Buy Online, Pick-Up in Store,” which enables members to shop on Warehouse’s website or mobile app and pick up their order,

assembled by Warehouse employees, at a store. The company does not plan to offer e-commerce or delivery itself or through third-party services.

- Membership is \$60 per year, paid upfront, and refundable on a prorated basis for days remaining in the year. Management has increased the price of membership every five years by \$5 and expects to continue to do so. The last price increase was three years ago.
- Management intends to open four new stores per year for the next five years.
- The company operates stores in urban and suburban areas in a single country. Over 90% of current members live within 10 kilometers of a store, and while membership has grown around existing stores over time, management believes the primary driver of membership growth is opening stores in new areas.
- Management does not plan to open stores in another country for the foreseeable future. An immaterial amount of merchandise is imported.

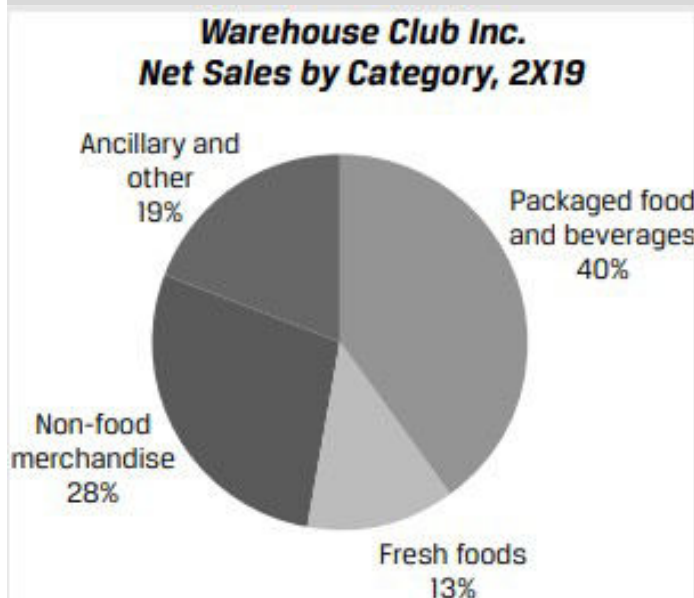
Example: Determining Warehouse Club Inc.'s Business Model

(This is Example 1 from the curriculum.)

Nguyen answers the following questions to determine Warehouse Club Inc.'s business model and to identify areas to conduct further research.

What is the firm selling?

Warehouse sells a range of consumer goods, mostly consumables and perishables, approximately 4,000 unique items. For customers to shop at the stores, they must hold a membership, which Warehouse sells for \$60 per year, per household. Nguyen graphs the following net sales by category data with descriptions supplied by Warehouse, and notes that the composition of net sales by category has not changed materially in the last 10 years.



“Non-food merchandise” includes a broad range of items such as appliances, electronics, health and personal care products, hardware, outdoor and sporting goods, jewelry, furniture, and housewares.

“Ancillary and other” are service businesses that operate in or next to Warehouse’s stores and include takeaway prepared foods, prescription pharmacies, petrol stations, optician centers, and hearing aids.

To whom?

Warehouse’s members are primarily consumer households that live within 10 kilometers of a store, are value-conscious, are willing to forgo a more extensive selection for lower unit prices on bulk-sized goods, and are able to afford the annual membership fee.

How?

Warehouse sells products and services both in and around its stores. Members can either shop at the store or shop online and pick up at the store (Nguyen notes that management did not disclose the size of this in terms of net sales). Management said that it does minimal advertising (besides promotional mail) in an area before it opens a store and also sends out membership renewal reminders; the company primarily relies on word of mouth and its stores’ physical presence to drive interest and foot traffic.

Memberships can be purchased online, by phone, or at any Warehouse store.

Management said that over three-quarters of members purchase and renew their memberships online.

For how much?

Warehouse has two revenue streams: goods and services (for which it earns net sales) and membership (for which it earns membership fees). Management sets the prices of its merchandise by item. Gross margin and markup on aggregate net sales for the last three fiscal years were as follows:

	31 December 2X17	31 December 2X18	31 December 2X19
Net sales	23,973	26,302	28,377
Merchandise costs	(21,258)	(23,399)	(25,248)
Gross margin	2,715	2,903	3,129
Gross margin % of net sales	11.33%	11.04%	11.03%
Markup	12.77%	12.41%	12.39%

Therefore, the average good or service is sold at a price that is a little over 12% of the cost for Warehouse to purchase and ready the item for sale. While management claims that the company is a cost and price leader, Nguyen will have to corroborate that.

Membership is \$60 per year, per household. Membership is required to shop at Warehouse stores, thus effectively raising the price of all its goods and services.

What does the company buy and rely on to operate effectively?

- Merchandise from a broad range of manufacturers and private-label goods from contract manufacturers
- Human capital in stores and in management
- Fixed assets including land in attractive locations, buildings, fixtures, refrigerators and freezers, and a range of information technology systems such as point-of-sale payment systems
- Credit and debit card payment networks and financial institutions to receive and make electronic payments

While these are all broadly available and the company has not had supply or service interruptions in the past, a key partner appears to be merchandise manufacturers, particularly those that manufacture Warehouse's private-label products. Since Warehouse is strict on cost and quality and carries a limited selection, the company does rely heavily on these contract manufacturers.

In summary, Warehouse's business model is aimed at creating a virtuous cycle whereby low prices drive memberships, which drive further sales, which drive negotiating leverage with suppliers to enable low prices at low costs.

4. Revenue Analysis

After understanding the company's business model, an analysis of historical financial statements will help the analyst identify key drivers of revenues, profitability, cash flows, and financial position.

Key drivers are causative factors that explain the level of and changes in an output variable (e.g. revenues).

Revenue Drivers

Revenue analysis can be done using a bottom-up or top-down approach. A bottom-up approach breaks down revenues into drivers such as sales volumes and prices, by product line or segment. A top-down approach expresses revenue as a function of drivers such as market share, market size, and GDP growth. The two approaches are often used together.

CASE STUDIES**Bottom-Up Approach to Revenue Analysis: Warehouse Club Inc.**

Warehouse's revenues are composed of net sales and membership fees. Based on her business model work, Nguyen knows that net sales are sales of merchandise and services in Warehouse stores (the company does not operate an e-commerce business). A logical driver

of net sales each year, therefore, is the number of stores open that year. Nguyen decomposes Warehouse's net sales into two drivers: the average number of stores open and net sales per average store. Nguyen also calculates the change in these drivers: the absolute change in stores each year (new-store openings; so far, Warehouse has never closed a store) and the annual percentage change in net sales per store.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Stores, year-end	108	112	116	120	126	130	136	141	145	149
New-store openings	5	4	4	4	6	4	6	5	4	4
Average stores (#)	106	110	114	118	123	128	133	139	143	147
Net sales per average store	138	150	162	166	170	169	166	173	184	193
Growth rate		9%	8%	2%	3%	-1%	-2%	4%	6%	5%
Net sales	14,488	16,539	18,442	19,545	20,940	21,597	22,054	23,973	26,302	28,377

Net sales in 2X19 were 96% higher than in 2X10, driven by similar growth contributions from the two drivers: the average number of stores increased by 39% and sales per average store increased by 40%.

An encouraging sign for forecasting this company's results is that each driver appears relatively stable, indicating that they might be somewhat predictable. Sometimes, however, key revenue drivers are volatile, such as interest rates and oil prices for banks and oil producers, respectively. Volatility does not mean that the analyst has selected an incorrect driver; drivers are based on the business model.

Nguyen then decomposes Warehouse's membership fee revenues into the annual price of membership and the average number of members.

Members		2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Members, end of year	6,004	6,707	7,011	7,410	7,980	8,474	9,044	9,386	9,804	10,241	
Members, annual average	5,900	6,360	6,855	7,218	7,691	8,218	8,764	9,217	9,600	10,017	
Avg. per member fee	50.00	50.00	55.00	55.00	55.00	55.00	55.00	60.00	60.00	60.00	
Membership fee revenues	295	318	377	397	423	452	482	553	576	601	

Membership growth has been a larger driver of membership fee revenue growth than price increases, as the number of members has grown by 70% from 2X10 to 2X19 while prices have increased by 20% over the same period.

Management indicated that over 90% of members live within 10 kilometers of a store and, from 2X10 to 2X19, the company opened 46 new stores. Nguyen adds to her membership analysis by calculating the average number of members per store; if this number has been falling over time, it could mean that existing stores are losing members and sales to new stores the company opens, an example of cannibalization.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Avg. members per store	57,282	57,818	60,128	61,171	62,528	64,205	65,892	66,546	67,133	68,141
Growth rate		0.9%	4.0%	1.7%	2.2%	2.7%	2.6%	1.0%	0.9%	1.5%

Not only has the average number of members per store not declined, but it has also grown by 19%, indicating no cannibalization and that new-store openings are not the only driver of membership growth. This is a positive sign, though Nguyen notes that additional analysis is needed to estimate a ceiling on the number of members a store can reasonably support, and to determine whether stores that opened many years ago have already hit this ceiling.

Pricing Power

Pricing power is defined as a company's ability to set prices without affecting its sales volumes. Pricing power is primarily a function of market structure and a company's competitive positioning in its market.

Firms selling nearly identical products in highly competitive markets lack pricing power. The price is dictated by the overall market supply and demand, and all firms generally sell at the same price (i.e. they are price takers). Highly competitive markets usually don't start that way, but rather become more competitive as new firms enter, the rate of innovation slows, and imitation spreads, a process known as *commoditization*.

Firms operating in less competitive markets (monopolistic competition, oligopoly, or monopoly) tend to have some pricing power. Analysts assess pricing power by comparing a firm's prices with its costs, i.e. its profit margins.

Rising profitability over time is an important indicator of pricing power because it demonstrates that competitors, new entrants, or substitutes are not driving down prices faster than costs.

Top-Down Revenue Analysis

CASE STUDIES

Top-Down Approach to Revenue Analysis: Warehouse Club Inc.

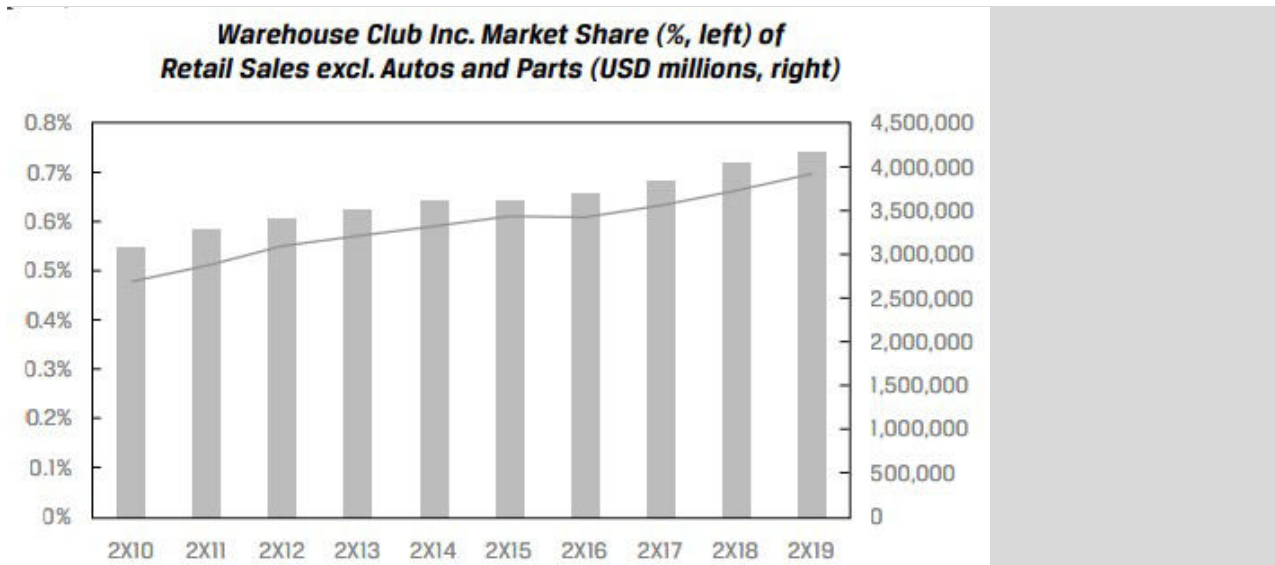
A government agency reports total national retail and food services sales each month for the country that Warehouse operates in (the United States). The data capture consumer spending at hundreds of thousands of retailers and restaurants. Sales are categorized by the type of retailer or restaurant, which reflects the primary type of product or service it sells. An excerpt of a monthly release is shown below:

National Retail Sales, January 2X19 (billions of USD)	
Total retail and food services sales	456,012
Total retail sales	399,359
Automobile and parts dealers	88,152
Petrol stations	36,539
Pharmacies and drug stores	24,273
Grocery stores	57,392
Total food services sales	56,653
Full-service restaurants	24,730
Limited-service restaurants	25,273
Pubs, bars, and other	6,650

Warehouse sells a broad range of retail goods in most cities, so US national retail spending is generally representative of Warehouse's market size—the existing demand for goods or services offered by a company. By expressing Warehouse's revenues as a percentage of the market, or its market share, over time, we can assess whether the company has been gaining or losing relevance with its customers relative to competitors and substitutes.

A challenge for analysts is determining market size. The data might be difficult to gather (not the case here), and judgment is required for determining what to include and exclude: only sales of products identical to the company's, sales of similar products, all sales of competitors, and so on. The common practice is to include all sales of similar products and to consider substitutes separately (discussed in the next module on industry analysis). A substitute serves the same function as a product but differs in form (e.g., movie theater and streaming video, grocery stores and restaurants, print and digital advertising). This practice is subject to a wide range of opinions, however, and some analysts include sales of substitutes in market size estimates. The share of market that a firm does not have (i.e., $100\% - x\%$ market share) represents sales potential.

Elaine Nguyen decides to use total US retail sales as the market size for Warehouse, with one exclusion: automobiles and parts. Warehouse does not sell automobiles or automobile parts, nor does it sell substitute means of transportation or repair. Nguyen considers food service substitutes (restaurants and pubs) for Warehouse.



Nguyen makes three observations:

- The market is large and growth has been positive, albeit at a slow (3.4%) rate. This finding aligns with Nguyen's expectations, because this is a developed market that represents a broad range of consumer spending (a large portion of which is for necessities) and the country has been in an economic expansion.
- Warehouse Club Inc. has a small (70 basis points in 2X19) but growing share of the market (up from 48 basis points in 2X10), indicating that its value proposition relative to other retailers has been resonating with consumers.
- Warehouse's market share increased in most years, with the exception of 2X16, when its share was flat from the prior year. Nguyen will have to investigate further.

5. Operating Profitability and Working Capital Analysis

Operating costs are costs incurred in generating current period revenue – for e.g., costs related to the production, sale and delivery of goods and services, the management of business activities etc.

Investing costs are costs related to the acquisition and construction of long-term assets like property and equipment, patents, trademarks etc.

Financing costs include payments to debt and equity investors e.g., dividend and interest payments.

Operating Costs and Their Classification

A company's operating costs can be categorized in three ways: by their behavior with output, their nature, or their function. This is shown in Exhibit 4 of the curriculum.

Behavior with Output <i>Whether the cost varies or not with output in the short run</i>	Nature <i>What the cost is</i>	Function <i>The purpose of the cost</i>
- Variable costs vary with output - Fixed costs do not vary with output	- Compensation of employees - Raw materials - Merchandise - Office supplies	- Cost of goods sold - Sales and Marketing - General and administrative - Research and development

Behavior with Output: Fixed and Variable Costs

Based on whether the cost varies or not with output, operating costs can be categorized into fixed or variable costs. The proportion of fixed to variable costs in the total operating costs affects the stability and predictability of operating profit of a company.

Operating profit can be expressed as:

$$[Q \times (P - VC)] - FC$$

where:

Q = units of outputs sold in a period

P = price per unit of output

VC = variable operating costs expressed per unit of output

FC = fixed operating costs stated on a total dollar, not per unit, basis

To be profitable, a company's contribution margin (P - VC) must be positive, and Q must be high enough to exceed FC.

Degree of operating leverage (DOL): Operating leverage refers to the amount of fixed costs in the operating cost structure of a company. It represents both benefits and risks. If operating costs are largely fixed operating profit can increase rapidly with increases in Q. However, if Q declines, operating profit will also fall rapidly.

The degree of operating leverage measures the sensitivity of operating profit to changes in sales.

$$DOL = \% \Delta \text{ Operating Profit} / \% \Delta \text{ Sale}$$

Natural and Functional Operating Cost Classifications and Measures of Operating Profitability

Instead of fixed and variable, IFRS and US GAAP require companies to disclose operating costs using either a natural or functional cost classification (most companies choose a

functional cost classification). Because of this the income statements of different companies appear similar even though their business models may differ significantly.

Exhibit 5 from the curriculum shows the key profitability measures that can be calculated from the functional classification of operating costs.

Gross Profit:	EBITDA:	Operating Profit:
Revenue	Revenue	Revenue
Cost of Sales	Cost of Sales	Cost of Sales
Gross Profit	Operating Expenses i.e. SG&A, R&D, etc.	Operating Expenses i.e. SG&A, R&D, etc.
	EBITDA	Depreciation and Amortization
		Operating Profit (EBIT)
Gross Margin:	EBITDA Margin:	Operating/EBIT Margin:
Gross Profit	EBITDA	Operating Profit (EBIT)
Revenue	Revenue	Revenue

Apart from these profitability measures, additional considerations used in analysis are industry profitability (covered in the next LM), economies of scale, and economies of scope.

Economies of scale refer to a decrease in costs per unit as output increases, which is typically the result of having fixed costs in the cost structure that are spread across more units of output.

Economies of scope refer to a decrease in costs per unit as the number of product or business lines increases, and are typically the result of shared costs between product lines. E.g., a financial services firm offering consumer banking, brokerage, asset management, and payment processing services.

CASE STUDIES

Warehouse Club Inc. Operating Profitability Analysis

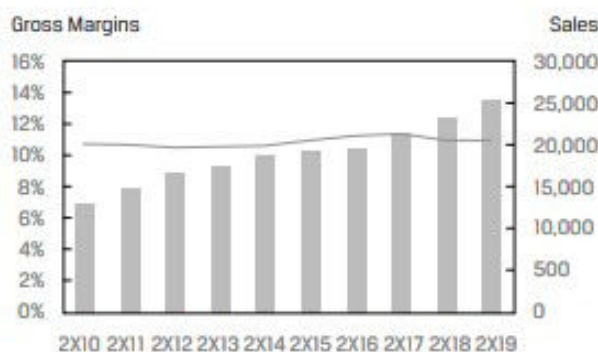
Warehouse Club Inc.'s operating costs are composed of merchandise costs (costs of sales), SG&A expenses, depreciation and amortization, and costs associated with opening new stores.

Based on her business model work, Elaine Nguyen believes the following about drivers for each of these operating costs.

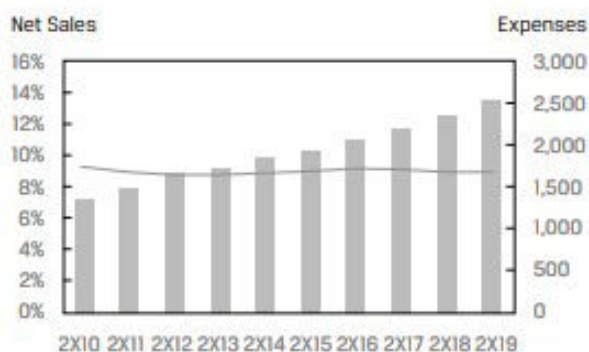
Operating Cost	Driver(s)
Merchandise costs (costs of sales)	Sales
SG&A expenses	Sales Number of employees and per-employee compensation
Depreciation and amortization	Long-term assets Useful lives
Store opening expenses	Store openings

Based on these drivers, Nguyen calculates Warehouse's gross margin ($1 - \text{cost of sales as \% of net sales}$), SG&A expenses as a percentage of net sales, gross fixed assets to depreciation and amortization (implied useful life), store opening expenses per new-store opening, and, finally, profit margins. Profit margins include membership fee revenues.

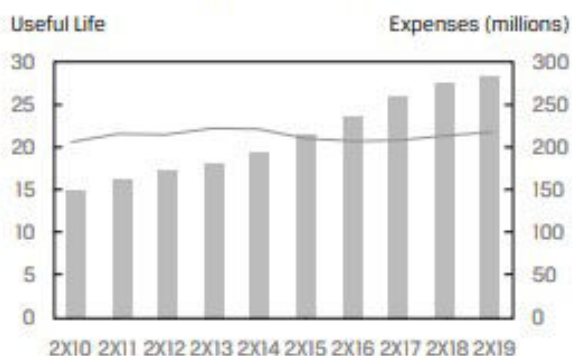
Warehouse Club Inc. Gross Margin (line, left) and Cost of Sales (bars, right, USD millions)



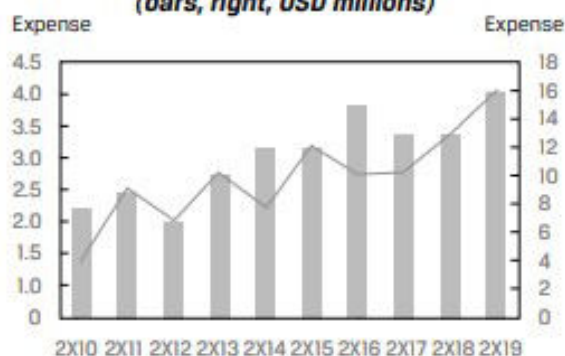
Warehouse Club Inc. SG&A % of Net Sales (line, left) & SG&A Expenses (bars, right, USD millions)

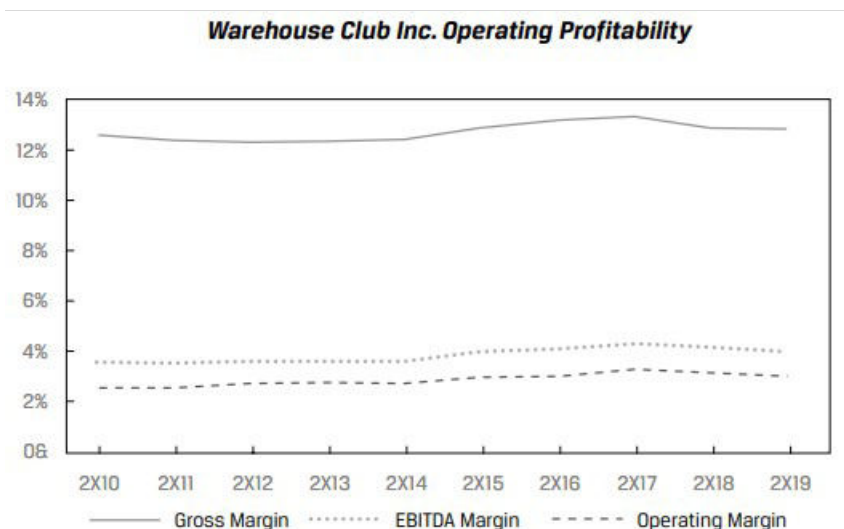


Warehouse Club Inc. Implied Useful Life (line, left) & D&A Expense (bars, right, USD millions)



Warehouse Club Inc. Store Opening Expense per Opening (line, left) & Store Opening Expense (bars, right, USD millions)





Nguyen makes the following observations:

- Operating costs have increased significantly on an absolute basis from 2X10 to 2X19, but on a relative-to-sales basis have remained stable over the same period, indicating that the driver of the increase has been net sales growth.
- One exception is store opening expenses per store opening, which have quadrupled from 2X10 to 2X19, though in aggregate remain a small cost (less than 1/100 of SG&A expenses). Note that these are operating costs related to store openings, such as employee training before opening, not the investing cost of land and fixed assets.
- SG&A expenses do not appear to be fixed relative to net sales, as they have remained roughly the same percentage of net sales over time.
- Margins have remained mostly stable, though they did increase from 2X15 to 2X17 before coming back down in 2X18 and 2X19. Margins are low for Warehouse, with EBITDA and operating margins of 4% and 3%, respectively, in 2X19.
- The amount of operating profit is similar to the amount of membership fee revenues each year. Nguyen creates the following chart:



This chart implies that net sales from its stores—after deducting costs of sales, SG&A expenses, D&A, and store opening expenses—account for the remaining 30% of operating profit. This percentage is likely an understatement, since some amount of SG&A expenses is associated with membership fee revenues (marketing, customer service, payment processing, management), but it does align with Warehouse management's assertion in its annual report that prices provide members with a significant return on membership: the company earns a thin margin on net sales, and it is membership fees that allow the company to run profitably.

Working Capital

The primary measures of a company's working capital are its cash conversion cycle and the ratio of net working capital to sales.

A short cash conversion cycle means that company requires less external financing to fund operations.

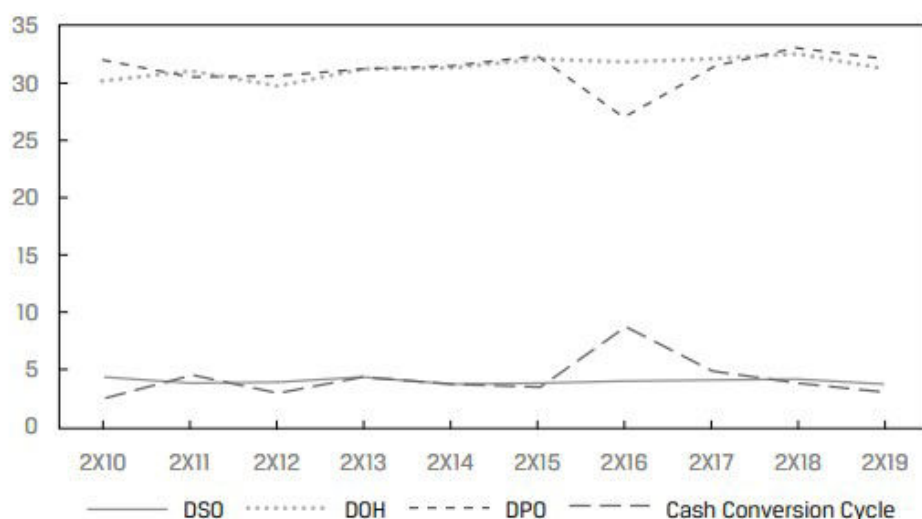
The net working capital requirement refers to a minimum level of investment, in addition to capital investments, that cannot be distributed to investors. A negative net working capital means that suppliers are a source of financing for the company.

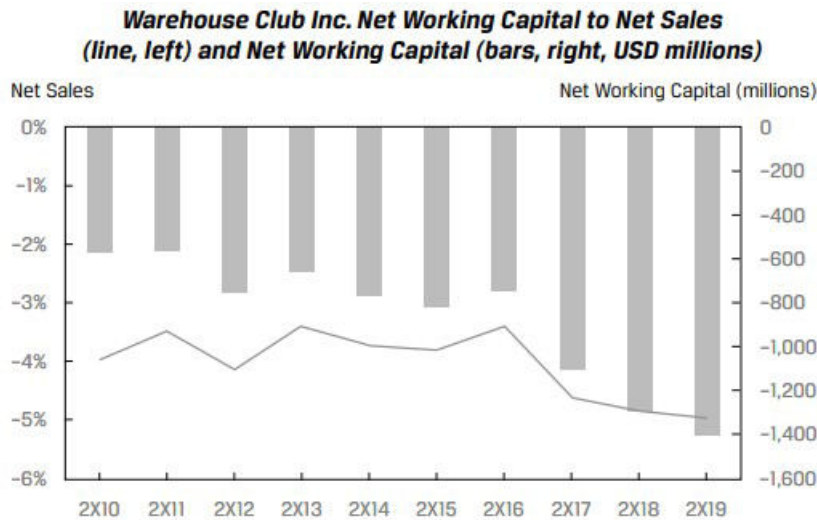
CASE STUDIES

Warehouse Club Inc. Working Capital Management

Elaine Nguyen calculates the working capital measures for Warehouse

Warehouse Club Inc. Activity Ratios and Cash Conversion Cycle (Days)





Nguyen makes the following observations:

- The cash conversion cycle is short, less than a week, primarily as a result of a DSO of 4, because Warehouse's customers pay in cash or with credit or debit cards that settle in less than five business days, and days of inventory on hand is nearly equal to days payable outstanding, indicating that inventory is financed by suppliers.
- Net working capital is negative, now down to -5.0% of net sales, indicating that it is a source of financing for the company.
- Generally, all the activity ratios have been stable since 2X10, except for a one-year increase in 2X16 from a decline in days payable outstanding. Nguyen looked into this increase in the notes to the financial statements in a historical annual report. The company indicated that it changed accounting information systems in early 2X17 and made accelerated payments to suppliers at the end of 2X16, in advance of the system change in case there were problems.

6. Capital Investments and Capital Structure

Sources and Uses of Capital

An important part of company analysis is to determine whether a company creates economic value for its investors, i.e., whether the required rates of returns of debt and equity investors have been met or exceeded. To evaluate this, an analyst first needs to understand a company's sources and uses of capital. A generic structure is shown in Exhibit 6 of the curriculum.

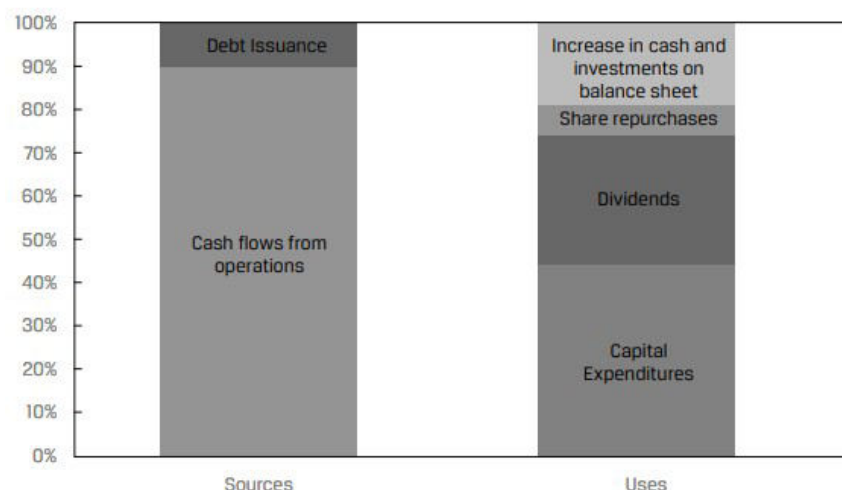
Sources of Capital	Uses of Capital
Cash flows from operations, including net working capital (if negative)	Cash and investments on hand
Debt issuance	Net working capital (if positive)
Equity issuance	Capital expenditures and additions to intangibles
Asset disposals	Acquisitions
	Debt paydown
	Dividends and share repurchases

CASE STUDIES

Warehouse Club Inc.'s Sources and Uses of Capital

Elaine Nguyen calculates the following information on Warehouse over 2X10–2X19. Cash flows from operations include the effect of additional net working capital financing (i.e., Warehouse's net working capital became more negative over time).

Warehouse Club Inc. Sources and Uses of Capital, 2X10–2X19



Nguyen makes the following observations:

- Capital expenditures and returns to shareholders (dividends and share repurchases) are almost equal, which is surprising considering that new-store openings performed well; Nguyen expected the company to devote a greater amount to capital expenditures for new stores.
- The company has made no acquisitions.
- Even after returns to shareholders, the company seems to prefer equity financing to debt financing. This preference is logical considering the company's thin operating margin, though the fact that it primarily sells a wide range of necessities reduces its sales risks (i.e., from changes in the business cycle and in technology or fashion) and membership fees are a source of predictable income.

- The fact that debt issuance is smaller than the increase in cash and investments on the balance sheet indicates that the company's net debt has decreased over time, implying a decline in financial leverage.

Evaluating Capital Investments and Capital Structure

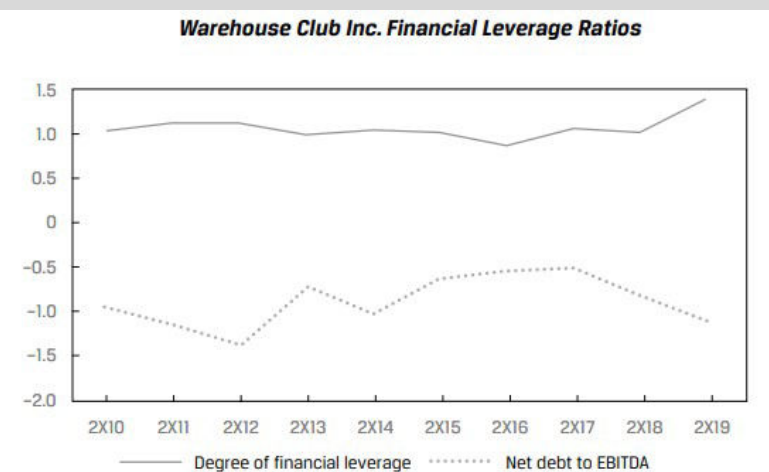
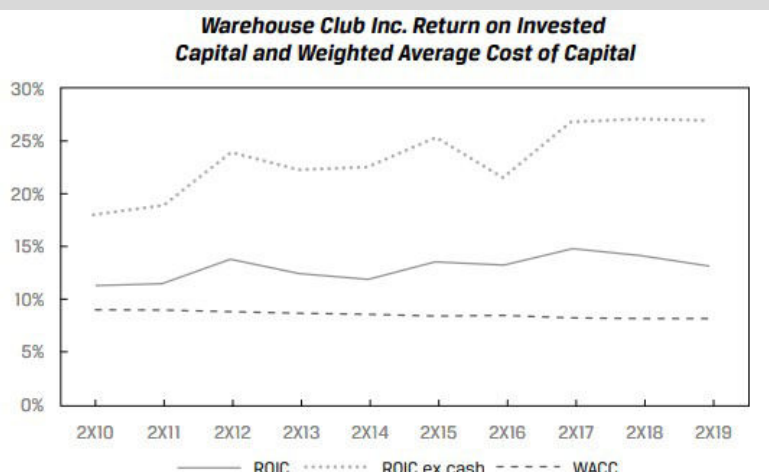
The degree of financial leverage measures net income's sensitivity to changes in operating income. The capital structure of the issuer is the primary driver of financial leverage.

$$DFL = \% \Delta \text{ Net income} / \% \Delta \text{ Operating income}$$

CASE STUDIES

Evaluating Warehouse Club Inc.'s Capital Investments and Capital Structure

Elaine Nguyen calculates the following information on Warehouse over 2X10–2X19. Additionally, the company's interest coverage ratio (EBIT to interest expense) has been over 20 times and recently reached 30 times in 2X19, and the company's credit is rated investment grade by major credit rating agencies.

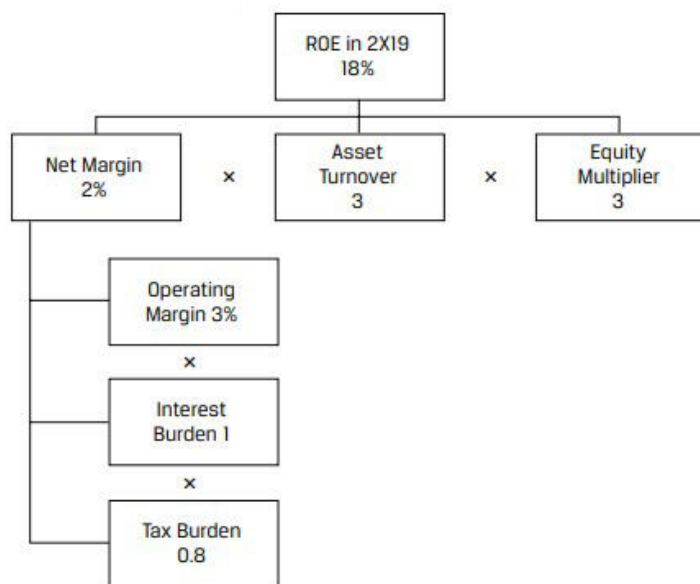


Nguyen makes the following observations:

- The company has created value for investors, with a spread of ROIC over WACC of 200–700 basis points since 2X10. WACC has declined slightly over time as interest rates have fallen; this calculation assumes a constant required rate of return on equity of 9%.
- The company's returns on capital are far higher than its low operating and net margins would suggest, owing to its high asset turnover and negative net working capital as a low-cost means of financing from suppliers, requiring less capital from investors.
- The company has a conservative capital structure, with negative net debt and a degree of financial leverage close to 1.0 because interest expense is largely covered by interest income on cash holdings (i.e., net and operating income move together).
- The capital structure conservatism is evidenced in the counterfactual ROIC, which excludes cash and investments on the balance sheet. While an exaggeration (the company would need some cash on hand), ROIC could be almost 15 percentage points higher if the company reduced net cash by investing in value-creating projects or by returning capital.

ROIC or return on assets measures unlevered returns, these returns are modified by the financial leverage to produce levered returns – return on equity. ROE decomposition can be used as a comprehensive measure of a company's profitability.

The following diagram from the curriculum illustrates the ROE decomposition for Warehouse Club Inc.



Summary

LO: Describe the elements that should be covered in a thorough company research report.

Company research reports examine a company's past and present financial statements, as well as its industry and competitors, and forecast its future financial statements. Reports conclude with a valuation, an investment recommendation, and a discussion of investment risks.

The structure content and tone of a research report can vary. Initial research reports are extensive. Subsequent research reports are usually shorter than the initial reports.

LO: Determine a company's business model.

The first step of company analysis is understanding the company's business model. A business model describes the company's operations and includes the elements listed below, which analysts investigate by answering several key questions.

<i>Business Model Element</i>	<i>Key Questions for Analysts</i>
The product(s) or service(s) the company sells	What is the firm selling?
Customers and key customer groups	To whom does the company sell?
Sales channels, including customer acquisition and product/service delivery mechanisms	How does the company reach potential and current customers and how does it deliver products?
How the product(s) or service(s) are priced and paid for	How much does the company charge and how are prices and payment terms structured?
Resource, supplier, and partner relationships need to operate effectively	What does the company buy and rely on?

LO: Evaluate a company's revenue and revenue drivers, including pricing power.

Revenue analysis can be done using a bottom-up or top-down approach. A bottom-up approach breaks down revenues into drivers such as sales volumes and prices, by product line or segment. A top-down approach expresses revenue as a function of drivers such as market share, market size, and GDP growth. The two approaches are often used together.

Pricing power is defined as a company's ability to set prices without affecting its sales volumes. Pricing power is primarily a function of market structure and a company's competitive positioning in its market.

LO: Evaluate a company's operating profitability and working capital using key measures.

Based on whether the cost varies or not with output, operating costs can be categorized into fixed or variable costs. The proportion of fixed to variable costs in the total operating costs affects the stability and predictability of operating profit of a company. Operating leverage refers to the amount of fixed costs in the operating cost structure of a company.

$$\text{DOL} = \% \Delta \text{ Operating Profit} / \% \Delta \text{ Sale}$$

Instead of fixed and variable, IFRS and US GAAP require companies to disclose operating costs using either a natural or functional cost classification. Key profitability measures that can be calculated from a functional cost classification are: gross, EBITDA, and operating margins.

The primary measures of a company's working capital are its cash conversion cycle and the ratio of net working capital to sales.

LO: Evaluate a company's capital investments and capital structure.

The degree of financial leverage measures net income's sensitivity to changes in operating income. The capital structure of the issuer is the primary driver of financial leverage.

$$\text{DFL} = \% \Delta \text{ Net income} / \% \Delta \text{ Operating income}$$

LM06 Industry and Competitive Analysis

1. Introduction.....	2
2. Uses of Industry Analysis	2
3. Industry Classification	4
4. Industry Survey	8
5. Industry Structure and External Influences.....	13
6. Competitive Positioning.....	16
Summary.....	18

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Version 1.0

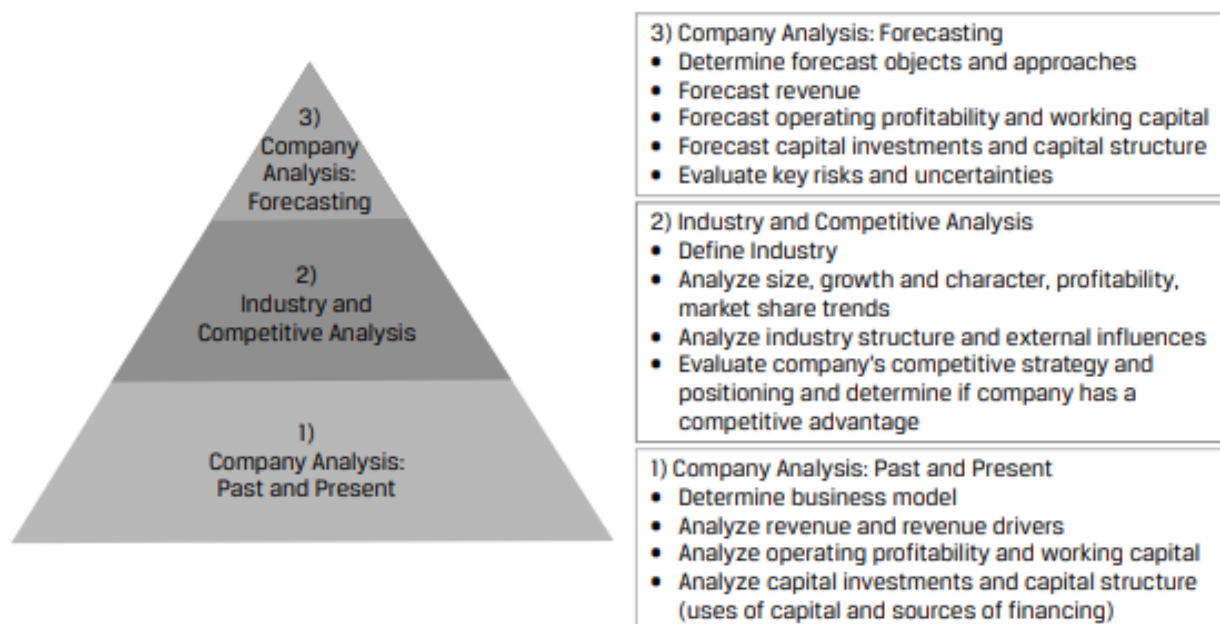
1. Introduction

This learning module covers:

- How an industry is defined and the challenges associated with grouping companies that operate in multiple industries.
- How to determine an industry's size, growth characteristics, profitability, and market share trends.
- How to analyze the competitiveness of an industry using Porter's Five Forces and PESTLE frameworks.
- Evaluating the competitive strategy and position of a company.

2. Uses of Industry Analysis

As seen in a prior exhibit, the next step in the company and industry analysis framework is 'industry and competitive analysis'. This involves the study of the drivers of an industry' size, profitability, and market share trends. We also evaluate a company's competitive positioning in its industry.



Why Analyze an Industry?

- Companies in the same industry have similar business models. They tend to be exposed to the same demand and supply opportunities and risk factors.
- In a study conducted on the relative importance of industry versus company-specific effects on profitability, it was found that industry was the most important factor in the sustainability of economic profit.

Instructor's Note: An average company in a growing industry may perform well than a good company in a dying industry.

Improve Forecasts

By taking a broader industry perspective, analysts can better understand the dynamics in which a company operates and make better earnings forecasts. Without this broader perspective, analysts may underestimate competitive forces and overestimate the degree to which a company controls its destiny.

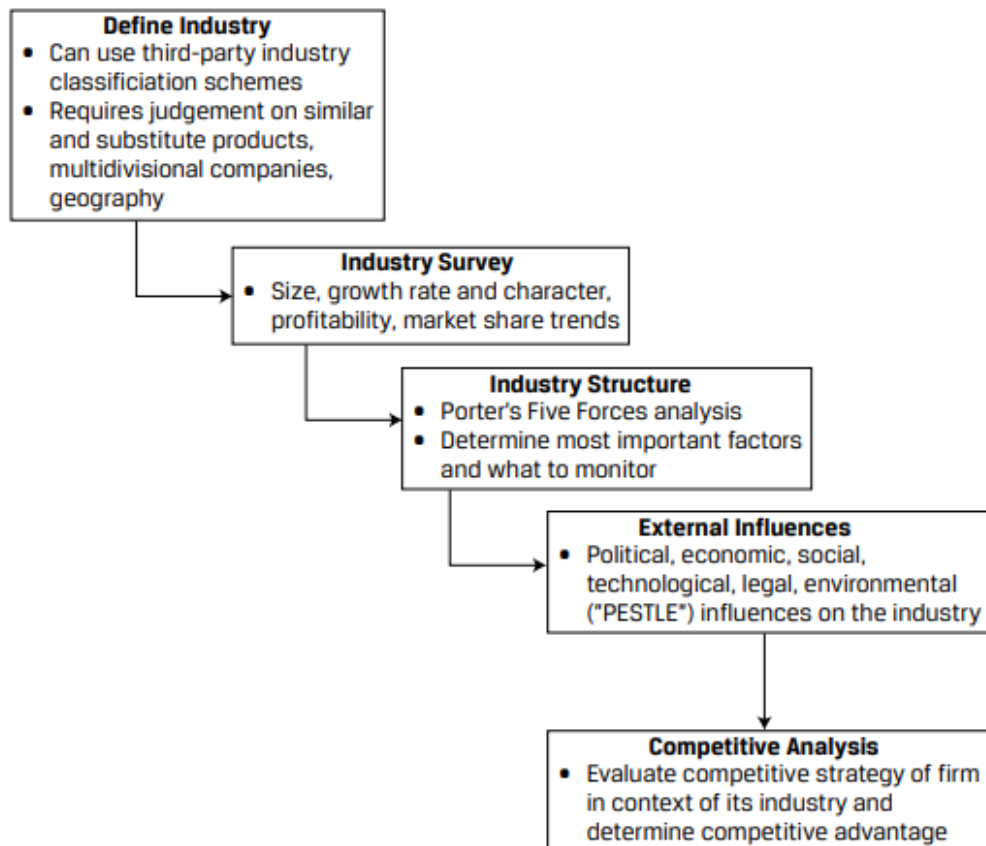
Identify Investment Opportunities

Assessing a company's strength and weaknesses relative to industry peers can help uncover attractive investment opportunities.

Some investors may want to gain exposure to a specific industry while diversifying away the company-specific risks. These investors may take a basket approach, investing in a number of companies from that industry with position sizes scaled according to size, liquidity, or relative attractiveness.

Industry and Competitive Analysis Steps

The steps in industry and competitive analysis are illustrated in Exhibit 2 of the curriculum.



3. Industry Classification

An industry is commonly defined as companies that sell similar products or services, from the perspective of a customer.

Defining industries and classifying companies can be challenging. To assist analysts, third party organizations maintain industry classification schemes that are widely used in investment management.

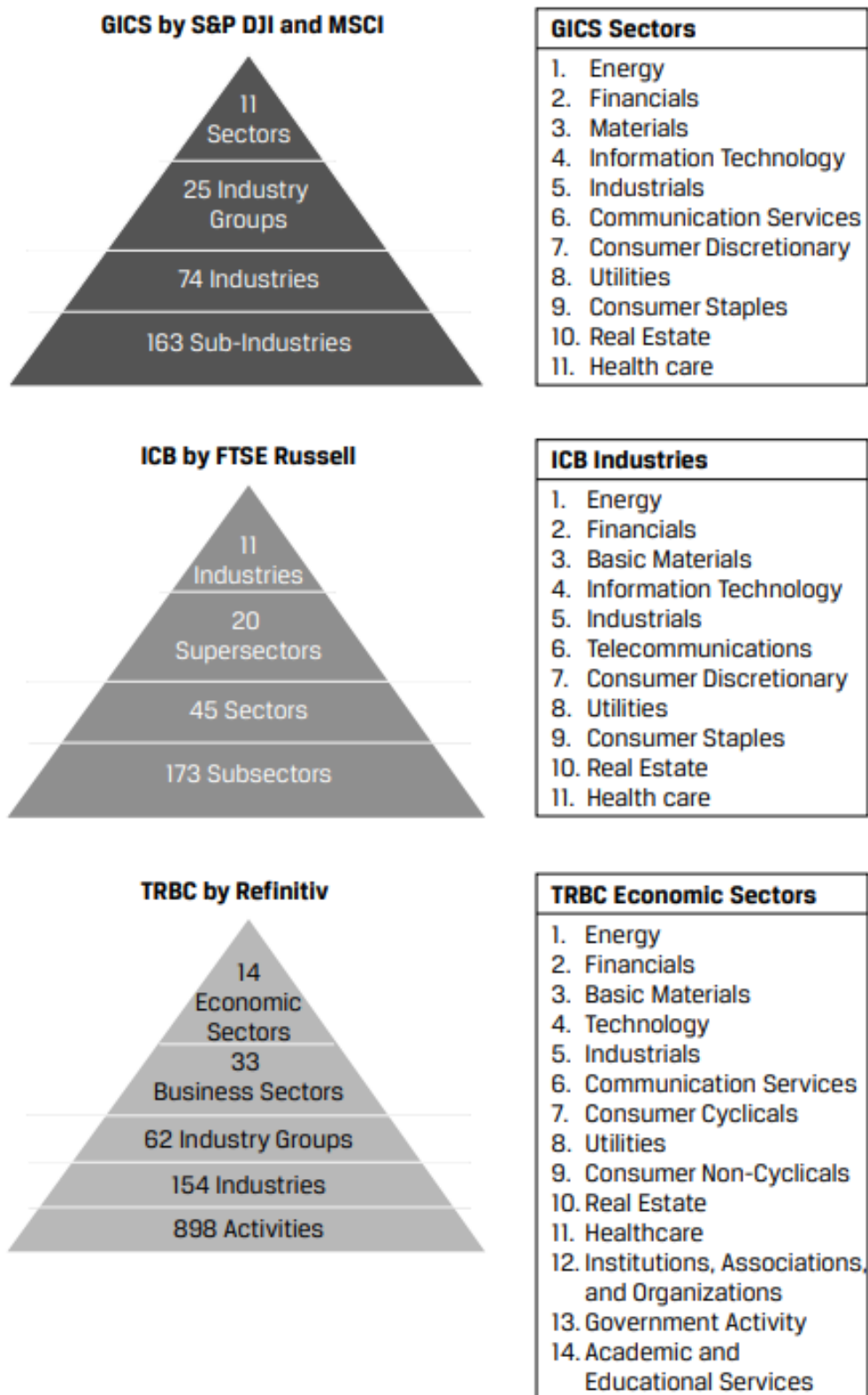
Third-Party Industry Classification Schemes

Classification systems are provided by both commercial entities and government agencies. However, commercial classification systems are commonly used in the investment industry because they are more frequently updated as compared to government classification systems.

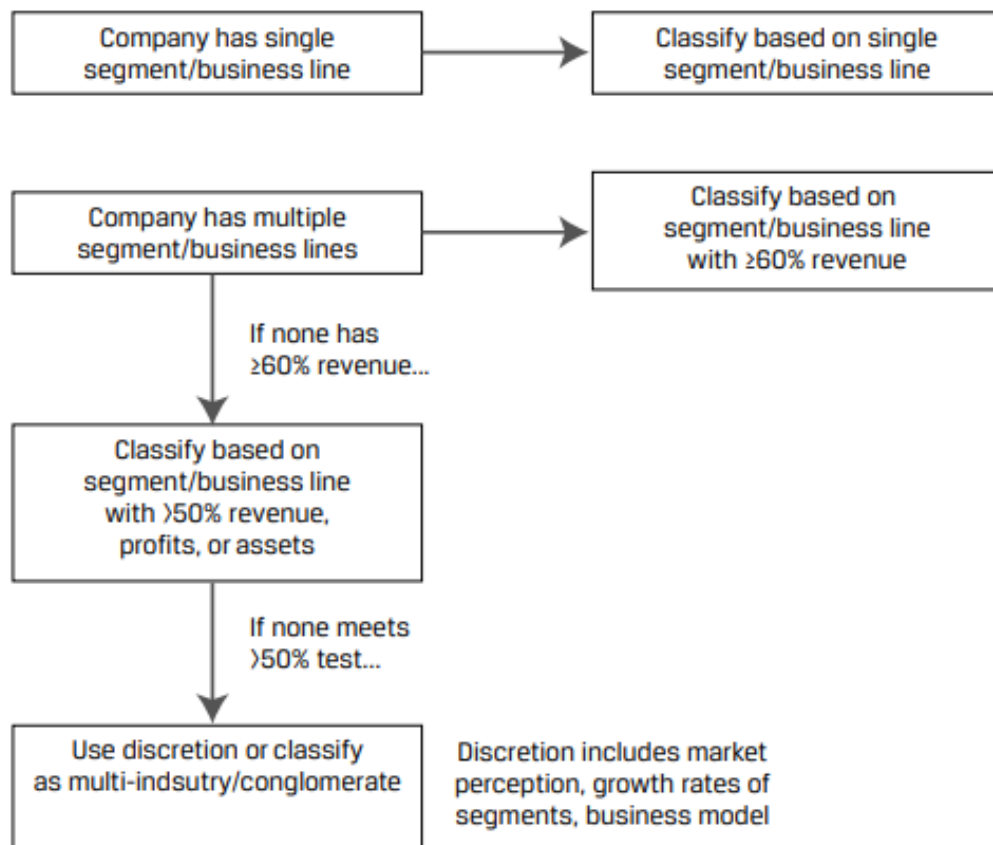
Three widely used commercial industry classification schemes are:

- Global Industry Classification Standard (GICS) by MSCI and S&P Dow Jones Indices
- Industry Classification Benchmark (ICB) by FTSE Russell
- The Refinitiv Business Classification (TRBC) by Refinitiv

These classification systems contain multiple levels: starting at the broadest level – a general sector grouping, that is then subdivided into more narrowly defined sub-industry groups. Exhibit 3 from the curriculum shows the names and hierarchy of each scheme, the number of groups in each tier, and the composition of the highest tier.



The general process of classifying companies is illustrated in Exhibit 4.



Limitations of Third-Party Industry Classification Schemes

Third party classification systems are a useful starting point; however, analysts need to be aware of the methodologies and the limitations of these systems. Four important limitations are:

1. Groupings of companies with business model variations or that sell substitute products
2. The classification of multi-product companies
3. Geographical considerations
4. Changes in groupings over time that affect prior-period comparability of industry statistics

Instructor's Note: We continue with the case study of 'Warehouse Club Inc' from the previous learning module. These cases studies are presented as-is from the curriculum.

CASE STUDIES

Defining Warehouse Club Inc.'s Industry

Warehouse Club Inc. has a single line of business and would be assigned to the "Consumer Staples Merchandise Retail" sub-industry in GICS.

Sector	Industry Group	Industry	Sub-Industry	Other Sub-Industry Constituents
Consumer Staples	Consumer Staples Distribution & Retail	Consumer Staples Distribution & Retail	Consumer Staples Merchandise Retail	Costco, Walmart, BJ's, Carrefour, Sun Art Retail Group, Coles Group, PriceSmart

While this classification is logical, the sub-industry includes companies outside Warehouse's domestic geography that are less relevant to its customers and does not include many companies in its domestic geography that sell similar products to the same types of customers. For example, grocers and drugstores are included in a different sub-industry (Food Retail and Drug Retail, respectively), and e-commerce retailers that it competes with are in a different sector entirely (Consumer Discretionary).

The challenge in placing Warehouse in an industry is that it sells several categories of consumer goods, both staples and discretionary goods like apparel, as do many other companies. Retailers also have varying business models—including membership-based warehouses, e-commerce, and co-locations with other services like pharmacies—and have varying location, assortment, and pricing strategies.

Elaine Nguyen chooses to define Warehouse's industry more broadly than its GICS sub-industry by taking a customer perspective: retailers of all kinds operating in Warehouse's domestic geography, except for automobile and auto parts retailers as Warehouse does not sell those products or even substitutes for them (and it is a large category of domestic retail sales). Nguyen also does not include restaurants and food services, instead considering them substitutes.

Alternative Methods of Grouping Companies

Apart from the similarity of product/service approach, alternative approaches to grouping companies are:

- Geography: Companies can be classified by country and then countries can be aggregated into categories such as developed, emerging, and frontier markets.
- Sensitivity to the business cycle: Depending on their sensitivity to the business cycle, companies can be classified into:
 - *Defensive*: Earnings are relatively stable over the business cycle. They produce goods or services for which is not affected much by the business cycle. Examples

of non-cyclical industries are food and beverage, household and personal care products, health care, and utilities.

- *Cyclical*: Earnings are highly dependent on the stage of the business cycle. They produce goods or services that are often expensive and/or represent purchases that can be delayed. Examples of cyclical industries are autos, housing, basic materials, industrials, and technology.
- Statistical similarities: Companies that historically have had highly correlated returns are grouped together.
- ESG characteristics: Companies are grouped together on some metrics such as the ratio of carbon emissions to revenues, board and executive personnel diversity, and exposure to certain business such as tobacco and gambling.

4. Industry Survey

After defining an industry, the next step is to survey the industry by estimating its size, calculating its historical growth rate, evaluating the nature of that growth rate, measuring profitability, and identifying major industry players and market share trends over time.

Industry Size and Historical Growth Rate

The size of an industry is typically measured by total annual sales from a product or customer perspective, which does not necessarily include all sales of each industry constituent. For example, Amazon's retail segment sales would be included in sizing the retail industry, but Amazon Web Services segment sales would not be.

Characterizing Industry Growth

An industry's growth rate can be characterized in a style box as shown in Exhibit 5.

Defensive	-Utilities -Beverages -Pharmaceuticals	-Biotechnology -Software -Gaming
Cyclical	-Crude oil -Natural gas -Freight transportation	-Semiconductors -Fintech -Digital advertising
	Mature	Growth

Industry Profitability Measures

The best measure of industry profitability is return on invested capital (ROIC) which measures after-tax operating profits for each dollar of invested capital and is agnostic to capital structure.

A profitable industry (high ROIC) is preferable to an unprofitable one. Rather than a point estimate of ROIC, a time series of ROIC gives a more meaningful picture. It helps answer the question – Is the industry profitability rising or falling?

Market Share Trends and Major Players

Market shares are measured by expressing industry participant's annual revenues as percentages of industry size each year.

The Herfindahl–Hirschman Index (HHI) is frequently used to measure industry concentration. It is calculated by squaring the market share of each firm competing in a market and then summing the resulting numbers. High numbers indicate high concentration and vice-versa.

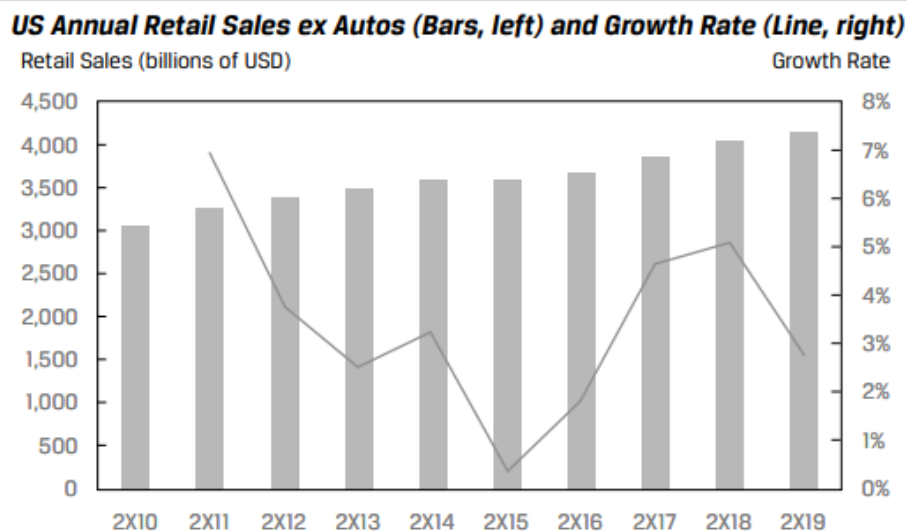
The HHI of a market composed of four firms with shares of 30, 30, 20, and 20 is $30^2 + 30^2 + 20^2 + 20^2 = 2,600$. The maximum HHI is for a monopoly: $100^2 = 10,000$. Some countries consider markets with an HHI between 1,500 and 2,500 to be moderately concentrated, while markets with an HHI greater than 2,500 are considered highly concentrated. Acquisitions in highly concentrated markets that increase the HHI by more than 200 points are frequently subject to regulatory challenges.

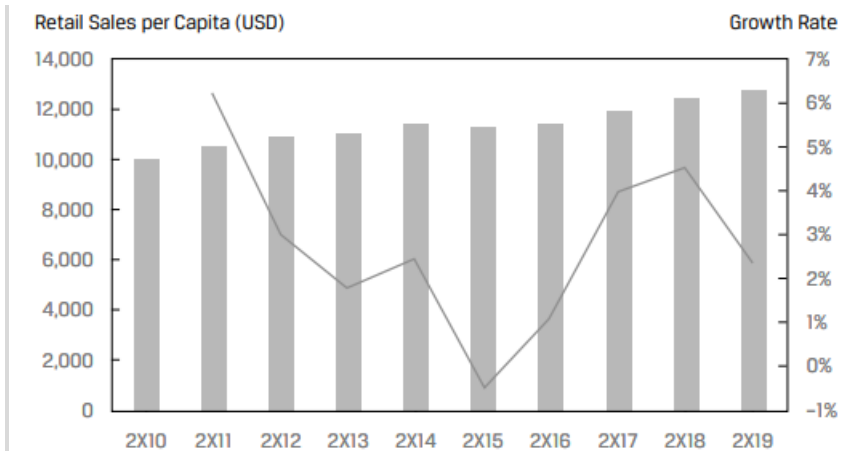
CASE STUDIES

Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd.

Industry Size and Growth Rate

Elaine Nguyen sizes the US retail (excluding autos) industry using three measures: total annual revenues, annual revenues on a per capita basis, and the number of retailers. For all these measures, Nguyen uses data from the US Census Bureau, since the industry consists mostly of small businesses with no public filings and the data are high quality; the Bureau has been collecting such data since 1952.



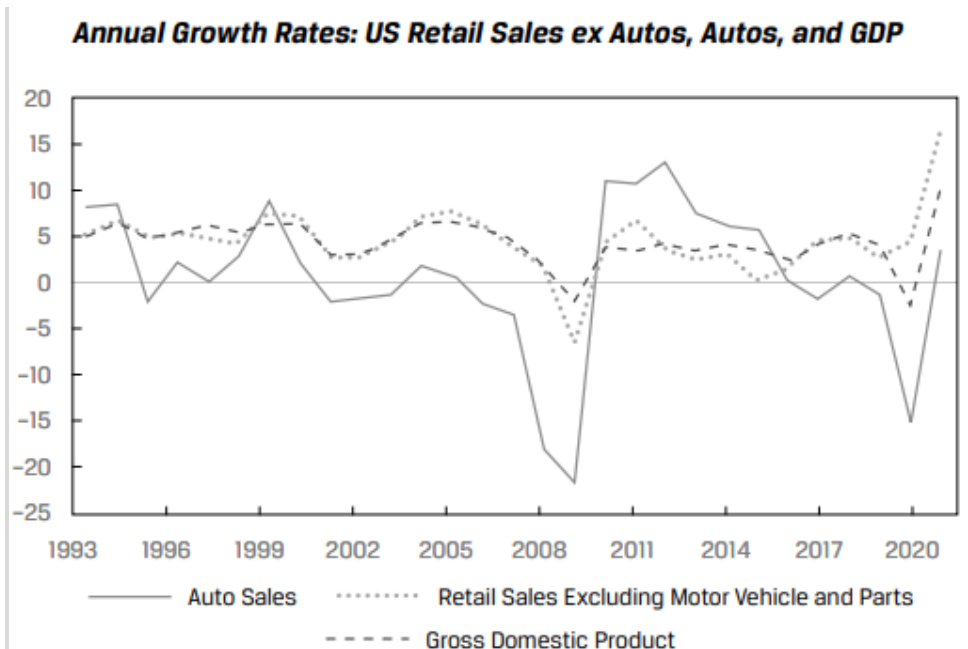


At the end of 2X19, there were 556,531 firms in the industry, but over 92% of them employed fewer than 20 people. Just 470 large firms with over 5,000 employees accounted for 62% of total industry employment. Of these large firms, 144 were publicly traded.

US Retailers (ex autos) as of 2X19	
< 20 employees	516,766
Between 20 and 5,000 employees	39,295
> 5,000 employees	470
Total	556,531
Note: public companies	144

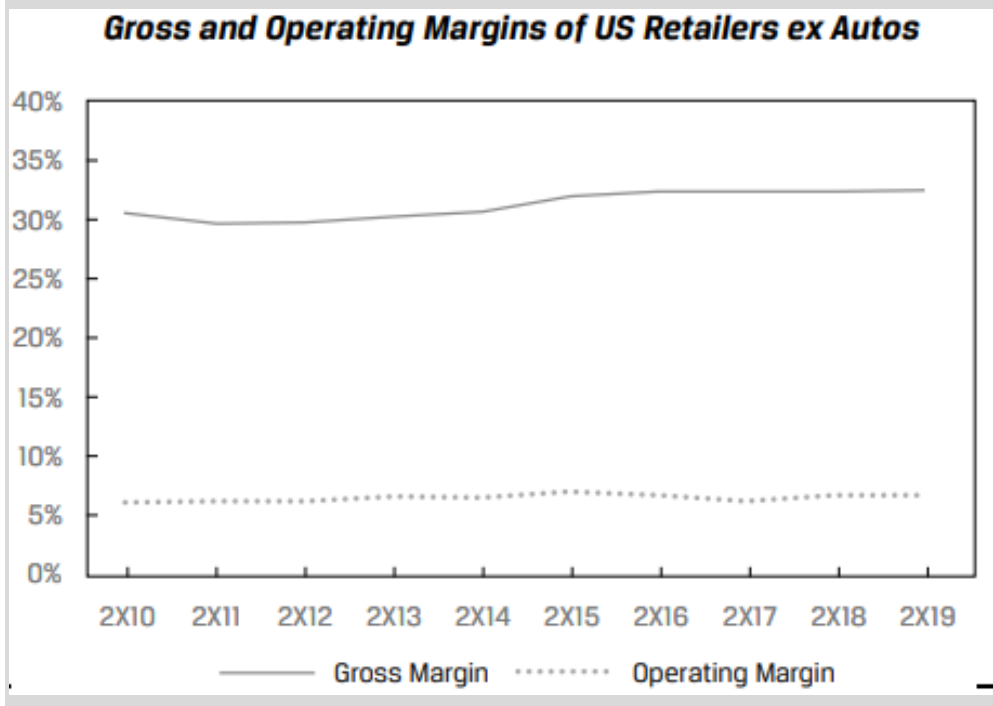
Character of Industry Growth

The retail industry has existed for hundreds of years in the USA, though there have been many changes in both form (e.g., department stores, discounters, e-commerce) and the products sold. Comparing industry sales growth with GDP growth since the early 1990s shows that the two are tightly related. However, the industry does not show extreme cyclicity, unlike the auto industry. Elaine Nguyen classifies the industry as mature and moderately cyclical.



Industry Profitability

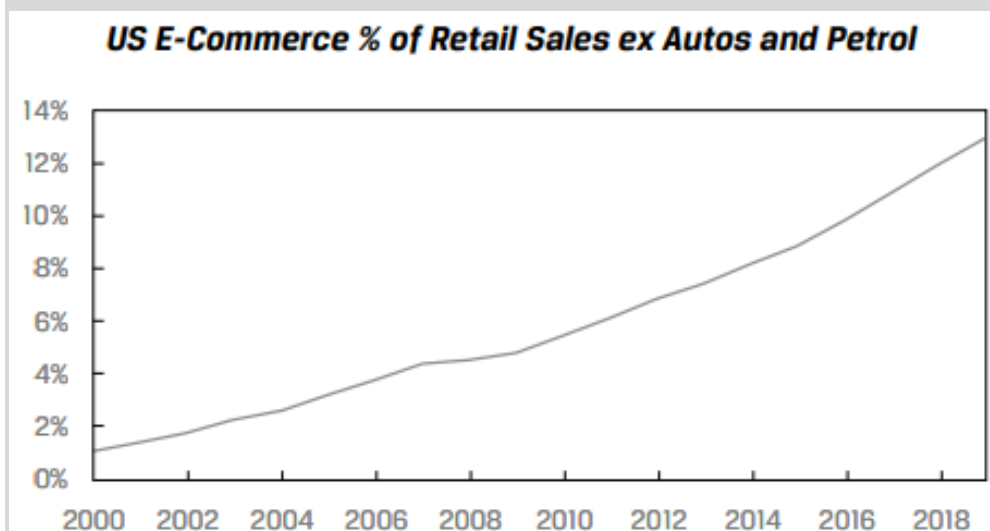
Nguyen analyzes industry profitability in two ways: (1) the gross and operating margins of all retailers (except autos) in the USA using Census Bureau data and (2) the profitability of the 10 largest publicly traded retailers in the USA by 2X19 sales. Because of segment disclosure changes, M&A, and other corporate transactions, Nguyen examines the last 5 rather than 10 years of data.



Operating Margins	2X15	2X16	2X17	2X18	2X19
Walmart (US segment)	6.4%	5.8%	5.3%	5.2%	5.1%
Walmart (Sam's Club segment)	3.2%	2.9%	1.5%	2.6%	2.8%
Amazon.com (North America segment)	2.2%	3.0%	2.7%	5.1%	4.1%
Costco Wholesale (US segment)	2.7%	2.7%	2.8%	2.7%	2.7%
The Home Depot, Inc.	13.3%	14.2%	14.5%	14.4%	14.4%
Kroger Co.	3.3%	3.0%	2.1%	2.1%	1.8%
Target Corp.	6.9%	7.1%	5.8%	5.5%	6.0%
Lowe's Companies, Inc.	8.4%	9.0%	9.6%	7.9%	8.8%
Albertsons Companies, Inc.	0.7%	1.1%	0.3%	1.0%	1.5%
Ahold Delhaize (US segment)	4.0%	3.9%	4.1%	4.5%	4.3%
The TJX Companies, Inc. (US segments)	13.0%	12.8%	11.8%	11.8%	11.6%

Market Share Trends

The most significant market share trend in US retail is the shift toward e-commerce, which has been ongoing since the late 1990s. This shift has accrued largely to Amazon.com, the pioneer in US e-commerce, with its share of e-commerce at roughly 40% (Amazon has a large third-party merchant business but does not disclose GMV, so its exact share is unknown).



The industry is fragmented, with the top 10 largest firms holding less than 30% market share, the 10th largest holding less than 1%, with a large amount of share held by over 500,000 small retail businesses. However, the top 10 largest firms have gained share (~2%, from 26.3% in 2X15 to 28.6% in 2X19).

US Net Sales	2X15	2X16	2X17	2X18	2X19
Walmart US and Sam's Club	355,206	365,198	377,693	389,505	399,796
Amazon.com*	63,708	79,785	106,110	141,366	170,773
Costco Wholesale (US segment)	84,351	86,579	93,889	102,286	111,751
The Home Depot, Inc.	80,550	86,615	92,413	99,386	101,333
Kroger Co.	109,830	115,337	123,280	121,852	122,286
Target Corp.	73,717	69,414	71,786	74,433	77,130
Lowe's Companies, Inc.	59,074	61,311	63,263	65,872	67,147
Albertsons Companies, Inc.	58,734	59,678	56,925	60,535	62,455
Ahold Delhaize (US segment)	42,482	42,946	43,194	44,174	44,841
The TJX Companies, Inc. (US segments)	23,899	25,651	27,365	29,845	32,021
Total top 10	951,551	992,514	1,055,918	1,129,254	1,189,533
Share of US retail sales ex autos	26.3%	26.9%	27.4%	27.9%	28.6%

5. Industry Structure and External Influences

Porter's Five Forces is a commonly used framework of assessing industry structure that helps determine an industry's long-run profitability.



The table below summarizes what each of these five forces means:

Porter's Five Forces	
Force	Description
Threat of substitute products	If substitutes to a company's products are easily available, then the threat is high and demand for the company's products will decrease. Customers may switch to alternative products if switching costs are low.

	Ex: Low-priced brands are close substitutes to premium brands; low-cost mobiles from China are substitutes to Samsung or iPhone; If coffee prices increase substantially, coffee drinkers may switch to tea; or during a recession, movie goers may prefer to watch movies at home, using substitute forms instead of going to the cinema. If this force is strong, it will weaken the pricing power of the market players.
Bargaining power of customers	Customers enjoy bargaining power in industries with large volumes and smaller number of buyers. The price competition and profitability is low as customers demand low prices. Ex: Airlines ordering numerous aircrafts from Boeing or Airbus. Since airlines typically order a large number of aircrafts, they have high bargaining power.
Bargaining power of suppliers	Suppliers enjoy pricing power in industries where suppliers are small and the supply of key inputs to a company is scarce. Ex: Consumer products companies have limited control over price.
Threat of new entrants	If barriers to entry are high, then the threat of new entrants is low. Conversely, if barriers to entry are low, then the threat of new entrants is high. Ex: The threat of new entrants is high in the mobile handset market.
Intensity of rivalry among existing competitors	Industries with high fixed costs, high exit barriers, little differentiation in products, and similar size experience intense rivalry. Ex: Boeing and Airbus.

CASE STUDIES

Porter's Five Forces Analysis for US Retail (ex autos)

Threat of New Entrants: Very High

Opening a retailer, especially an e-commerce retailer that utilizes third-party merchant services, is relatively easy and common. In the USA, retailers are the most common type of business formed—by a wide margin, with over 40,000 new firms filing formation papers each month (the next most common is transportation and warehousing, with around half the number of retailer formations). Customers can easily switch retailers, as most do not have an ongoing relationship such as a subscription fee or contract, and there are minimal regulatory barriers such as licenses and patents.

Threat of Substitutes: Low

Broadly speaking, the substitute for retail is consumer services (e.g., restaurants, travel, healthcare), which also includes digital services (such as streaming video subscriptions and gaming). Most categories of goods, however, are not easily replaceable with services (e.g.,

apparel and home décor), or they enjoy a cost advantage because of lower labor intensity (e.g., fresh food). There have been periods when services grew faster than goods, but as shown in an earlier exhibit, retail sales have grown essentially in line with US nominal output/income. Retail is arguably one of the oldest industries, and while it has evolved over time, it has yet to be replaced.

Bargaining Power of Customers: Moderate

Retail customers are highly fragmented, with each consumer representing a distinct decision-maker, as there are generally no group purchasing organizations. However, many products are sold by many retailers, and the internet has enabled easy comparison shopping at retailers. Customers are price sensitive with respect to identical products sold by different retailers.

Bargaining Power of Suppliers: Low to Moderate

Key suppliers for retailers include manufacturers of goods, employees, and lessors of retail or fulfillment space. In most cases, numerous options are available. However, branded goods are sold exclusively by a sole manufacturer that may impose high prices and other economic terms, like shelf space and visibility, and may want to sell only to certain retailers (e.g., makers of luxury goods may not sell to discount retailers in order to maintain exclusivity).

Rivalry among Existing Competitors: High

Given the sheer number of similarly sized firms selling similar or identical products, retailers compete fiercely, often with price promotions and discounts as price is one of the few ways they can lure customers away from competitors.

External Influences on Industry Growth

Apart from internal factors, it is also important to look at factors outside the industry that influence the industry's profitability. A framework commonly used for this purpose is PESTLE analysis: political, economic, social, technological, legal and environmental influences.

Political influences: Examples of political influences include changing fiscal and monetary policy, government's direct selling and purchasing activities, regulatory changes, and geopolitical conditions and actions.

Economic influences: Demand for products and services are affected by overall economic activity at any point in time. Economic variables that affect an industry's revenues and profits are: GDP, level of interest rates, inflation, and how easily money is available to businesses. *Example:* People cut down on discretionary spending during the festive/holiday season if inflation is very high (emerging economies), or if the economy is in a recession leading to job cuts.

Social Influences: How people work, spend their money and leisure time pursuing hobbies, and travel affect various industries. For example, more women entering the workforce

worldwide has spun many new industries, while boosting others. Restaurants, work wear for women, home and child care services, and demand for more cars are some of the effects of this trend.

Technological Influences: New technologies can rapidly change an industry or push them into the decline stage faster. *Examples*: Invention of the microchip and the evolution of the computer hardware industry; impact of digital imaging technology on the photographic film industry, USBs on DVD/CD, digital music on cassette player industry.

Legal Influences: Refers to changes in laws and regulations that alter an industry's business practices. Examples of industries strongly affected by legal influences are tobacco, alcohol, and gambling.

Environmental Influences: In recent times, the need to evaluate and mitigate environmental impact has become an important consideration for industries. Climate change poses a real threat to the growth and profitability of many industries. For example, public awareness about the environmental impact of livestock and protection of animal rights has been increasing. Many people are shifting towards healthier and plant-based diets. These factors will impact the agriculture industry.

6. Competitive Positioning

An analyst should assess a company's competitive strategy along three dimensions:

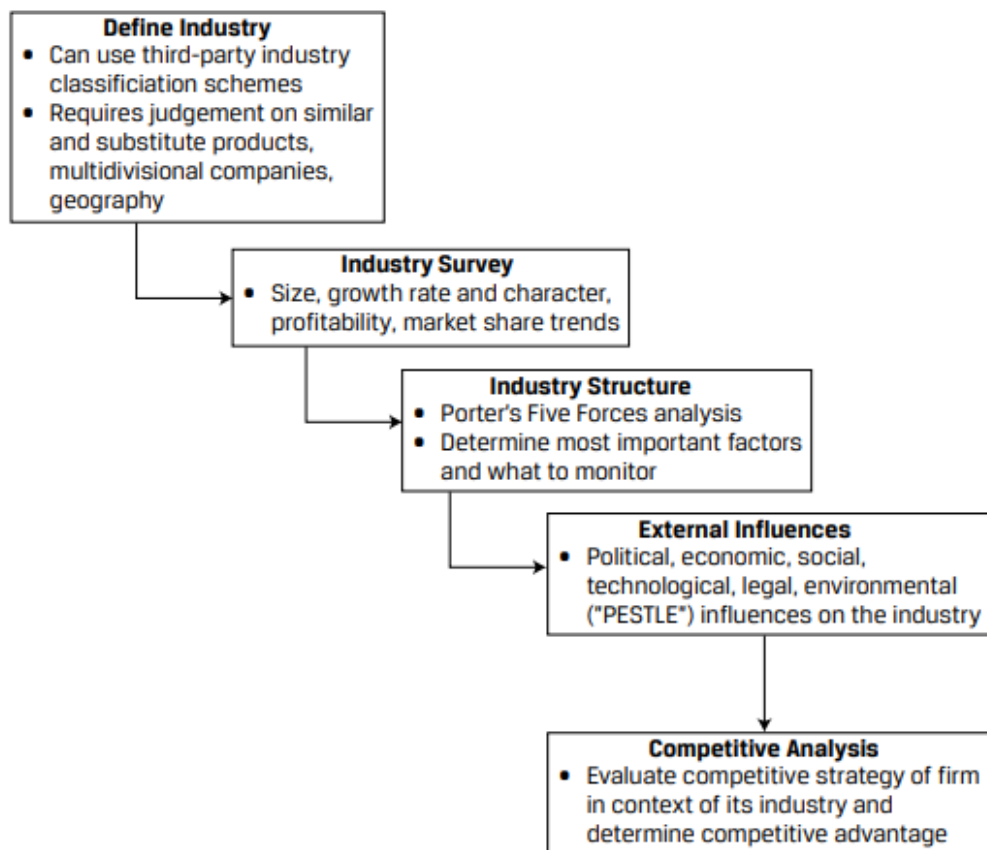
- Does the strategy create a defense against the five industry forces?
- Does the strategy benefit from, or is at least not at odds with, the expected external industry influences identified in the PESTLE analysis?
- Does the company have the resources and capabilities to execute the strategy?

Three well known competitive strategies that have worked in a variety of industries are: cost leadership, differentiation, and focus. Exhibit 6 from the curriculum compares the three strategies.

	Cost Leadership	Differentiation	Focus
Means of executing strategy	▪ Economies of scale from fixed costs ▪ Favorable access to raw materials ▪ Culture of strict cost control ▪ Aggressive pricing to gain high volume ▪ Low-cost distribution ▪ Economies of scope	▪ Investments in advertising, brand, customer service, proprietary distribution channels ▪ Protection using trademarks, copyright, patents ▪ Superior quality, unique features ▪ Culture of strong customer experience ▪ Premium pricing ▪ Integration of services, software, and hardware	▪ Proximity to customers and strong understanding of their needs ▪ May incorporate elements of strategy from both cost leadership and differentiation, but focused on particular group
Which of the Five Forces it defends against (why it works)	▪ Threat of new entrants: Capital requirements and scale advantages deter entrants ▪ Bargaining power of customers: Customers can only bring prices down to the costs of the marginal producer, leaving margin for the cost leaders ▪ Industry rivalry: Rivals may not be able to compete on price with cost leaders	▪ Threat of new entrants and of substitutions: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable to unwilling to comparison shop or switch ▪ Bargaining power of suppliers: The company may have the ability to pass along price increases to customers and/or margin to absorb cost increases	▪ Threat of new entrants and of substitutes: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable or unwilling to comparison shop or switch
Industry appropriateness	▪ Capital intensive ▪ Price-conscious customers ▪ Customers do not value or notice product differences ▪ Minimal innovation in industry	▪ Price is not foremost concern for customers ▪ Customers value distinctiveness ▪ Innovation in industry, with products varying in features and forms	▪ Difficult (or uneconomical) to serve customer group, product, or geography for other players
Risks to the strategy	▪ Cost inflation, loss of discipline ▪ Technological change that results in loss of cost leadership or market share ▪ Desire for premiumization among customers	▪ Imitation by competitors ▪ Buyers become sophisticated, no longer demand level of service ▪ Pricing premium becomes too high for customers to bear ▪ May preclude high market share, as customers value exclusivity	▪ Larger competitors outcompete on price ▪ The differences in demand between the narrow group and industry as a whole narrow ▪ Buyers become sophisticated, no longer demand level of service

Summary

LO: Describe the purposes of, and steps involved in, industry and competitive analysis.



LO: Describe industry classification methods and compare methods by which companies can be grouped.

An industry is commonly defined as companies that sell similar products or services, from the perspective of a customer.

Analysts usually use third-party classification systems such as GICS, but they need to be aware of the methodologies and limitations of these systems.

Apart from the similarity of product/service approach, alternative approaches to grouping companies are: geography, sensitivity to business cycle, statistical similarities, and ESG characteristics.

LO: Determine an industry's size, growth characteristics, profitability, and market share trends.

After defining an industry, the next step is to survey the industry by estimating its size, calculating its historical growth rate, evaluating the nature of that growth rate, measuring profitability, and identifying major industry players and market share trends over time.

The size of an industry is typically measured by total annual sales from a product or customer perspective, which does not necessarily include all sales of each industry constituent.

An industry's growth rate can be characterized in a style box: Cyclical – Defensive vs. Mature – Growth.

The best measure of industry profitability is return on invested capital (ROIC) which measures after-tax operating profits for each dollar of invested capital and is agnostic to capital structure.

Market shares are measured by expressing industry participant's annual revenues as percentages of industry size each year. The Herfindahl–Hirschman Index (HHI) is frequently used to measure industry concentration.

LO: Analyze an industry's structure and external influences using Porter's Five Forces and PESTLE frameworks.

Porter's Five Forces is a commonly used framework of assessing industry structure that helps determine an industry's long-run profitability.



To examine external factors, analysts can use the PESTLE framework: political, economic, social, technological, legal and environmental influences.

LO: Evaluate the competitive strategy and position of a company.

An analyst should assess a company's competitive strategy along three dimensions:

- Does the strategy create a defense against the five industry forces?
- Does the strategy benefit from, or is at least not at odds with, the expected external industry influences identified in the PESTLE analysis?
- Does the company have the resources and capabilities to execute the strategy?

Three well known competitive strategies that have worked in a variety of industries are: cost leadership, differentiation, and focus.

LM07 Company Analysis: Forecasting

1. Introduction.....	2
2. Forecast Objects, Principles, and Approaches.....	2
3. Forecasting Revenues	4
4. Forecasting Operating Expenses and Working Capital.....	6
5. Forecasting Capital Investments and Capital Structure	8
6. Scenario Analysis.....	9
Summary.....	12

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Version 1.0

1. Introduction

This learning module covers:

- What to forecast, approaches to forecasting, and selecting a forecast horizon.
- Forecasting particular items: revenues; operating expenses and working capital; capital investments and capital structure.
- Use of scenario analysis in considering multiple outcomes.

2. Forecast Objects, Principles, and Approaches

What to Forecast?

Analysts can focus on different forecast objects such as:

- Drivers of financial statement lines: This concept was covered in the previous learning module. The net sales for Warehouse Club Inc were analyzed using two bottom-up drivers: the average number of stores opened and the average net sales per store. Net sales can be forecasted by forecasting these drivers individually and multiplying them. The advantage of this approach is that it improves the explanatory value of the forecast and may also improve accuracy.
- Individual financial statement lines: Instead of forecasting drivers, we can directly forecast individual financial statement lines. This approach is often used for lines without clear drivers, for less-material items, and for items that the analyst does not have a perspective on. For example: amortization expense, analysts may use the estimate provided by the management or simply assume that the quantity will remain the same in future periods.
- Summary measures: This includes items like free cash flow, earning per share, and total assets. The advantage of using these as forecast objects is efficiency. The disadvantage is reduced transparency, which makes it difficult to audit the forecast. This method is best suited when the summary measure is stable and predictable, or when issuer disclosures are severely restricted.
- Ad hoc objects: This includes items that may not have been reported on previous financial statements. Examples include the outcomes of a significant legal proceeding, a government regulatory action, a tax dispute, or a natural disaster.

An analyst's choice of forecast object depends on available information, efficiency, accuracy, explanatory value, and verifiability.

Focus on Objects That Are Regularly Disclosed

CFAI recommends using forecast objects that are either regularly disclosed or can be directly calculated using what is regularly disclosed.

Non-regularly disclosed information can be used to supplement forecasts but is problematic for direct use because forecasts cannot be confirmed in a timely manner.

CFAI also cautions against overly complex models, because they require more forecasts and take more time to update, often with no improvement in accuracy.

Forecast Approaches

For any object there are four general forecast approaches:

- Historical results (assume past is precedent): This approach uses past observed or calculated values as a forecast. This is an easy to implement approach and may be appropriate for companies operating in industries where the industry structure is not expected to change. The approach can also be used to forecast objects that are not material or that the analyst does not hold an opinion on. The approach is less appropriate for companies in cyclical industries because the future period could be at a different point in the business cycle.
- Historical Base Rates and Convergence: This approach uses an industry or peer group average or median, calculated over a long time period as a “base rate” for forecasting where an object will converge to over time. This approach is more complex than the previous approach because analyst discretion is required to select the object, the sample to calculate the base rate, and a time frame for convergence to the base rate. The approach may be appropriate for companies operating in well established industries such as banks, airlines, retailers etc. It is less appropriate for companies in changing or new industries.
- Management guidance: A company's management may publicly announce earnings, revenue, and other targets for the next quarter, year, or longer (known as management guidance or simply guidance). This guidance is often provided as a range (e.g. sales growth of 3%-5%) and is based on many sub-forecasts and assumptions by the management about macroeconomic growth, cost inflation, market share changes, exchange rates etc. Using guidance for forecasts may be appropriate when management has demonstrated a track record of reliable estimates.
- Analyst's discretionary forecasts: All other forecast approaches can be grouped under this category. This can include approaches based on surveys, quantitative models, probability distributions, and other unobservable inputs. Discretionary forecasts approaches are commonly used for companies in cyclical industries, companies that have no comparables, or companies undergoing a fundamental change like a shift in the competitive or regulatory environment.

An analyst's choice of forecast approach depends on the company's industry structure, sensitivity to the business cycle, and business model, as well as the reliability and availability of information.

Selecting a Forecast Horizon

The forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclicity of the industry, company-specific factors, and the preferences of the analyst's employer.

Long-term fund managers may focus their forecasting primarily on the next three to five years, whereas shorter-term managers may focus more on the next one or two quarters.

3. Forecasting Revenues

Forecast Objects for Revenues

Revenues can be forecasted using a top-down or bottom-up approach. Common top-down forecast objects include “growth relative to GDP growth” and “market growth and market share”.

Growth relative to GDP growth approach: In this approach, we first forecast the growth rate of nominal GDP. We then forecast the company's revenue growth relative to GDP growth. For example, we can assume that the company's revenue will grow at a rate of 150 bps above the nominal GDP growth rate.

The forecast may also be in relative terms. For example, if we forecast that the GDP will grow at 4%, and we believe that the company's revenue will grow at a 20% faster rate, then the forecasted increase in the company's revenue is $4\% \times (1 + 0.20) = 4.8\%$.

Market growth and market share: In this approach, we combine forecasts of growth in particular markets with forecasts of a company's market share. For example, assume Tesla is expected to maintain a market share of 1% in the automobile market. If the automobile market is expected to grow to \$30 billion in annual revenue, then Tesla's annual revenue is forecasted to grow to $1\% \times \$30 \text{ billion} = \300 million .

Examples of bottom-up drivers for revenue forecast include:

- *Volumes and average selling price:* Volumes and prices of the company's products are forecasted separately and multiplied to get a revenue forecast. For example: The revenues for a mutual fund can be forecasted based on the AUM and management fee rates.
- *Product-line or segment revenues:* Forecasts for individual products, product or business lines, geographic areas, or reporting segments are made and then aggregated into a total revenue forecast.
- *Capacity-based measure:* For example, a retailer's revenue may be forecasted based on same-store sales growth and sales related to new stores.
- *Return or yield based measure:* Forecasts based on balance sheet accounts. For example, a bank's interest revenue can be calculated as loans multiplied by the average interest rate.

Separating Recurring and Non-Recurring Revenue or Revenue Growth

While making forecasts, non-recurring items and effects should be excluded. There are two types of non-recurring items and effects: those that are disclosed by a company's management and those that are not, which require analyst judgment to identify.

Examples of the first type include: the effect of changes in exchange rates, acquisitions/divestitures etc.

An example of the second type is the rapid increase in e-commerce sales during the COVID-19 pandemic. After things normalized, e-commerce sales as a percentage of retail sales began to decline.

Forecast Approaches for Revenues

Instructor's Note: The curriculum illustrates the four approaches to forecasting revenues through the 'Warehouse Club Inc' example. This information is presented as-is from the curriculum.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Top-down drivers (e.g., market size, market share)	Historical results	US retail sales excluding autos has grown on average by 3.4% each year for the last 10 years and the company's share of that (70 basis points in 2X19) has increased by 2 basis points per year. Nguyen can forecast revenues by assuming 3.4% market growth and 2 basis points of market share gain each year going forward.
	Historical base rates and convergence	Rather than assuming continued share gains, Nguyen may forecast that Warehouse's sales growth rate will converge to the market growth rate over the next five years, and forecast revenues by assuming a smooth decline in annual market share gains to 0 basis points by 2X25.
	Management guidance	If Warehouse provides annual or longer-term expectations for market size, market share, or sales, Nguyen can use these in her revenue forecast.
	Analyst's discretionary forecast	Warehouse has operated stores in some regional geographies for a long time. Nguyen may commission a study or survey to determine Warehouse's market share in its most established markets (which could be multiples of its all-in market share) and develop market size and share estimates by established/growing/new regions for Warehouse and aggregate them into a sales forecast.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Bottom-up drivers (e.g., stores, sales per store, membership, membership annual price)	Historical results	Warehouse has opened 4–6 new stores, grown net sales per store by 3.8%, and added ~450,000 members per year for the last 10 years. Nguyen could assume these same quantities going forward as the revenue forecast.
	Historical base rates and convergence	There are several mature retailers in the USA that ceased opening new stores several years ago. Nguyen may slow the rate of new-store openings gradually to zero in the USA.
	Management guidance	Management has provided guidance on new-store openings per year and the annual price of membership, which are bottom-up drivers for the revenue forecast.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for stores and members by determining stores per capita and the ratio of members to stores in Warehouse's more mature regional geographies and then applying these quantities, or fractions of them, to different regions where Warehouse is opening stores today. This approach would require a geographic analysis of stores and commissioning surveys to determine regional membership counts or the use of third-party or government data, if available.

Analysts must explicitly or implicitly incorporate their view on key risk factors in their revenue forecasts. Four key risk factors to consider for all companies are competition, changes in business cycle, inflation and deflation, and technological developments.

4. Forecasting Operating Expenses and Working Capital

Operating costs include cost of goods sold (COGS) and selling, general, and administrative expenses (SG&A). Disclosures about operating costs are often less detailed than revenue. Therefore, analysts are often forced to use more aggregated forecasted objects on the company level (rather than forecasting costs separately for different geographic regions, or business segments) or summary measures like EBITDA margins. However, forecasts for operating expenses should still be coherent with revenue forecasts. The choice of forecast object can vary depending on the forecast horizon.

Cost of Sales and Gross Margins

Since COGS are directly related to sales, we can forecast future COGS as a percentage of future sales. Analysts should understand the historical relationship between COGS and sales, and determine if this relationship is likely to decrease, increase, or remain unchanged.

$\text{Sales} - \text{COGS} = \text{gross margin}$. Therefore, COGS and gross margin are inversely related.

Some factors that analysts should consider while forecasting COGS are:

- Forecasting accuracy can be improved by forecasting COGS for the company's various segments or product categories separately.
- Determine if a company has employed hedging strategies to protect its gross margins. When input prices increase, COGS increase and gross margin decreases. However, hedging strategies can help mitigate this impact.

- Examine gross profit margins of competitors. This can be a useful cross check for estimating a realistic gross margin. Any difference in gross margins for companies in the same segment must relate to differences in their business operations.

SG&A Expenses

As compared to COGS, SG&A expenses are less directly related to revenue.

SG&A can be broken down into two components – fixed and variable. The fixed components such as R&D expenses, management salaries, head office expenses, supporting IT and administrative operations tend to increase and decrease gradually over time. They do not fluctuate with sales.

On the other hand, the variable components such as selling and distribution costs are more directly related to sales. For example, when sales increase, the company may pay out higher bonuses to its salesforce, it may also hire additional salespersons.

Working Capital Forecasts

Working capital forecasts are typically made using efficiency ratios as the forecast objects, which are combined with sales and cost forecasts to project accounts receivables, inventories, accounts payable, and other current asset and liabilities.

Exhibit 9 from the curriculum illustrates the forecast of operating costs and working capital for Warehouse Club Inc.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross margin (on net sales)	Historical results	From 2X10 to 2X19, Warehouse's gross margin increased by an average of 2 bps per year. Such an increase may reflect increasing bargaining power over suppliers with scale, and Nguyen can use the same 2 bps per year increase in gross margin as a forecast.
	Historical base rates and convergence	Both the industry average gross margin and that of larger, more mature retailers such as Walmart are much greater than Warehouse's gross margin. But Warehouse's business model is to use a low gross margin on net sales to drive high-margin membership revenues. Therefore, Nguyen probably will not use this approach.
	Management guidance	If Warehouse management provides guidance on margins, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may formulate a discretionary forecast for gross margin by exploring the relationship between it, inflation/deflation in Warehouse's product categories, and management's desire for member savings vs. competitors to equal 5× the annual price of membership.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Selling, general, and administrative expenses (SG&A) as a percentage of sales	Historical results	SG&A, which captures the cost of operating stores and some merchandise distribution, is primarily composed of employee payroll. Relative to sales, it has remained mostly stable at or around 9%. Nguyen may forecast it to remain the same.
	Historical base rates and convergence	Warehouse's SG&A as a percentage of sales is already lower than that of larger retailer competitors (and substantially lower than the retail industry average), owing to its business model differences that emphasize store productivity and cost leadership. Nguyen will therefore probably not use this approach to forecast SG&A.
	Management guidance	If Warehouse management provides guidance on SG&A, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Since SG&A is primarily composed of employee payroll, Nguyen may forecast it based on projected employee productivity (e.g., employees to stores) and projected wage rates, which may be based on macroeconomic variables like projected economy-wide wage growth.
Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Working capital ratios (e.g., DSO, DOH, DPO)	Historical results	As of 2X19, Warehouse's DSO, DOH, and DPO were 3, 30, and 28, respectively. DSO and DPO have been stable for the last 10 years, while DOH has decreased by 7 days since 2X10. Nguyen might forecast a stable DSO and DPO and a modestly declining DOH over the next several years.
	Historical base rates and convergence	Warehouse's working capital ratios and cash conversion cycle are materially better than the industry average and some larger peers and have been for many years, owing to its business model differences. Nguyen will therefore probably not use this forecast approach, unless Warehouse changes its business model (e.g., it launches an e-commerce business).
	Management guidance	If Warehouse management provides guidance on working capital (e.g., a projection for the net working capital to sales ratio), Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for working capital by, for example, estimating net working capital to sales ratios by product category (e.g., fresh foods, packaged foods, non-food merchandise) and applying them to a net sales forecast also broken down by category.

5. Forecasting Capital Investments and Capital Structure

Forecasts for capital expenditure may differentiate between maintenance and growth capital expenditures. Forecasts for maintenance capital expenditures are usually based on depreciation and amortization expenses. Forecasts for growth capital expenditures are linked to a company's strategy, expansion plans, and revenue growth.

Forecasts for a company's capital structure take into account the historical leverage ratios and capital structure, the company's financial strategy, and capital expenditure forecasts.

Exhibit 12 from the curriculum illustrate these approaches for Warehouse Club Inc.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Capital expenditures as a percentage of net sales	Historical results	Warehouse has spent 2% of net sales on capital expenditures for the last 10 years, which Nguyen can use as the forecast going forward.
	Historical base rates and convergence	More mature retailers (e.g., Walmart in the USA) have all but stopped opening new stores, which has reduced their capital requirements to maintenance and new initiatives like e-commerce. If Warehouse does not intend to go to new geographies or launch an e-commerce business, Nguyen might forecast capital expenditures trending down to maintenance level, along with a decline in new-store openings per year, as Warehouse matures.
	Management guidance	Warehouse management provides guidance on new-store openings, which Nguyen can use to forecast capital expenditures (along with a maintenance capital expenditure forecast and a forecast for growth capital expenditures per new store).
	Analyst's discretionary forecast	Nguyen might instead use a discretionary forecast for capital expenditures by projecting a maintenance CAPEX amount (not disclosed by management) and growth CAPEX associated with new-store openings.
Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross debt to EBITDA	Historical results	For the last five years, Warehouse's ratio of debt to trailing twelve months EBITDA has been 1.10, which Nguyen can project going forward. The product of this ratio and projected EBITDA (from earlier forecasts of sales and operating expenses) is Warehouse's gross debt.
	Historical base rates and convergence	Warehouse's more mature competitors and peers use a greater amount of debt in their capital structure, and return more capital to shareholders with buybacks and dividends. Nguyen might forecast the same for Warehouse over time, with an increasing ratio of debt to EBITDA and percentage of free cash flow returned to shareholders.
	Management guidance	Warehouse's management may provide a target capital structure through a target debt to EBITDA ratio or credit rating, which Nguyen can use as the forecast.
	Analyst's discretionary forecast	Nguyen might instead make a discretionary forecast linked to her free cash flow forecast, projecting a debt to EBITDA ratio and return of capital to shareholders. Depending on Fyleton Investments' strategy, Nguyen might engage with Warehouse's management to advocate for such a capital structure.

6. Scenario Analysis

The final step in forecasting involves considering the possibility of various outcomes based on key risk factors and assessing their likelihood of occurrence.

Rather than develop single point estimate forecasts, analysts make several forecast scenarios that vary based on different outcomes with respect to key risk factors. To make investment

decisions, investors compare these scenarios to other analysts' forecasts for a company, as well as forecasts implied by current valuations.

Technological developments can change the economics of individual companies and entire industries. Sometimes a technological development results in a new product that threatens to cannibalize demand for an existing product. For example, introduction of tablets affected the demand for PCs. To estimate the impact on future demand for the existing product, analysts forecast the unit sales of the new product and multiply it by an expected cannibalization factor. The cannibalization factor can be different for different market segments. For example, the cannibalization factor of tablets is higher in the consumer segment than the non-consumer segment.

Technological developments can affect the demand for a product, the quantity supplied, or both. If technological development leads to lower manufacturing costs, then the supply curve shifts to the right as more quantity is produced at the same price. But if technology results in better substitute products, then the demand curve will shift to the left.

Example:

(This is based on Example 4 from the curriculum)

The pre-cannibalization PC shipment projections (in thousands) for next year are:

Consumer PC shipments	170,022
Non-consumer PC shipments	180,881
Total PC shipments	350,903

The forecasted tablet shipments (in thousands) for next year are:

Consumer tablet shipments	36,785
Non-consumer tablet shipments	1,686
Total tablet shipments	38,471

Assuming a cannibalization factor of 30% for the consumer segment and 10% for the non-consumer segment, calculate the post-cannibalization PC projections.

Solution:

Cannibalization in the consumer segment = $0.3 \times 36,785 = 11,036$

Cannibalization in the non-consumer segment = $0.1 \times 1,686 = 169$

Therefore, the post-cannibalization PC projections are:

Consumer PC shipments	$170,022 - 11,036 = 158,986$
Non-consumer PC shipments	$180,881 - 169 = 180,712$
Total PC shipments	339,698

To evaluate the impact of technological developments on operating costs and margins, we first need to understand the breakdown between fixed and variable costs. If sales decrease

because of cannibalization, then variable costs also decrease. However, the fixed costs remain unchanged.

Since very few companies provide a breakdown of fixed versus variable costs, we can estimate them using the following formulas:

$$\% \text{ Variable Cost} \approx \frac{\% \Delta (\text{cost of revenue} + \text{operating expense})}{\% \Delta \text{revenue}}$$

$$\% \text{ Fixed Cost} = 1 - \% \text{ Variable cost}$$

Summary

LO: Explain principles and approaches to forecasting a company's financial results and position.

Analysts can focus on different forecast objects such as:

- Drivers of financial statement lines
- Individual financial statement lines
- Summary measures
- Ad hoc objects

An analyst's choice of forecast object depends on available information, efficiency, accuracy, explanatory value, and verifiability.

LO: Explain approaches to forecasting a company's revenues.

For any forecast object there are four general forecast approaches:

- Historical results
- Historical base rates and convergence
- Management guidance
- Analyst discretion

An analyst's choice of forecast approach depends on the company's industry structure, sensitivity to the business cycle, and business model, as well as the reliability and availability of information.

Revenue can be forecasted using a top-down or bottom-up approach. Common top-down forecast objects include "growth relative to GDP growth" and "market growth and market share". Bottom-up drivers for revenue forecast include: volumes and average selling price, product-line or segment revenues, capacity-based measures, return or yield based measures.

LO: Explain approaches to forecasting a company's operating expenses and working capital.

Operating costs include cost of goods sold (COGS) and selling, general, and administrative expenses (SG&A). Disclosures about operating costs are often less detailed than revenue. Therefore, analysts are often forced to use more aggregated forecasted objects on the company level (rather than forecasting costs separately for different geographic regions, or business segments) or summary measures like EBITDA margins. However, forecasts for operating expenses should still be coherent with revenue forecasts.

Working capital forecasts are typically made using efficiency ratios as the forecast objects, which are combined with sales and cost forecasts to project accounts receivables, inventories, accounts payable, and other current asset and liabilities.

LO: Explain approaches to forecasting a company's capital investments and capital structure.

Forecasts for capital expenditure may differentiate between maintenance and growth capital expenditures. Forecasts for maintenance capital expenditures are usually based on depreciation and amortization expenses. Forecasts for growth capital expenditures are linked to a company's strategy, expansion plans, and revenue growth.

Forecasts for a company's capital structure take into account the historical leverage ratios and capital structure, the company's financial strategy, and capital expenditure forecasts.

LO: Describe the use of scenario analysis in forecasting.

Rather than develop single point estimate forecasts, analysts make several forecast scenarios that vary based on different outcomes with respect to key risk factors. To make investment decisions, investors compare these scenarios to other analysts' forecasts for a company, as well as forecasts implied by current valuations.

LM08 Equity Valuation: Concepts and Basic Tools

1. Introduction	2
2. Estimated Value and Market Price	2
3. Categories of Equity Valuation Models.....	2
4. The Background for the Dividend Discount Model.....	3
4.1 Dividends: Background for the Dividend Discount Model.....	3
5. Dividend Discount Model (DDM) and Free-Cash-Flow-to-Equity Model (FCFE).....	4
6. Preferred Stock Valuation.....	5
7. The Gordon Growth Model	6
8. Multistage Dividend Discount Models	8
9. Multiplier Models and Relationship Among Price Multiples, Present Value Models, and Fundamentals.....	9
9.1 Relationships among Price Multiples, Present Value Models, and Fundamentals	10
10. Method of Comparables and Valuation Based on Price Multiples.....	11
10.1 Illustration of a Valuation Based on Price Multiples.....	12
11. Enterprise Value	12
12. Asset-Based Valuation	13
Summary	15

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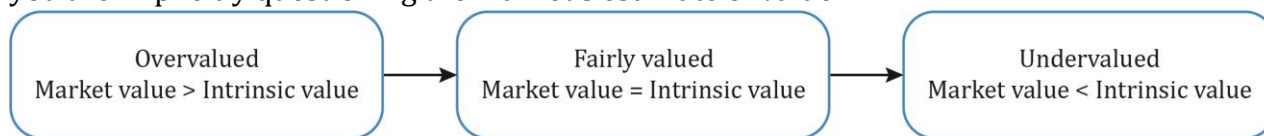
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1. Introduction

We began the equities section with a discussion on how securities markets are organized, how efficient markets are, the different types of equity securities, and how to analyze an industry and a company. The focus of this reading is on determining the intrinsic value of the security.

2. Estimated Value and Market Price

The **intrinsic value** of a security is based on its fundamentals and characteristics. It is also called the fundamental value or estimated value as it is based on the fundamentals such as earnings, sales, and dividends. If the intrinsic value is different from the market price, then you are implicitly questioning the market's estimate of value.



Assume, Caterpillar Inc. is trading on NYSE at \$84.53. An analyst estimates its intrinsic value as \$88.21. Is it overvalued, fairly valued, or undervalued? Going by the relationships given above, the security is undervalued. In reality, making this decision is not that straightforward. It depends on an analyst's input values and assumptions in the model. Some factors to consider when market value $\neq$ intrinsic value:

- *Percentage difference between the market price and intrinsic value.* Assume you calculate the intrinsic value of a security to be \$95, but it is currently trading at \$180. Since the percentage difference is large, it is prudent to calculate the intrinsic price once again because the assumptions or input data to the model may be incorrect.
- *Confidence in your model.* High confidence means the market price will converge to the intrinsic value over the time horizon considered. If your confidence is low, you might see the two prices diverging substantially.
- *Model sensitivity to assumptions.* If many securities appear to be under- or overvalued, analysts should check the model's sensitivity to their inputs.
- *Number of analysts.* The more the number of analysts covering a security, the less the mispricing. Recollect what we read about efficient markets. The market price, in this case, is likely to reflect intrinsic value. Securities neglected by analysts are often mispriced.

3. Categories of Equity Valuation Models

Three major categories of equity valuation model are:

Present value models

- They estimate value as present value of expected future benefits.
- Future benefits are defined as either cash distributed to shareholders (dividend discount models) or cash available to shareholders after meeting the necessary

capital expenditure and working capital expenses (free-cash-flow-to-equity models).

Multiplier models

- They estimate intrinsic value based on a multiple of some fundamental variable.
- For example, either Stock price / earnings (or sales, book value, cash flow).
- Or Enterprise value / EBITDA (or sales).

Asset-based valuation models

- They estimate the value of equity as the value of assets less the value of liabilities.
- Book values of assets and liabilities are typically adjusted to their fair values when using these models.

The choice of model depends on availability of information and the analyst's confidence in the appropriateness of the model. Generally, analysts will try to use more than one model.

4. The Background for the Dividend Discount Model

4.1 Dividends: Background for the Dividend Discount Model

A dividend is a distribution made to shareholders based on the number of shares owned.

Cash dividends are payments made to shareholders in cash. The three types of cash dividends are:

1. Regular cash dividends: They are paid out on a consistent basis. A stable or increasing dividend is viewed as a sign of financial stability.
2. Special dividends: They are one-time cash payments when the situation is favorable (Also called as extra dividends or irregular dividends; used by cyclical firms).
3. Liquidating dividend: This is distributed to shareholders when a company goes out of business.

Stock dividend: Company distributes additional shares instead of cash. A stock dividend simply divides the 'pie' (the market value of equity) into smaller pieces without affecting the value of the pie. Since the market value of equity is unaffected, stock dividends are not relevant for valuation purposes.

Stock split: Increases the number of shares outstanding. For example, in a 2 for 1 split, each shareholder is issued an additional share for each share currently owned.

Reverse stock split: Reduces the number of shares outstanding. For example, in a 1 for two reverse stock split, each shareholder would receive one share for every two old shares.

Stock splits and reverse stock split are similar to stock dividends. They do not change the market value of equity hence they are not relevant for valuation purposes.

Share repurchase: This is an alternative to cash dividends. Here the company uses cash to buy back its own shares. An important point to note is that, as compared to stock dividends and stock splits, share repurchases affect the market value of equity. The effect on

shareholders' wealth is equivalent to a cash dividend. Some key reasons why companies engage in share repurchases instead of cash dividends are:

1. to support share prices.
2. flexibility in the amount and timing of cash distribution.
3. when tax rates on capital gains are lower than tax rates on dividends.
4. to offset the impact of employee stock options.

Dividend payment chronology

A dividend payment schedule is as follows:

1. Declaration date: Company declares the dividend.
2. Ex-dividend date: Cutoff date on or after which buyers of a stock are not eligible for the dividend. Also is the first date when the stock trades without dividend.
3. Holder-of-record date: A record of shareholders who are eligible to receive the dividend is made (usually two days after the ex-dividend date).
4. Payment date: Dividend payment made to the shareholders.

5. Dividend Discount Model (DDM) and Free-Cash-Flow-to-Equity Model (FCFE)

This model is based on the principle that the value of an asset should be equal to the present value of the expected future benefits. The simplest present value model is the dividend discount model (DDM). According to the DDM, the intrinsic value of a stock is the present value of future dividends, plus the present value of terminal value.

Intrinsic value = PV of future dividends + PV of terminal value

$$V_0 = \sum_{t=1}^n \frac{D^t}{(1+r)^t} + \frac{P^n}{(1+r)^n}$$

Example

For the next three years, the annual dividends of stock X are expected to be 1.0, 1.1, and 1.2. The expected stock price at the end of year 3 is expected to be \$20.00. The required rate of return on the shares is 10%. What is the estimated value?

Solution:

Calculate the present value of each of the future dividends at the required rate of return of 10%.

$$\text{PV of cash flow 1} = \frac{1}{1.1} = 0.909$$

$$\text{PV of cash flow 2} = \frac{1.1}{(1.1)^2} = 0.909$$

$$\text{PV of cash flow 3} = \frac{20+1.2}{(1.1)^3} = 15.928$$

$$\text{Estimated value} = 0.909 + 0.909 + 15.92 = 17.74$$

In the exam, use a financial calculator with the following keystrokes:

CF0 = 0; CF1 = 1; CF2 = 1.1; CF3 = 21.2; I = 10%, CPT NPV

NPV = 17.7

Free cash flow to equity (FCFE) is the residual cash flow available to be distributed as dividends to common shareholders. In practice, the FCFE model is often used because:

- FCFE is a measure of a firm's dividend-paying capacity.
- It can be used for a non-dividend paying stock (unlike DDM which requires the timing and the amount of the first dividend to be paid).
- It can also be used for a company that pays dividends which are extremely small or the dividends being paid are not an indication of a company's ability to pay dividends.
- Not all of the available cash flow is distributed to shareholders because a company retains some part of it for future investments as a going concern.

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

where: FCInv = fixed capital investment

Net borrowing = borrowings – repayments

$$V_0 = \sum_{t=1}^{\infty} \frac{\text{FCFE}^t}{(1+r)^t}$$

Required Rate of Return on a share

Analysts generally use CAPM (capital asset pricing model) to calculate the required return on a share.

Required rate of return on share = current expected risk free rate + Beta_i [market risk premium]

In addition to CAPM, there are other methods to calculate the required return like the bond yield plus risk premium method which we will see later.

6. Preferred Stock Valuation

For a non-callable, non-convertible perpetual preferred share paying a level dividend and assuming a constant required rate of return, the value is given by the equation below:

$$V_0 = \frac{D_0}{r}$$

where: V_0 = present value of the perpetuity; D_0 = dividend and r = rate of return

Example

A \$100 par value, non-callable, non-convertible perpetual preferred stock pays a 5% dividend. The discount rate is 8%. Calculate the intrinsic value of the preferred share.

Solution:

Expected annual dividend = $0.05 \times 100 = 5$

Value of the preferred share = $5.00/0.08 = 62.50$

Other types of preferred shares to consider are:

- *Shares which mature on a given date*: In the earlier example, instead of being a perpetual share, assume the share matured after four years. To calculate the value of this share, calculate the present value of the four dividends with the last one paid at the end of the fourth year at the required rate of 8%. Input these values in your financial calculator: $N = 4$; $I = 8$; $PMT = 5$; $FV = 100$; $PV = ?$ Present value of this share = 90.06.
- *Callable (redeemable) shares*: These shares are callable by the issuer at some point before maturity. Assuming all the conditions are the same as the shares which mature on a given date, will investors pay 90.06, or less or more for this share? They will pay less for this share as investors stand the risk of the issuer calling the share when it trades above the par value.
- *Shares with retraction option (putable shares)*: Here, the holder of the preferred stock has an option to sell the share to the issuer at a specified price before the maturity date. Unlike callable shares, putable shares will trade at a value above 90.06 as the put option is valuable to investors. If the share trades below the par value, investors can sell it back to the issuer.

7. The Gordon Growth Model

One of the disadvantages of the dividend discount model is that it is difficult to accurately estimate the amount of dividends for a long period of time. The Gordon growth model simplifies this by assuming that **dividends grow indefinitely at a constant rate**; it is also called the constant-growth dividend discount model. According to this model, the intrinsic value of a security can be calculated as:

$$V_0 = \frac{D_1}{r - g}$$

where:

D_1 = next period's dividend

r = required rate of return

g = dividend growth rate

In the equation above, if the growth rate is zero, then the equation reduces to the present value of a perpetuity.

To estimate a long-term growth rate of dividends, analysts use various methods such as:

- Using the historic growth rate for the firm
- Using the industry median growth rate
- Estimating the sustainable growth rate using the formula: $g = b \times ROE$
where: b = earnings retention rate = $(1 - \text{dividend payout ratio})$

ROE = return on equity

Assumptions of the Gordon Growth model:

- Dividends are the correct metric to use for valuation purposes. Dividends are a reflection of a company's earnings.
- Dividend growth rate is perpetual.
- Required rate of return is constant throughout the life of the security.
- Dividend growth rate < required rate of return.

When is it not appropriate to use the Gordon Growth Model?

- If the company is currently not paying a dividend as it may reinvest earnings in attractive opportunities.
- If the company is not profitable enough currently to pay a dividend. An analyst may still use the model by assuming that the company will pay a dividend in the future.

What happens to the value if dividend value is increased?

Let us look at the formula again. $V_0 = \frac{D_1}{r-g}$

When dividend increases, numerator increases. If the payout ratio increases, retention rate decreases and value of g decreases. If g decreases, the denominator increases. As a result, the impact on value, if the dividend is increased cannot be determined with certainty.

Example

Estimate the intrinsic value of a stock given the following data:

Beta = 1.5; RFR = 3%; market risk premium = 5%; dividend just paid = \$1.00; dividend payout ratio = 0.4; return on equity = 15%.

Solution:

$$V_0 = \frac{D_1}{r-g} = \frac{D_0 \cdot (1+g)}{r-g}$$

Note: the values of r, g and expected dividend are not given. So, first calculate these values.

$$r = \text{RFR} + \text{Beta} \times \text{market risk premium} = 3 + 1.5 \times 5 = 10.5\%$$

$$g = b \times \text{ROE} = (1 - 0.4) \times 0.15 = 0.09$$

$$\text{Applying the Gordon growth model, } V_0 = 1 \times \frac{1.09}{0.105 - 0.09} = 72.67$$

Example

A company does not currently pay dividend but is expected to begin to do so in 4 years. The first dividend is expected to be \$2.00 and to be received at the end of year 4. The dividend is expected to grow at 5% into perpetuity. The required return is 10%. What is the estimated current intrinsic value?

Solution:

To calculate the intrinsic value, first calculate the value of dividend at the end of period 3 and then discount it to $t=0$ using the Gordon growth model.

$$V_3 = \frac{D_4}{r - g} = \frac{2}{0.10 - 0.05} = 40$$

$$V_0 = \frac{40}{(1.1)^3} = 30.05$$

Instructor's Note: Do not forget to discount 40 to the present value. The undiscounted value is commonly presented as one of the answer options as a trap.

8. Multistage Dividend Discount Models

It is an ideal situation to assume that all companies grow at a constant rate indefinitely and pay a constant dividend. The assumption is true to an extent only for stable companies. In reality, companies go through a finite rapid growth phase followed by an infinite period of sustainable growth.

A two-stage DDM can be used to calculate the value of such companies transitioning from growth to mature stage. The Gordon growth model may be used to calculate the terminal value at the beginning of the second stage which represents the present value of dividends during the sustainable growth phase.

$$V_0 = \sum_{t=1}^n \frac{D_0(1 + g_s)^t}{(1 + r)^t} + \frac{V_n}{(1 + r)^n}$$

The first term is discounting the dividends during the high growth period. The second term is calculating the terminal value for the second sustainable growth period and then discounting it to the present value where V_n = terminal value at time n estimated using the Gordon growth model.

Example

Let us understand the concept better with the help of an example. The current dividend for a company is \$4.00. The dividends are expected to grow at 20% a year for 4 years and then at 10% after that. The required rate of return is 18%. Estimate the intrinsic value.

Solution:

First draw a timeline.



We will use this formula:

$$V_0 = \sum_{t=1}^n \frac{D_0(1+g_s)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n}$$

where $n = 4$ (high growth period)

Solve for the second term: $\frac{V_n}{(1+r)^n}; V_4 = \frac{D_5}{r-g} = \frac{D_4(1+g_L)}{r-g} = \frac{8.29 * 1.1}{0.18 - 0.1} = \frac{9.12}{0.08} = 114$

Using the financial calculator, we can calculate the present value of dividends and terminal value by entering the following values: $CF_0 = 0$; $CF_1 = 4.8$; $CF_2 = 5.76$; $CF_3 = 6.91$; $CF_4 = 8.29 + 114$; $I = 18$; $NPV = 75.48$

Note: while calculating V_4 , you need to use 10% as growth rate since it is the long-term growth rate.

Three Stage Models

The concept of a two-stage model can be extended to as many stages as a company goes through. Often, companies go through three stages beyond the startup phase: growth, transition, and maturity.

9. Multiplier Models and Relationship Among Price Multiples, Present Value Models, and Fundamentals

Price multiple is a ratio that uses a company's share price with some monetary flow/value for evaluating the relative worth of a company's stock. Commonly used price multiple ratios are listed below:

Price multiples	
Ratio	What it measures
Price-to-earnings ratio (P/E)	Trailing P/E: $\frac{\text{Price per share}}{\text{Trailing 12 month earnings per share}}$ For example, price = 50, EPS = 5; P/E = 10 Forward/leading/estimated P/E: $\frac{\text{Stock price}}{\text{Leading 12 month earnings per share}}$ Most commonly used ratio. Analysts prefer stocks with low P/E to high P/E.
Price-to-book ratio P/B	$P/B = \frac{\text{Price per share}}{\text{Book value per share}}$ Book value per share = $\frac{\text{Assets} - \text{Liabilities}}{\text{Shares outstanding}}$ Evidence suggests that companies with low P/B tend to outperform stocks with high P/B (expensive stock).
Price-to-sales ratio	$P/S = \frac{\text{Price per share}}{\text{Sales per share}}$

	Like P/E ratio, this can be trailing or leading ratio. One advantage of P/S ratio is that it can never be negative unlike P/E as earnings can be negative. It is useful during periods of economic slowdown or extraordinary growth.
Price-to-cash-flow ratio	$P/CF = \frac{\text{Price per share}}{\text{Cash flow per share}}$ One aspect to note here is what cash flow measure has been used by the analyst. The cash flow measure may be operating cash flow, free cash flow, etc.
Common criticism: These ratios do not consider the future. When forecasts of fundamental values are used, such as estimated EPS in leading P/E, the P/E value may differ substantially from the trailing P/E. When comparing companies, the multiples should be consistently used. For example, you cannot compare the trailing P/E of one company with a leading P/E of another company.	

Instructor's Note:

For a growing company, what will be higher: the leading P/E or the trailing P/E?

The trailing P/E will be higher as the earnings are higher in the future periods. So the leading P/E will be lower.

9.1 Relationships among Price Multiples, Present Value Models, and Fundamentals

We can link price multiple to fundamentals through a discounted cash flow model such as the Gordon Growth Model. How? By assuming that the intrinsic value of a security is equal to its market price, i.e., the security is fairly valued.

$$V_0 = \frac{D_1}{r-g} \text{ becomes } P_0 = \frac{D_1}{r-g} \text{ if we assume that: } V_0 = P_0$$

$$\text{Forward P/E} = \frac{P_0}{E_1} = \frac{\frac{D_1}{r-g}}{E_1} = \frac{\text{Dividend payout ratio}}{r-g}$$

The multiple you see above is related to the fundamentals as both dividend payout ratio and growth rate represent the fundamentals of a company. Some interpretations based on the formula:

- The forward P/E and payout ratio appear to be positively related. But, it does not necessarily mean a higher dividend payout increases the P/E.
- A higher payout ratio may mean the company is retaining less for reinvestment, which in turn means, a slower growth rate. Since P/E and growth rate are positively related, if g slows (denominator increases), then P/E decreases. This is known as dividend displacement of earnings.
- P/E is inversely related to the required rate of return.

Example

Between 2008 and 2012, a company's dividend payout ratio has been 40% on average. In 2008, the dividend was \$1.00 and has grown steadily to \$1.8 for 2012. This growth rate is expected to continue in the future. Using a discount rate of 20%, estimate the company's justified forward P/E.

Solution:

$$\frac{P}{E_1} = \frac{\text{Payout ratio}}{r - g}$$

The growth rate is not given. So calculate g with the information given about dividends. The growth rate is expected to continue; so it will be the long-term constant growth rate.

$$g = \left(\frac{1.8}{1}\right)^{\frac{1}{4}} - 1 = 0.16$$

$$\frac{P}{E_1} = \frac{0.4}{0.2 - 0.16} = 10$$

10. Method of Comparables and Valuation Based on Price Multiples

This method compares relative values estimated using multiples. The objective is to determine if a stock or asset is fairly valued, undervalued, or overvalued relative to the benchmark value of the multiple. For example, if the average P/B value for private sector banks is 1.1, and the P/B for the bank under consideration is 0.65, then it is relatively undervalued, all else equal. This method is based on the principle that similar assets should be priced the same: **the law of one price**.

For example, assume that there are two companies the data for which is given below:

	Company A	Company B
P	100	50
E_1	10	6
P/E	10	8.3

On a relative value basis, company B is a better buy.

The primary difference between P/E multiples based on comparables and P/E multiples based on fundamentals:

- P/E multiple based on comparables uses the law of one price. For example, if the trailing P/E of Caterpillar is 13.2, Komatsu is 15.5, and Deere is 9.6. Which one of these is undervalued? Given this data, Deere is undervalued relative to the other stocks.
- P/E multiple based on fundamentals is calculated as payout ratio/($r - g$). With this method we only need information about a target company.

10.1 Illustration of a Valuation Based on Price Multiples

In this section, we will see through examples how price multiples are used in cross-sectional analysis, time-series analysis, and in valuing a private company.

Example

The table below presents the current P/E ratio of a few automobile companies.

Company	P/E
Volkswagen	12.01
Ferrari	24.57
Lamborghini	13.42
Pagani	14.1

Solution:

This is a cross-sectional analysis as different companies are compared at a specific point in time. According to the data, Volkswagen is the most undervalued as it has the lowest P/E. For every \$ of earnings, we are paying \$12.01. It must be noted that several other factors in conjunction with relative value analysis must be performed before making a buy decision. Share prices plunge if a company is on the verge of bankruptcy.

Example

The table below computes the P/E ratio for Nikon over a five period 2012 - 2016. Determine if the stock is overvalued or undervalued relative to historic levels?

Year	Price (in \$)	EPS	P/E = Price/EPS
2012	17.52	1.71	10.25
2013	29.19	1.42	20.56
2014	35.7	1.2	29.75
2015	7.55	0.61	12.38
2016	5.42	0.48	11.3

Solution:

This is a time series analysis. The 2012 P/E level for Nikon indicates it is undervalued relative to the historic high of 29.75 in 2014. Analysts may recommend buying the stock if it were to return to the historic high levels provided the increase in P/E is not due to a decrease in EPS, which is not the case here. Other fundamental factors should also be considered such as slowing revenues, the growing popularity of alternative cameras and smartphones affecting Nikon's business, slowing economy, etc.

11. Enterprise Value

Enterprise value is used as an alternate measure for equity; it measures the market value of the whole company (debt and equity).

Enterprise value = market value of debt + market value of equity + market value of preferred stock – cash and investments

$EV = MVE + MVD + MVP - \text{cash and cash equivalents}$

Note: use estimates if market values are not available

The most commonly used EV multiple is EV/EBITDA. EBITDA is earnings before interest, taxes, depreciation, and amortization. It is a proxy for cash flow, or how much cash the company is generating. However, it may include other non-cash expenses and revenues.

When is EV/EBITDA used?

- When earnings are negative, making P/E useless. EBITDA is usually positive.
- For comparing companies with significant differences in capital structure.
- To evaluate the cost of a takeover.

A major limitation of the enterprise value model is that it is difficult to obtain the market value of debt.

Example

The EV/EBITDA ratio for a company is 10. EBITDA is 20 million. Market value of debt is 50 million. Cash is 2 million. What is the value of equity?

Solution:

$$\frac{EV}{EBITDA} = \frac{MVD + MVE - \text{Cash}}{EBITDA}$$

$$10 = \frac{50 + MVE - 2}{20}$$

MVE = 152 million

12. Asset-Based Valuation

An asset-based valuation of a company uses the estimates of the market or the fair value of the company's assets and liabilities. This valuation method is appropriate for companies that have low proportion of intangible or off-the-books assets. It is commonly used for valuing private enterprises.

Other factors to consider:

- Book values may be very different from market values.
- Some intangible assets are not reported; asset-based value could be considered a 'floor' value.
- Asset values are hard to estimate in a hyper-inflationary environment.

Some examples when this method is not appropriate:

- A hugely popular restaurant in a rented space. The restaurant is popular because of the proprietor's cooking skills and secret recipes. The proprietor would like to sell the

business and retire. This method is not appropriate as setting a value for the proprietor's cooking skills is challenging. Only the restaurant's equipment, inventory, and furniture can be valued.

- In the case of a laundry business, the equipment and inventory can be valued at depreciated value or at replacement cost. But intangibles such as convenience due to location, clever marketing, etc. cannot be assigned a value.

The tables below list the pros and cons of the different valuation models we have seen so far.

Comparables Valuation Using Multiples	
Advantages	Disadvantages
Good predictor of future returns.	Lagging numbers tell about past.
Widely used.	Not always comparable across firms.
Easily available.	Impacted by economic conditions.
Time-series comparison.	Might conflict with fundamental method.
Cross-sectional comparison.	Sensitive to different accounting methods.
Allows us to identify relatively underpriced securities.	Negative denominator.

DCF	
Advantages	Disadvantages
Based on PV of future cash flows.	Inputs have to be estimated.
Widely accepted and used.	Estimates sensitive to inputs.

Asset-Based Model	
Advantages	Disadvantages
Floor values.	Market values hard to determine.
Works when assets have easily determinable market values.	Market values often different from book values.
Works well for companies that report fair values.	Do not account for intangible assets.
	Asset values hard to determine during hyperinflation.

Summary

LO: Evaluate whether a security, given its current market price and a value estimate, is overvalued, fairly valued, or undervalued by the market.

Market value > Intrinsic value - Overvalued

Market value = Intrinsic value - Fairly valued

Market value < Intrinsic value - Undervalued

Factors to consider when market value $\neq$ intrinsic value:

- Percentage difference between the market price and intrinsic value.
- Confidence in your model.
- Model sensitivity to assumptions.
- Number of analysts.

LO: Describe major categories of equity valuation models.

Type of Model	Characteristics
Present Value Models	• Estimate intrinsic value as the present value of expected future benefits. • Future benefits defined as cash to be paid to shareholders, or cash flows available to be distributed to shareholders. • Ex: Gordon growth model, two-stage dividend discount model, free cashflow to equity model.
Multiplier Models, also known as market multiple models	• Based on share price multiples or enterprise value multiples. • The share price multiple model estimates intrinsic value based on a multiple of some fundamental variable such as revenues, earnings, cash flows, or book value. • Ex: P/E, P/S • Enterprise value multiple models are of the form: enterprise value/some fundamental variable. Here, the fundamental variable is usually EBITDA or revenue.
Asset-Based Models	• Estimate intrinsic value based on the estimated value of assets and liabilities.

LO: Describe regular cash dividends, extra dividends, stock dividends, stock splits, reverse stock splits, and share repurchases

Cash dividends are payments made to shareholders in cash. The three types of cash dividends are:

1. Regular cash dividends are paid out on a consistent basis. Stable or increasing dividend is viewed as a sign of financial stability.

2. Special dividends are one-time cash payments when the situation is favorable (also called as extra dividends or irregular dividends; used by cyclical firms).
3. Liquidating dividend is distributed to shareholders when a company goes out of business.

Stock dividends are payments made to shareholders in additional shares instead of cash.

Stock Splits divides each existing share into multiple shares.

Reverse stock splits are the opposite of stock splits and decreases the total number of outstanding shares.

Share repurchase is when a company buys back its own outstanding shares using cash.

LO: Describe dividend payment chronology

A dividend payment schedule is as follows:

1. Declaration date: Board of directors approves dividend.
2. Ex-dividend date: Cutoff date on or after which buyers of a stock are not eligible for the dividend. It is also the first date when the stock trades without dividend.
3. Holder-of-record date: An entry of shareholders eligible for the dividend is made (usually two days after the ex-dividend date).
4. Payment date: Dividend payment made to the shareholders.

LO: Explain the rationale for using present value models to value equity and describe the dividend discount and free-cash-flow-to-equity models.

This model is based on the principle that the value of an asset should be equal to the present value of the expected future benefits. The simplest present value model is the dividend discount model (DDM).

According to the DDM, the intrinsic value of a stock is the present value of future dividends, plus the present value of the terminal value. It can be calculated using the formula:

$$V_0 = \sum_{t=1}^n \frac{D^t}{(1+r)^t} + \frac{P^n}{(1+r)^n}$$

Free-cash-flow-to-equity (FCFE) is the residual cash flow available to be distributed as dividend to common shareholders. FCFE model is used because it is a measure of a firm's dividend-paying capacity and can be used for stocks with small dividends or no dividend.

$$FCFE = CFO - FCInv + \text{Net borrowing}$$

$$V_0 = \sum_{t=1}^{\infty} \left(\frac{FCFE^t}{(1+r)^t} \right)$$

$$\text{Required return on share} = \text{risk free rate} + \text{Beta}_i [\text{market risk premium}]$$

LO: Calculate the intrinsic value of a non-callable, non-convertible preferred stock.

$$V = \frac{D}{r}$$

where:

V = present value of the perpetuity

D = dividend and r = rate of return

LO: Calculate and interpret the intrinsic value of an equity security based on the Gordon (constant) growth dividend discount model or a two-stage dividend discount model, as appropriate.

The Gordon growth model assumes that dividends grow indefinitely at a constant rate. It is also called the constant growth dividend discount model.

$$V_0 = \frac{D_1}{r - g}$$

where: g = retention rate * ROE

For a multi staged dividend discount model:

$$V_0 = \sum_{t=1}^n \frac{D_0(1 + g_s)^t}{(1 + r)^t} + \frac{V_n}{(1 + r)^n}$$

The first term is discounting the dividends during the high growth period. The second term is calculating the terminal value for the second sustainable growth period and then discounting it to the present value.

LO: Identify characteristics of companies for which the constant growth or a multistage dividend discount model is appropriate.

The constant growth model is appropriate for companies that pay dividends growing at a constant rate. These are usually mature and stable firms (e.g., producer of a staple food product).

A two-stage DDM can be used to calculate the value of companies transitioning from growth to mature stage.

A three-stage model is used for companies that go through three stages beyond the startup phase: growth, transition, and maturity.

LO: Explain the rationale for using price multiples to value equity and distinguish between multiples based on comparables versus multiples based on fundamentals.

Price multiple is a ratio that uses a company's share price with some monetary flow/value for evaluating the relative worth of a company's stock.

The method of comparables compares relative values estimated using multiples. The objective is to determine if a stock or asset is fairly valued, undervalued, or overvalued relative to the benchmark value of the multiple. It is based on the law of one price.

Price multiple can be linked to fundamentals through a discounted cash flow model such as the Gordon Growth Model by assuming that the intrinsic value of a security is equal to its market price, i.e., the security is fairly valued.

$$\text{Forward P/E} = P_0/E_1 = \frac{\frac{D_1}{E_1}}{r - g} = \frac{\text{dividend payout ratio}}{r - g}$$

LO: Calculate and interpret the following multiples: price to earnings, price to an estimate of operating cash flow, price to sales, and price to book value.

Price-to-earnings ratio (P/E):

$$\text{Trailing P/E} = \frac{\text{price per share}}{\text{trailing 12 month earnings per share}}$$

$$\text{Forward P/E} = \frac{\text{stock price}}{\text{leading 12 month earnings per share}}$$

$$\text{Price – to – book ratio P/B} = \frac{\text{price per share}}{\text{book value per share}}$$

$$\text{Price – to – sales ratio P/S} = \frac{\text{price per share}}{\text{sales per share}}$$

$$\text{Price – to – cashflow ratio P/CF} = \frac{\text{price per share}}{\text{cash flow per share}}$$

LO: Describe enterprise value multiples and their use in estimating equity value.

Enterprise value is used as an alternate measure for equity. It measures the market value of the whole company (debt and equity).

$$\text{Enterprise value} = \text{market value of debt} + \text{market value of equity} + \text{market value of preferred stock} - \text{cash and investments}$$

The most commonly used EV multiple is EV/EBITDA. It is used in the following situations:

- When earnings are negative making P/E useless.
- For comparing companies with significant differences in capital structure.
- To evaluate the cost of a takeover.

LO: Describe asset-based valuation models and their use in estimating equity value.

An asset-based valuation of a company uses the estimates of the market or the fair value of the company's assets and liabilities. This valuation method is appropriate for companies that have low proportion of intangible or off-the-books assets. It is commonly used for valuating private enterprises.

LO: Explain advantages and disadvantages of each category of valuation model.

Advantages	Disadvantages
Comparables Valuation Using Multiples	
Good predictor of future returns.	Lagging numbers tell about past.
Widely used.	Not always comparable across firms.
Easily available.	Impacted by economic conditions.
Time-series comparison.	Might conflict with fundamental method.
Cross-sectional comparison.	Sensitive to different accounting methods.
Allows us to identify relatively underpriced securities.	Negative denominator.
DCF	
Based on PV of future cash flows.	Inputs have to be estimated.
Widely accepted and used.	Estimates sensitive to inputs.
Asset-Based Model	
Floor values.	Market values hard to determine.
Works when assets have easily determinable market values.	Market values often different from book values.
Works well for companies that report fair values.	Does not account for intangible assets.
	Asset values hard to determine during hyperinflation.